Emergence

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Part I

Symmetry

1 Abelian Harmonic Analysis

1.1 Fourier series and Fourier transform

Definition 1.1 (Fourier series)**.** With normalized Haar measure on $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x)e^{-ikx}\,dx, \qquad k \in \mathbb{Z}.$$

The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}$.

Definition 1.2 (Fourier transform)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, the semiclassical Fourier transform is

$$(\mathcal{F}_\hbar f)(p) = (2\pi\hbar)^{-n/2} \int_{\mathbb{R}^n} e^{-ip\cdot x/\hbar} f(x)\,dx.$$

Exercise 1.1 (Fourier inversion and Plancherel)**.** Prove that $(e^{ikx})_{k\in\mathbb{Z}}$ is an orthonormal basis of $L^2(\mathbb{T})$ and derive

$$f = \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}, \qquad \|f\|_2^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Prove Fourier inversion on $\mathcal{S}(\mathbb{R}^n)$ and prove that $\mathcal{F}_\hbar$ extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$. Show that convolution becomes multiplication and differentiation becomes multiplication by frequency.

2 Harmonic and Representation-Theoretic Applications

2.1 Lattices, Poisson summation, and diffraction

Definition 2.1 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi\cdot\gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Definition 2.2 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma\in\Gamma} f(x+\gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b\in B} c_b e^{-iq\cdot b}.$$

Exercise 2.1 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi\in\Gamma^*} \mathcal{F}f(\xi)e^{i\xi\cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 2.2 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma\in\Gamma_N} e^{-iq\cdot\gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d \sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

2.2 Groups, representations, and abelian duality

Definition 2.3 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Definition 2.4 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Definition 2.5 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 2.3 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 2.4 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 2.6 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|} \sum_{g\in G} \overline{f(g)}h(g).$$

Exercise 2.5 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V \rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Exercise 2.6 (Molecular symmetry)**.** For the equilibrium geometry of NH_3, determine the C_{3v}-action on nuclear displacements, remove translations and rotations, and prove

$$V_{\mathrm{vib}} \cong 2A_1 \oplus 2E.$$

Deduce four fundamental frequencies, two of them doubly degenerate. For NH_2D, restrict to C_s and prove $E|_{C_s} \cong A' \oplus A''$. Deduce that symmetry no longer enforces either degeneracy and compare with the observed band splitting.

Definition 2.7 (Topological spaces)**.** A *topology* on X is a family of subsets containing $\varnothing, X$ and closed under arbitrary unions and finite intersections. A map is *continuous* when inverse images of open sets are open. A space is *Hausdorff* when distinct points have disjoint neighborhoods, *second countable* when its topology is generated by a countable family of open sets, *compact* when every open cover has a finite subcover, and *locally compact* when every point has a neighborhood with compact closure. The *Borel sets* form the σ-algebra generated by the open sets. A Borel measure is *regular* when it is inner regular on open sets and outer regular on Borel sets.

Definition 2.8 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

Definition 2.9 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$. The disjoint union $TM = \coprod_{p \in M} T_pM$ has a canonical smooth vector-bundle structure: if $x = (x^1, \ldots, x^n)$ is a coordinate chart on U, its bundle chart is

$$TM|_U \longrightarrow x(U) \times \mathbb{R}^n, \qquad v \in T_pM \longmapsto (x(p), v(x^1), \ldots, v(x^n)).$$

The cotangent bundle T^*M has the dual bundle charts.

Definition 2.10 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. A *flow* of X is a smooth local action Φ of $\mathbb{R}$ on M satisfying $\frac{d}{dt}\Big|_0 \Phi_t(p) = X_p$. The *Lie bracket* is the vector field $[X, Y](f) = X(Yf) - Y(Xf)$, and the *Lie derivative* of Y along X is $\mathcal{L}_X Y = [X, Y]$.

Exercise 2.7 (Flows and brackets)**.** Prove local existence and uniqueness of flows. Prove that $[X, Y] = 0$ if and only if their local flows commute, and that the Lie derivative is natural under diffeomorphisms.

Definition 2.11 (Lie groups)**.** A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 2.8 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 2.12 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \frac{d}{dt}\Big|_0 \rho(\exp tX).$$

Definition 2.13 (Convolution)**.** Let G be a locally compact group with left Haar measure μ. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x)\, d\mu(y).$$

Exercise 2.9 (Haar measure and the regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \le \|f\|_1 \|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 2.14 (Pontryagin dual)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, d\mu(x).$$

The *double-dual map* sends $x \in G$ to the character $\chi \mapsto \chi(x)$ of $\widehat{G}$.

Exercise 2.10 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on $\mathbb{R}^n$ and Fourier series on $\mathbb{T}^n$. For a closed subgroup $H \le G$, define its annihilator $H^\perp \subseteq \widehat{G}$ and prove $\widehat{G/H} \cong H^\perp$.

Exercise 2.11 (Interference and Fourier optics)**.** For coherent scalar waves with complex amplitudes a_1, a_2, derive $|a_1 + a_2|^2$ and the fringe spacing in Young's double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

2.3 Compact groups, rotations, and spin

Exercise 2.12 (Peter–Weyl theorem)**.** Let G be compact with normalized Haar measure. Prove that matrix coefficients of irreducible unitary representations span a dense subspace of $C(G)$ and $L^2(G)$ and that

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}}} H_\pi \otimes H_\pi^*.$$

For $f \in L^2(G) \subseteq L^1(G)$, define $\widehat{f}(\pi) = \int_G f(g)\pi(g)^* \, dg$ and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \operatorname{Tr}(\widehat{f}(\pi)\pi(\,\cdot\,)), \qquad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \|\widehat{f}(\pi)\|_{\mathrm{HS}}^2$$

, where the first sum converges in $L^2(G)$. For $f \in C(G)$, construct finite matrix-coefficient approximate identities and prove uniform convergence of the corresponding summability means to f.

Definition 2.15 (Rotations and spin)**.** Define

$$SO(3) = \{R \in \operatorname{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},$$
$$SU(2) = \{U \in \operatorname{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}.$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 2.13 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm \mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 2.14 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \dots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \dots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 2.15 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 2.16 (Quantum symmetries)**.** Let a compact group G act continuously and unitarily by $U : G \to U(H)$ on a Hilbert space with self-adjoint Hamiltonian $\mathcal{H}$. It is a *symmetry group* when $U_g \operatorname{Dom}(\mathcal{H}) = \operatorname{Dom}(\mathcal{H})$ and $\mathcal{H}U_g\psi = U_g\mathcal{H}\psi$ for every $g \in G$. A *quantum number* is an eigenvalue of a self-adjoint operator whose spectral projections commute with those of $\mathcal{H}$. A bounded operator multiplet A has *symmetry type* W when its components define an intertwiner from W to $B(H)$ under conjugation by G.

Exercise 2.16 (Symmetry, degeneracy, and selection rules)**.** Prove that every finite-dimensional eigenspace of $\mathcal{H}$ is symmetry-invariant and decomposes into irreducible G-representations. Deduce symmetry-forced degeneracies. Prove that a transition matrix element $\langle f, Ai \rangle$ can be nonzero only if the trivial representation occurs in $V_f^* \otimes W \otimes V_i$, and derive the angular-momentum rules for an electric-dipole operator.

Exercise 2.17 (Atomic spectra)**.** For the Coulomb Hamiltonian, separate the Schrödinger equation in spherical coordinates, obtain the quantum numbers n, ℓ, m, and derive

$$E_n = -\frac{m_e e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}.$$

Recover the Balmer series and compare its predicted energy differences with the empirical values computed in Exercise 31.5. Apply the electric-dipole rule to identify forbidden transitions.

Exercise 2.18 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

and identify a basis of V_ℓ with $Y_{\ell m} \propto D^{\ell}_{m0}$. For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\phi^2.$$

Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$ and completeness of the spherical harmonics. Using the Coulomb separation from Exercise 2.17, identify its angular factor and recover $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$. Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

Exercise 2.19 (Molecular rotational spectra)**.** Model a rigid molecule with orientation $R \in SO(3)$ and principal moments I_1, I_2, I_3. On $L^2(SO(3))$, derive

$$H_{\mathrm{rot}} = \frac{J_1^2}{2I_1} + \frac{J_2^2}{2I_2} + \frac{J_3^2}{2I_3}.$$

Use the Peter–Weyl basis to reduce H_{rot} to finite-dimensional blocks. For a symmetric top, derive

$$E_{JK} = \frac{\hbar^2 J(J+1)}{2I_\perp} + \frac{\hbar^2 K^2}{2}\left(\frac{1}{I_\parallel} - \frac{1}{I_\perp}\right).$$

Derive the electric-dipole selection rules and the splitting in a static electric field. Compare the predicted components with microwave measurements of the symmetric-top molecule s-trioxane reported in NBS Special Publication 343 (1971).

2.4 Semisimple harmonic analysis

Definition 2.17 (Cartan and Iwasawa decompositions)**.** Let G be a connected real semisimple Lie group with finite center and let $\mathfrak{g}$ be its Lie algebra. A *Cartan involution* gives a vector space decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

into its $+1$ and -1 eigenspaces; the subgroup K with Lie algebra $\mathfrak{k}$ is maximal compact, and G/K is a noncompact Riemannian symmetric space. Choose a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$, put $A = \exp\mathfrak{a}$, choose positive restricted roots, and let N be the subgroup generated by their root spaces. The *Iwasawa decomposition* is the diffeomorphism

$$K \times A \times N \longrightarrow G, \qquad (k,a,n) \longmapsto kan.$$

If $\mathfrak{a}^+$ is a positive Weyl chamber, the *Cartan decomposition* is $G = K\overline{A^+}K$. Write $M = Z_K(A)$ and $W = N_K(A)/M$, and let ρ be half the sum of the positive restricted roots, counted with multiplicity.

Definition 2.18 (Spherical principal series and spherical transform)**.** For $\lambda \in \mathfrak{a}^*$, the normalized *spherical principal series* is

$$\pi_\lambda = \mathrm{Ind}_{MAN}^{G}(1_M \otimes e^{i\lambda} \otimes 1_N),$$

where normalized induction incorporates the modular factor determined by ρ. It contains a unit K-fixed vector v_λ, and its *zonal spherical function* is

$$\varphi_\lambda(g) = \langle v_\lambda, \pi_\lambda(g) v_\lambda \rangle.$$

It is K-bi-invariant, satisfies $\varphi_\lambda(e) = 1$, and is a joint eigenfunction of the G-invariant differential operators on G/K.

For $f \in C_c^\infty(K\backslash G/K)$, its *spherical transform* is

$$\widetilde{f}(\lambda) = \int_G f(g)\varphi_{-\lambda}(g)\, dg.$$

Harish-Chandra's c-function is the coefficient governing the leading asymptotics of $\varphi_\lambda(\exp H)$ as $H \to \infty$ in a positive chamber.

Exercise 2.20 (Harish-Chandra transform on $\mathbb{H}^2$)**.** For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\mathrm{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For a smooth compactly supported radial function, define

$$\widetilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r) \sinh r\, dr.$$

Prove directly from the differential equation that distinct real values of λ^2 give orthogonal generalized eigenfunctions after truncation to $[0, R]$ with a common self-adjoint boundary condition. Compare the large-r equation with a constant-coefficient wave equation and show that its oscillation parameter is continuous. Contrast this with the discrete Dirichlet spectrum in the finite hyperbolic-drum exercise below.

Exercise 2.21 (A finite hyperbolic drum)**.** Discretize the Laplace–Beltrami operator by a real symmetric weighted graph Laplacian on a finite regular hyperbolic tiling with Dirichlet boundary. Prove that the finite operator has a discrete orthogonal eigenbasis, and explain why those eigenvalues approximate a bounded domain with a boundary rather than the continuous spectrum of all $\mathbb{H}^2$. Compare the predicted mode ordering and propagation along discrete geodesics with the circuit measurements in Lenggenhager et al., *Nature Communications* **13** (2022). State which observations support the effective hyperbolic metric and which do not by themselves establish the infinite-space Plancherel theorem.

Definition 2.19 (Harish-Chandra characters and discrete series)**.** A *distribution* on a Lie group G is a continuous linear functional on $C_c^\infty(G)$ with its usual test-function topology. Let π be an irreducible admissible unitary representation of a connected real reductive group. Although the infinite-dimensional operator $\pi(g)$ usually has no trace, for $f \in C_c^\infty(G)$ the integrated operator

$$\pi(f) = \int_G f(g)\pi(g)\,dg$$

is trace class. The *Harish-Chandra character* is the conjugation-invariant distribution

$$\Theta_\pi(f) = \mathrm{Tr}(\pi(f)).$$

An irreducible unitary representation belongs to the *discrete series* if it has a nonzero matrix coefficient square-integrable modulo the center.

Exercise 2.22 (From compact characters to the semisimple Plancherel formula)**.** For a compact group, prove

$$\Theta_\pi(f) = \int_G f(g)\,\mathrm{Tr}(\pi(g))\,dg$$

and recover the ordinary character used in Peter–Weyl theory. Show that a unitary operator on an infinite-dimensional Hilbert space is not trace class, which explains the distributional definition above. Prove directly that Θ_π is invariant under conjugation.

For $SL_2(\mathbb{R})$, identify the K-types of the spherical principal series and show that they contain a K-fixed vector. Contrast them with the holomorphic and antiholomorphic discrete series, whose nonzero one-sided K-types imply that they have no K-fixed vector. Deduce that spherical analysis on G/K sees the class-one principal series but not the discrete series. Compare the two families tabulated in Harish-Chandra, *Proceedings of the National Academy of Sciences* **43** (1957) for spherical functions and Harish-Chandra, *Transactions of the American Mathematical Society* **119** (1965) for invariant eigendistributions.

3 Unitary Coefficient Spaces

Definition 3.1 (Unitary representations and coefficient spaces)**.** Let G be a locally compact group and let $\pi : G \to U(H_\pi)$ be a strongly continuous unitary representation. Strong continuity means that $s \mapsto \pi(s)x$ is continuous for every $x \in H_\pi$. Define

$$\Phi_\pi : S_1(H_\pi) \longrightarrow C_b(G), \qquad \Phi_\pi(A)(s) = \mathrm{Tr}(A\pi(s)).$$

The *coefficient space* of π is

$$A_\pi(G) = \Phi_\pi(S_1(H_\pi))$$

with the quotient operator-space structure from $S_1(H_\pi)/\ker\Phi_\pi$.

Exercise 3.1 (Coefficient-space factorization)**.** Prove that $A_\pi(G)$ consists of the functions

$$u(s) = \sum_{j=1}^{\infty} \langle y_j, \pi(s)x_j\rangle, \qquad \sum_{j=1}^{\infty} \|x_j\|\,\|y_j\| < \infty,$$

and that its norm is the infimum of the displayed sums. For unitary representations π and ρ, prove that pointwise multiplication induces a complete contraction

$$A_\pi(G) \mathbin{\widehat{\otimes}} A_\rho(G) \longrightarrow A_{\pi\otimes\rho}(G).$$

4 The Fourier Algebra

Definition 4.1 (Left regular representation and Fourier algebra)**.** Let G be a locally compact group with left Haar measure. Its *left regular representation* on $L^2(G)$ is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

The *Fourier algebra* is

$$A(G) = A_\lambda(G).$$

Definition 4.2 (Pontryagin dual)**.** For a locally compact abelian group G, its *Pontryagin dual* is

$$\widehat{G} = \{\chi : G \to \mathbb{T} : \chi \text{ is a continuous homomorphism}\}$$

with pointwise multiplication and the compact-open topology.

Exercise 4.1 (Fell absorption and the Fourier algebra)**.** Prove Fell's absorption principle

$$\lambda \otimes \pi \cong \lambda \otimes I_{H_\pi}$$

for every unitary representation π of G. Deduce that $A(G)$ is a completely contractive Banach algebra under pointwise multiplication and that $A(G) \subseteq C_0(G)$. For abelian G, prove the completely isometric algebra identification

$$A(G) \cong L^1(\widehat{G})$$

under the inverse Fourier transform.

5 Dual Algebras of Coefficient Spaces

Definition 5.1 (Operator topologies and commutants)**.** The *weak operator topology* on $B(H)$ is generated by $T \mapsto |\langle x, Ty\rangle|$, and the *strong operator topology* is generated by $T \mapsto \|Tx\|$. For $S \subseteq B(H)$, define

$$S' = \{T \in B(H) : Ts = sT \text{ for every } s \in S\}, \qquad S'' = (S')'.$$

A *von Neumann algebra* is a unital $*$-subalgebra $M \subseteq B(H)$ satisfying $M = M''$.

Definition 5.2 (Coefficient von Neumann algebras)**.** For a unitary representation $\pi : G \to U(H_\pi)$, define

$$VN_\pi(G) = \{\pi(s) : s \in G\}''.$$

The *group von Neumann algebra* is

$$VN(G) = VN_\lambda(G).$$

For concrete von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, define

$$M \mathbin{\overline{\otimes}} N = (M \odot N)'' \subseteq B(H \otimes K).$$

Definition 5.3 (Normal functionals and maps)**.** A positive linear functional $\varphi : M \to \mathbb{C}$ on a von Neumann algebra is *normal* when

$$a_i \uparrow a \quad \Longrightarrow \quad \varphi(a_i) \uparrow \varphi(a)$$

for every bounded increasing net in M_+. The linear span of the normal positive functionals is the *predual* M_*. The *ultraweak topology* is $\sigma(M, M_*)$. A bounded linear map between von Neumann algebras is *normal* when it is ultraweakly continuous.

Exercise 5.1 (Coefficient-space duality)**.** Prove the bicommutant theorem. Prove the completely isometric identifications

$$A_\pi(G)^* \cong VN_\pi(G), \qquad A(G)^* \cong VN(G),$$

with pairing induced by $(A, T) \mapsto \mathrm{Tr}(AT)$. Prove that the induced weak-$*$ topology on $VN_\pi(G)$ is its ultraweak operator topology and that its predual is $A_\pi(G)$. Prove $M \cong (M_*)^*$ isometrically for every von Neumann algebra M.

6 Group and Twisted Group C^*-Algebras

Definition 6.1 (Twisted convolution and involution)**.** Let G be locally compact with left Haar measure, modular function Δ_G, and continuous normalized multiplier σ. For $f, g \in C_c(G)$, define

$$(f *_\sigma g)(x) = \int_G f(y)g(y^{-1}x)\sigma(y, y^{-1}x)\, dy$$

and

$$f^{*_\sigma}(x) = \Delta_G(x^{-1})\overline{\sigma(x, x^{-1})}\,\overline{f(x^{-1})}.$$

For a σ-projective unitary representation U, define

$$U(f) = \int_G f(x)U_x\, dx.$$

Definition 6.2 (Full and reduced twisted group C^*-algebras)**.** Define

$$C^*(G, \sigma) = \overline{C_c(G)}^{\|\cdot\|_{\max,\sigma}}, \qquad \|f\|_{\max,\sigma} = \sup_U \|U(f)\|,$$

where U ranges over σ-projective unitary representations. The *twisted left regular representation* is

$$(\lambda_\sigma(x)\xi)(t) = \sigma(x, x^{-1}t)\xi(x^{-1}t),$$

and

$$C^*_r(G, \sigma) = \overline{\lambda_\sigma(C_c(G))}^{\|\cdot\|}.$$

For $\sigma = 1$, write $C^*(G)$ and $C^*_r(G)$.

Exercise 6.1 (Twisted integrated representations)**.** Prove that $C_c(G)$ is a $*$-algebra under $*_\sigma$ and $*_\sigma$ and that

$$U(f *_\sigma g) = U(f)U(g), \qquad U(f^{*_\sigma}) = U(f)^*.$$

Prove that nondegenerate $*$-representations of $C^*(G, \sigma)$ correspond to strongly continuous σ-projective unitary representations of G. Prove that cohomologous multipliers define isomorphic full and reduced twisted group C^*-algebras.

Exercise 6.2 (Group operator-algebra completions)**.** Prove

$$VN(G) = C^*_r(G)'' = \overline{C^*_r(G)}^{\mathrm{SOT}} = \overline{C^*_r(G)}^{\mathrm{WOT}}.$$

For abelian G, prove

$$C^*(G) \cong C^*_r(G) \cong C_0(\widehat{G}).$$

For a finite-dimensional symplectic space (E, ω) and $\lambda \neq 0$, use the Stone–von Neumann theorem to prove

$$C^*(E, \sigma_{\lambda\omega}) \cong C^*_r(E, \sigma_{\lambda\omega}) \cong \mathcal{K}(L^2(V))$$

when $E = V \oplus V^*$.

Exercise 6.3 (SNAG theorem)**.** For a locally compact abelian group G, use $C^*(G) \cong C_0(\widehat{G})$ and the representation theorem for $C_0(X)$ to prove that every strongly continuous unitary representation $U : G \to U(H)$ has a unique projection-valued measure E_U on $\widehat{G}$ satisfying

$$U_s = \int_{\widehat{G}} \chi(s)\, dE_U(\chi), \qquad s \in G.$$

7 Fourier–Wigner Analysis

Definition 7.1 (Unitary systems and coefficient transforms)**.** Let (X, μ) be a measure space and let $U : X \to U(H)$ be a weakly measurable map. The *coefficient transform* of the unitary system U is

$$\mathcal{V}_U : S_1(H) \longrightarrow L^\infty(X), \qquad \mathcal{V}_U(A)(x) = \operatorname{Tr}(AU_x).$$

Definition 7.2 (Weyl system and Fourier–Wigner transform)**.** Let V be an n-dimensional real linear space with Lebesgue measure dq, and equip V^* with the dual measure dp for which $(2\pi)^{-n/2} \int_V e^{-ip(q)} f(q)\, dq$ is unitary on L^2. For $q \in V$ and $p \in V^*$, define

$$(T_q\xi)(x) = \xi(x - q), \qquad (M_p\xi)(x) = e^{ip(x)}\xi(x).$$

For $z = (q, p) \in V \oplus V^*$, define

$$W(z) = e^{-ip(q)/2} M_p T_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W} : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \qquad \mathcal{W}(A)(z) = (2\pi)^{-n/2} \operatorname{Tr}(AW(z)).$$

Exercise 7.1 (Weyl relations and the Moyal identity)**.** Prove

$$M_p T_q = e^{ip(q)} T_q M_p, \qquad W(z)^* = W(-z),$$

and

$$W(q, p)W(q', p') = \exp\left(\frac{i}{2}(p(q') - p'(q))\right) W(q + q', p + p').$$

For $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$, prove the Moyal identity

$$\int_{V \oplus V^*} \overline{\mathcal{W}(A)(z)} \mathcal{W}(B)(z)\, dz = \operatorname{Tr}(A^* B).$$

Deduce that $\mathcal{W}$ extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever $\mathcal{W}(A) \in L^1(V \oplus V^*)$, prove

$$A = (2\pi)^{-n/2} \int_{V \oplus V^*} \mathcal{W}(A)(z) W(z)^*\, dz$$

in the weak operator topology.

8 Symplectic Geometry from the Weyl System

8.1 Projective representations and multipliers

Definition 8.1 (Multipliers and projective representations)**.** A *normalized multiplier* on a locally compact group G is a continuous map $\sigma : G \times G \to \mathbb{T}$ satisfying

$$\sigma(x,e) = \sigma(e,x) = 1, \qquad \sigma(x,y)\sigma(xy,z) = \sigma(y,z)\sigma(x,yz).$$

A *σ-projective unitary representation* is a strongly continuous map $U : G \to U(H)$ satisfying

$$U_x U_y = \sigma(x,y) U_{xy}, \qquad U_e = I.$$

Multipliers σ and σ' are *cohomologous* when there is a continuous map $b : G \to \mathbb{T}$, $b(e) = 1$, such that

$$\sigma'(x,y) = b(x)b(y)\overline{b(xy)}\,\sigma(x,y).$$

Exercise 8.1 (The Weyl multiplier)**.** Prove that the phase in the Weyl product is a normalized multiplier on $V \oplus V^*$.

8.2 Alternating commutator form

Definition 8.2 (Commutator bicharacter)**.** For a multiplier σ on an abelian group A, define

$$c_\sigma(x,y) = \sigma(x,y)\overline{\sigma(y,x)}.$$

Exercise 8.2 (Commutator bicharacter)**.** Prove that c_σ is an alternating bicharacter and depends only on the cohomology class of σ. For the Weyl multiplier on $V \oplus V^*$, prove

$$c_\sigma\big((q,p),(q',p')\big) = e^{i(p(q')-p'(q))}.$$

8.3 Linear symplectic geometry

Definition 8.3 (Symplectic phase space)**.** A *symplectic form* on a finite-dimensional real linear space E is a nondegenerate alternating bilinear form $\omega : E \times E \to \mathbb{R}$. On $E = V \oplus V^*$, define

$$\omega\big((q,p),(q',p')\big) = p(q') - p'(q).$$

For $\lambda \in \mathbb{R}$, define

$$\sigma_{\lambda\omega}(z,z') = e^{i\lambda\omega(z,z')/2}.$$

The Weyl multiplier is σ_ω.

Exercise 8.3 (Linear Darboux theorem)**.** Prove that every symplectic space has a basis in which its form has matrix

$$\begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}.$$

Prove that $\sigma_{\lambda\omega}$ is a multiplier and that $c_{\sigma_{\lambda\omega}}(z,z') = e^{i\lambda\omega(z,z')}$.

9 Differential Geometry

9.1 Bundles, sections, and geometric structures

Definition 9.1 (Smooth bundles and sections)**.** A *smooth fiber bundle* with fiber F is a smooth map $\pi : E \to M$ locally isomorphic over M to $U \times F \to U$. A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal G-bundle* has a free right G-action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map $s : M \to E$ with $\pi s = \mathrm{id}_M$. For a representation $G \to \mathrm{GL}(V)$, the *associated vector bundle* is $P \times_G V = (P \times V)/((pg, v) \sim (p, gv))$.

Definition 9.2 (Frame bundles and G-structures)**.** The *frame bundle* of an n-manifold is the principal $\mathrm{GL}(n, \mathbb{R})$-bundle

$$\mathrm{Fr}(M)_p = \mathrm{Iso}(\mathbb{R}^n, T_pM).$$

For a closed subgroup $G \leq \mathrm{GL}(n, \mathbb{R})$, a *$G$-structure* is a principal G-subbundle of $\mathrm{Fr}(M)$. It is a choice of the frames related by the transformations in G. Riemannian, Lorentzian, and almost symplectic structures correspond respectively to reductions to $O(n)$, $O(1, n-1)$, and $\mathrm{Sp}(2n, \mathbb{R})$; an oriented, time-oriented spacetime has structure group $SO^+(1, 3)$.

Definition 9.3 (Tensor fields and metrics)**.** A *(r, s)-tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. Equivalently, an $O(p, q)$-structure is a smooth nondegenerate symmetric tensor g of signature (p, q). The cases $(n, 0)$ and $(1, n-1)$ give Riemannian and Lorentzian metrics. The length of a Riemannian curve is $\int \sqrt{g(\dot\gamma, \dot\gamma)}\, dt$; the proper time of a timelike curve is $c^{-1} \int \sqrt{-g(\dot\gamma, \dot\gamma)}\, dt$. A metric and an orientation define a volume form vol_g.

Exercise 9.1 (Proper time)**.** Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, prove that the inertial path maximizes proper time among piecewise smooth future-directed timelike paths with fixed timelike-separated endpoints, and derive special-relativistic time dilation. Using the muon rest lifetime $\tau_\mu \simeq 2.2\,\mu\mathrm{s}$, derive the mean laboratory travel distance $\gamma v \tau_\mu$ and compare it with the distance $v\tau_\mu$ predicted without time dilation.

9.2 Differential forms, cohomology, and Hodge theory

Definition 9.4 (Integration of forms)**.** An *orientation* of an n-manifold is a reduction of its frame bundle to $\mathrm{GL}^+(n, \mathbb{R})$, equivalently a continuous choice of component of nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 9.2 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt} \int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

Definition 9.5 (de Rham cohomology)**.** A differential form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d^2 = 0$, the *de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes $H^*_{\mathrm{dR}}(M)$ a graded-commutative algebra.

Exercise 9.3 (de Rham classes and periods)**.** On a star-shaped open subset of $\mathbb{R}^n$, construct a homotopy operator and prove the Poincaré lemma: every closed form of positive degree is exact. Use Stokes' theorem to prove that an exact k-form has zero integral over every closed oriented k-manifold mapped smoothly into M. Prove $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with the isomorphism given by integration.

On $\mathbb{R}^2 \setminus \{0\}$, show that

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed and has integral 2π around the unit circle, hence is not exact. Compute $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$. Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential $F = dA$ exactly when $[F] = 0$ in de Rham cohomology.

Definition 9.6 (Hodge star)**.** On an oriented pseudo-Riemannian n-manifold, the metric induces a nondegenerate pairing $\langle \cdot, \cdot \rangle_g$ on differential forms. The *Hodge star* is the bundle map

$$* : \Omega^p(M) \longrightarrow \Omega^{n-p}(M)$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle_g \ \mathrm{vol}_g$$

for p-forms α, β.

Exercise 9.4 (Metric volume and Hodge star)**.** Construct vol_g and the Hodge star intrinsically and derive their coordinate expressions. For a metric with q negative directions, prove on p-forms that

$$*^2 = (-1)^{p(n-p)+q}\,\mathrm{id}\,.$$

Compute $*$ on the standard coordinate forms of Euclidean three-space and Minkowski four-space, stating the orientation and signature conventions.

9.3 Connections, curvature, and gauge fields

Definition 9.7 (Connections and parallel transport)**.** A *connection* on a vector bundle $E \to M$ is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fs) = X(f)s + f\nabla_X s$. Along a curve γ it induces a covariant derivative D/dt; its solutions $Ds/dt = 0$ define *parallel transport* between endpoint fibers. On TM, the torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$, and a *geodesic* satisfies $D\dot\gamma/dt = 0$.

Definition 9.8 (Curvature and holonomy)**.** The *curvature* of ∇ is the $\mathrm{End}(E)$-valued two-form

$$R^\nabla(X,Y)s = \nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s.$$

The *holonomy group* at p consists of the automorphisms of E_p obtained by parallel transport around loops based at p.

Exercise 9.5 (Parallel transport and curvature)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

Definition 9.9 (Principal connections and gauge fields)**.** Let $P \to M$ be a principal G-bundle. A *principal connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M. A reduction of the frame bundle describes spacetime geometry; a separate principal bundle describes an internal gauge symmetry.

Exercise 9.6 (Gauge covariance and Bianchi identity)**.** Choose local sections s_i and derive transition maps $g_{ij} : U_i \cap U_j \to G$ and the transformation laws for the local potentials $A_i = s_i^* A$ and curvatures $F_i = s_i^* F_A$. Prove $d_A F_A = 0$ and gauge invariance of holonomy up to conjugacy. For $G = U(1)$, recover $A \mapsto A + d\chi$ and $F = dA$. Show that a flat potential is locally pure gauge, and explain why removing it globally can fail on a nonsimply connected base. The global obstruction, holonomy, and measured phase are analyzed in Exercise 23.2.

Definition 9.10 (Electromagnetic field)**.** An *electromagnetic gauge field* is a connection on an internal principal $U(1)$-bundle over an oriented Lorentzian four-manifold. Its field strength is the curvature two-form F; locally $F = dA$. Given a current one-form J, Maxwell's equations are

$$dF = 0, \qquad d{*}F = \mu_0 {*}J.$$

For the SI vector decomposition on Minkowski spacetime, fix

$$(x^0, x^1, x^2, x^3) = (ct, x, y, z), \qquad g = -(dx^0)^2 + dx^2 + dy^2 + dz^2,$$

with orientation $dx^0 \wedge dx \wedge dy \wedge dz$, and set

$$\begin{aligned} F &= \frac{1}{c}\, dx^0 \wedge (E_x\, dx + E_y\, dy + E_z\, dz) \\ &\quad - (B_x\, dy \wedge dz + B_y\, dz \wedge dx + B_z\, dx \wedge dy), \\ J &= c\rho\, dx^0 - j_x\, dx - j_y\, dy - j_z\, dz. \end{aligned}$$

With these sign conventions, the force law for a particle of charge q and four-velocity u is $m\nabla_u u = q(\iota_u F)^\sharp$.

Exercise 9.7 (Maxwell theory and electromagnetic waves)**.** Derive charge conservation from Maxwell's equations. Using the stated Minkowski conventions, recover the electric and magnetic vector equations and verify that the covariant force law reduces to $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Define the experimentally measured vacuum permittivity ε_0 by the electrostatic Gauss law $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$; comparison with the covariant equation should give $\mu_0 \varepsilon_0 c^2 = 1$. In vacuum, impose Lorenz gauge and derive wave equations with speed c. Prove transverse polarization and compare $c = (\mu_0 \varepsilon_0)^{-1/2}$ with the measured speed of light.

Exercise 9.8 (Yang–Mills equations and instantons)**.** For $G = SU(r)$, use $\langle X, Y \rangle = -2\,\mathrm{Tr}(XY)$ and define

$$S_{\mathrm{YM}}(A) = \frac{1}{2g^2} \int_M \langle F_A \wedge {*}F_A \rangle$$

on a closed oriented Riemannian four-manifold. Prove gauge invariance and derive $d_A^* F_A = 0$, where $d_A^* = - * d_A *$. Decompose $F_A = F_A^+ + F_A^-$ and, for

$$k(A) = -\frac{1}{8\pi^2} \int_M \operatorname{Tr}(F_A \wedge F_A),$$

prove $S_{\mathrm{YM}}(A) \geq 8\pi^2 |k(A)|/g^2$, with equality precisely for $F_A = \pm * F_A$. Use the Bianchi identity to show that these instantons solve the Yang–Mills equation.

9.4 Riemannian geometry and gravitation

Definition 9.11 (Levi–Civita geometry)**.** A connection on TM is *metric-compatible* when $\nabla g = 0$. A *Levi–Civita connection* is a metric-compatible torsion-free connection. Its curvature defines the *Riemann tensor* $\operatorname{Riem}(X, Y, Z, W) = g(R^\nabla(X, Y)Z, W)$, whose contractions are the *Ricci tensor* Ric and scalar curvature S.

Exercise 9.9 (Levi–Civita connection and holonomy)**.** Prove existence and uniqueness of the Levi–Civita connection and derive the Koszul formula. On the orthonormal frame bundle, identify it with a principal $O(p, q)$-connection and use the solder form to identify the associated vector bundle with TM. Prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints and that its parallel transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

Exercise 9.10 (Curvature identities and geodesic deviation)**.** For the Levi–Civita connection, prove the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field J, prove

$$\frac{D^2 J}{dt^2} + R^\nabla(J, \dot\gamma)\dot\gamma = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

Definition 9.12 (Einstein equation)**.** The *Einstein tensor* of a Lorentzian metric is $G = \operatorname{Ric} - \frac{1}{2} S g$. A *stress–energy tensor* is a symmetric two-tensor T whose contraction with an observer's velocity gives the observer's energy–momentum current. The *Einstein equation* with cosmological constant Λ is

$$G + \Lambda g = \frac{8\pi G_N}{c^4} T.$$

Exercise 9.11 (Gravitational-wave detection)**.** For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. For GW150914, use the reported detector-frame component-mass medians $36 M_\odot$ and $29 M_\odot$ to infer the chirp mass. Compare with the model-independent reconstruction $\mathcal{M} \simeq 30 M_\odot$ and with the reported peak strain $h \simeq 10^{-21}$ near 150 Hz in Abbott et al., *Physical Review D* **93** (2016) and the LIGO GW150914 fact sheet. Limit the conclusion to agreement of the observed waveform with general relativity in that regime.

10 Symplectic Manifolds

Definition 10.1 (Differential forms and exterior derivative)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^* M$; write $\Omega^k(M) = \Gamma(\Lambda^k T^* M)$. For a smooth map $F : M \to N$, define

$$(F^*\alpha)_p(v_1, \ldots, v_k) = \alpha_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

The *exterior derivative* is the family $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and $F^*d = dF^*$.

Exercise 10.1 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

Definition 10.2 (Symplectic manifolds and canonical transformations)**.** A *symplectic form* is a closed nondegenerate two-form ω. For $f \in C^\infty(M)$, define

$$\iota_{X_f}\omega = -df, \qquad \{f, g\} = -\omega(X_f, X_g).$$

A diffeomorphism $F : (M, \omega_M) \to (N, \omega_N)$ is a *canonical transformation* when $F^*\omega_N = \omega_M$. For a manifold Q, the tautological one-form on T^*Q is

$$\theta_\alpha(v) = \alpha(d\pi(v)),$$

and the canonical symplectic form is $\omega = d\theta$.

Definition 10.3 (Classical states and observables)**.** A *pure classical state* is a point of M, a *classical observable* is a function $f \in C^\infty(M, \mathbb{R})$, and a *statistical state* is a Borel probability measure μ with expectation

$$\mathbb{E}_\mu[f] = \int_M f \, d\mu.$$

For Hamiltonian flow Φ_t, define

$$\mu_t = (\Phi_t)_*\mu_0, \qquad f_t = f \circ \Phi_t.$$

Exercise 10.2 (Darboux theorem and Hamiltonian mechanics)**.** Prove the Darboux theorem and identify the canonical form on T^*Q in local coordinates. Derive Hamilton's equations and prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. Prove

$$\frac{d}{dt} f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

Definition 10.4 (Hamiltonian actions and reduction)**.** A symplectic action of a Lie group G on (M, ω) is *Hamiltonian* when it has an equivariant map $J : M \to \mathfrak{g}^*$ satisfying

$$d\langle J, \xi\rangle = -\iota_{\xi_M}\omega \qquad (\xi \in \mathfrak{g}).$$

Let $\eta \in \mathfrak{g}^*$ be a regular value and let $G_\eta = \{g : \mathrm{Ad}_g^* \eta = \eta\}$. If G_η acts freely and properly on $J^{-1}(\eta)$, define

$$M/\!/_\eta G = J^{-1}(\eta)/G_\eta, \qquad q^*\omega_\eta = i^*\omega,$$

where $i : J^{-1}(\eta) \hookrightarrow M$ and $q : J^{-1}(\eta) \to M/\!/_\eta G$.

Exercise 10.3 (Noether and Marsden–Weinstein theorems)**.** Prove that the components of J Poisson-commute with every invariant Hamiltonian. Prove that ω_η exists uniquely and is symplectic and that invariant Hamiltonian flows descend to $M/\!/_\eta G$. Compute the momentum maps for translations and rotations on $T^*\mathbb{R}^3$.

Exercise 10.4 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the volume form $\omega^n/n!$, and every density depending only on the Hamiltonian.

11 Quantum Topology

11.1 Subfactors, Temperley–Lieb algebras, and the Jones polynomial

Definition 11.1 (Subfactors, conditional expectations, and index)**.** A *subfactor* is an inclusion $N \subseteq M$ of factors with common unit. Conditional expectations $E : M \to N$ are understood in the sense of Definition 33.5. For finite factors with normalized trace, the *Jones index* $[M : N]$ is the Murray–von Neumann dimension of $L^2(M)$ as a left N-module, and $E_N : M \to N$ denotes the unique trace-preserving normal conditional expectation.

For arbitrary factors and a faithful normal conditional expectation E, set

$$[M : N]_E = \lambda_E^{-1}, \qquad \lambda_E = \sup\big(\{0\} \cup \{\lambda > 0 : E(x) \geq \lambda x \text{ for every } x \in M_+\}\big).$$

The *minimal index* is $[M : N]_{\min} = \inf_E [M : N]_E$, with value $+\infty$ if no finite-index expectation exists. For the type-III inclusions below, $[M : N]$ denotes this minimal index.

Definition 11.2 (Basic construction)**.** For finite factors $N \subseteq M$, the *Jones projection* e_N on $L^2(M)$ is the orthogonal projection onto $L^2(N)$. The *basic construction* is $M_1 = \langle M, e_N \rangle''$. Iteration gives the *Jones tower* $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \cdots$.

Exercise 11.1 (Jones tower and Temperley–Lieb relations)**.** For a finite-index inclusion $N \subseteq M$ of finite factors, derive $e_N x e_N = E_N(x) e_N$ and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \qquad e_i e_j = e_j e_i \ (|i - j| > 1), \qquad e_i e_{i\pm 1} e_i = [M : N]^{-1} e_i.$$

Definition 11.3 (Braid groups and Markov traces)**.** The *braid group* B_n has generators σ_i with relations $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ and $\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i - j| > 1$. A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

Exercise 11.2 (Jones polynomial)**.** Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

11.2 Conformal nets and superselection sectors

Definition 11.4 (Local conformal nets)**.** Let $\mathcal{I}$ be the proper open intervals of S^1. A *local conformal net* assigns a von Neumann algebra $\mathcal{A}(I) \subset B(H)$ to each $I \in \mathcal{I}$ such that

$$I \subset J \Longrightarrow \mathcal{A}(I) \subset \mathcal{A}(J), \qquad I \cap J = \varnothing \Longrightarrow [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

The assignment is covariant under orientation-preserving diffeomorphisms of S^1, represented projectively on H, and the rotation generator is positive. Its vacuum vector Ω is invariant under the Möbius subgroup and cyclic for the local algebras, and $\bigvee_I \mathcal{A}(I) = B(H)$. Physically, S^1 is the compactified chiral light ray and $\mathcal{A}(I)$ consists of observables localized in I. The norm-closed C^*-algebra

$$\mathfrak{A} = C^*\left(\bigcup_{I \in \mathcal{I}} \mathcal{A}(I)\right)$$

generated by the local algebras is the *quasilocal algebra*.

Definition 11.5 (Duality and regularity for conformal nets)**.** Write I' for the interior of the complement of I. A net has *Haag duality* when $\mathcal{A}(I) = \mathcal{A}(I')'$. It has the *split property* when $\overline{I} \subset J$ implies $\mathcal{A}(I) \subset F \subset \mathcal{A}(J)$ for some type-I factor F. It is *strongly additive* when removing an interior point from I to obtain $I_1 \cup I_2$ gives $\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. A representation of the quasilocal algebra is *locally normal* when its restriction to every $\mathcal{A}(I)$ is normal.

Exercise 11.3 (The two-interval subfactor)**.** For $E = I_1 \cup I_2$ a union of disjoint intervals, set $\mathcal{A}(E) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. First use locality and vacuum cyclicity for $\mathcal{A}(I')$ to prove that the vacuum is separating for $\mathcal{A}(I)$. Use the split property and the spatial tensor product to identify separated local algebras as independent subsystems. Then use locality to obtain the inclusion

$$\mathcal{A}(E) \subseteq \mathcal{A}(E')'.$$

Under the split and irreducibility hypotheses, show that this is a subfactor. Define the *μ-index* $\mu(\mathcal{A}) = [\mathcal{A}(E')' : \mathcal{A}(E)]$, and prove that it is independent of the two-interval configuration by conformal covariance. Contrast single-interval Haag duality $\mathcal{A}(I) = \mathcal{A}(I')'$ with the possible failure of duality for E, measured by this Jones index.

Definition 11.6 (DHR sectors)**.** A *DHR sector* of $\mathcal{A}$ is represented by a locally normal, transportable endomorphism ρ of the quasilocal algebra that acts identically on $\mathcal{A}(I')$ for some interval I; transportability means that an equivalent representative can be localized in any interval. Composition is the tensor product. For a finite-index irreducible sector localized in I, its statistical dimension satisfies

$$d(\rho)^2 = [\mathcal{A}(I) : \rho(\mathcal{A}(I))].$$

Exercise 11.4 (Critical Ising chain, Rydberg spectroscopy, and the Ising net)**.** For $J, h > 0$, consider the transverse-field Ising chain

$$H_{\mathrm{I}} = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i.$$

Use the Jordan–Wigner transformation and a Bogoliubov rotation to obtain

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}.$$

Show that the gap is $2|J - h|$ and closes at $h = J$, where the long-wavelength modes have linear dispersion. For open chains of lengths $L = 8, 10, 12, 14$, diagonalize the finite Hamiltonian at $h = J$, separate the $\mathbb{Z}/2\mathbb{Z}$ parity sectors, and rescale gaps by the lowest even-parity gap.

The Rydberg experiment realizes a different microscopic chain,

$$H_{\mathrm{R}} = \sum_i \left(\frac{\Omega}{2} P_{i-1} X_i P_{i+1} - \Delta n_i \right) + \sum_{j-i \geq 2} V_{ij} n_i n_j,$$

in the nearest-neighbor blockade regime, where n_i projects onto the Rydberg state and $P_i = 1 - n_i$. At finite L, numerically diagonalize the blockaded Hilbert space, separate reflection and parity sectors, and compare the rescaled gaps with the ratios $2 : 4 : 6 : 8$ and $3 : 5 : 7$ reported in the modulation-spectroscopy peaks of Sun et al., *Experimental observation of conformal field theory spectra* (2026). Plot the deviations against $1/L$ and distinguish a finite-size extrapolation from an exact finite-chain identity.

11.3 Chern–Simons theory and quantum invariants

Definition 11.7 (Chern–Simons theory and Wilson loops)**.** Let G have an invariant inner product on $\mathfrak{g}$, and fix a trivialization of a principal G-bundle over a closed oriented three-manifold. For the resulting $\mathfrak{g}$-valued connection potential A, the *Chern–Simons functional* at level k is

$$S_{CS}(A) = \frac{k}{4\pi}\int_M \left(\langle A \wedge dA\rangle + \frac{1}{3}\langle A \wedge [A \wedge A]\rangle\right).$$

For a representation V and closed curve K, the *Wilson loop* is $W_V(K) = \mathrm{Tr}_V(\mathrm{Hol}_A(K))$.

Exercise 11.5 (Gauge invariance, level, and knot invariants)**.** For compact, connected, simply connected simple G, normalize the inner product so that the winding term of a gauge transformation lies in $2\pi\mathbb{Z}$. Compute the change of S_{CS} and derive $k \in \mathbb{Z}$ as the condition for $e^{iS_{CS}}$ to be gauge invariant. Prove that its critical points are flat connections.

Exercise 11.6 (Reshetikhin–Turaev links and Jones evaluations)**.** Use the Temperley–Lieb realization of $\mathcal{C}_k$ and its ribbon functor to show that the Z_k^{RT} invariant of a framed link in S^3, with a component colored by V_j, is the corresponding j-colored Jones evaluation at

$$Q = q^2 = e^{2\pi i/(k+2)}.$$

In particular, identify V_1 with the fundamental color giving the ordinary Jones polynomial. Compute the normalized invariants of the unknot, Hopf link, and trefoil, state the normalization of the colored unknot, and track the framing factor separately from the ambient-isotopy invariant.

11.4 Anyons, topological order, and quantum Hall phases

Definition 11.8 (Anyons and topological degeneracy)**.** In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

Exercise 11.7 (The toric code: error correction from topological order)**.** Give an $L \times L$ square cellulation of the torus, with $L \geq 2$, a qubit on every edge, and denote the resulting physical Hilbert space by $\mathcal{H}$. Let X and Z be self-adjoint unitaries satisfying $X^2 = Z^2 = 1$ and $XZ = -ZX$ on one qubit. Operators on distinct edges commute. For each vertex v and plaquette p, set

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \subset \partial p} Z_e, \qquad H = -\sum_v A_v - \sum_p B_p.$$

Prove that all A_v and B_p commute and that the ground space is their common $+1$ spectral subspace. Establish the two relations $\prod_v A_v = 1$ and $\prod_p B_p = 1$, prove that the remaining stabilizers are independent, find $E_0 = -2L^2$, and compute

$$\dim\ker(H - E_0) = 2^{2L^2-(L^2+L^2-2)} = 4.$$

Show that an open Z-string creates violated vertex terms only at its endpoints, while an open dual X-string creates violated plaquette terms only at its endpoints. Prove that contractible closed strings are products of stabilizers. Show that strings around the two noncontractible cycles give two pairs of logical operators, with a primal Z-loop and a dual X-loop anticommuting exactly when their intersection number is odd. Deduce that the code encodes two logical qubits and has distance L.

Let $\mathcal{S}$ be the stabilizer group generated by the A_v and B_p, let $\mathcal{S}' \subseteq B(\mathcal{H})$ be its commutant, and let P be the ground space projection. Prove that the logical observable algebra is

$$\mathcal{A}_{\log} = P\mathcal{S}'P = \operatorname{span}\{P\overline{X}_1^{i_1}\overline{Z}_1^{j_1}\overline{X}_2^{i_2}\overline{Z}_2^{j_2}P : i_r, j_r \in \{0,1\}\} \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}),$$

where the barred operators are noncontractible logical loops. Equivalently, identify the logical Pauli group as the Pauli centralizer of $\mathcal{S}$ modulo $\mathcal{S}$. If an error channel has Kraus operators E_a for which every $E_a^* E_b$ is supported on fewer than L edges, prove that

$$PE_a^* E_b P \in \mathbb{C}P = \mathcal{A}'_{\log}$$

where, as in the definition of protected observable algebras, the prime is taken inside the code corner $PB(\mathcal{H})P$, and hence verify the exact OAQEC condition for $\mathcal{A}_{\log}$. Deduce in particular that the toric code corrects arbitrary errors on at most $\lfloor (L-1)/2 \rfloor$ edges.

Interpret the endpoint excitations as anyons e and m. Derive $e^2 = m^2 = 1$, set $\varepsilon = em$, and show that taking e once around m produces the phase -1. Explain how local error syndromes, global logical Wilson loops, and the fourfold TQFT degeneracy express the same topological structure; see Kitaev, *Fault-tolerant quantum computation by anyons*, and Dauphinais–Kribs–Vasmer, *Stabilizer Formalism for Operator Algebra Quantum Error Correction*, Quantum **8** (2024).

Exercise 11.8 (Quantum Hall phases and measured topological order)**.** Use flux insertion for a Laughlin state at filling $\nu = 1/m$ to derive quasiparticle charge e/m, exchange phase π/m, Hall conductance $\nu e^2/h$, and genus-g degeneracy m^g. Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.

12 Theta Functions and Modular Forms

12.1 Lattice theta series and Poisson duality

Definition 12.1 (Integral lattices and theta series)**.** A *Euclidean lattice* $L \subset \mathbb{R}^d$ is the integer span of a basis of $\mathbb{R}^d$. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y\rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if $L \subseteq L^*$, *unimodular* if $L = L^*$, and *even* if $\|x\|^2 \in 2\mathbb{Z}$ for every $x \in L$. For τ in the upper half-plane and $q = e^{2\pi i\tau}$, its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of q^n counts lattice vectors of squared length $2n$.

Exercise 12.1 (Poisson duality for lattice theta series)**.** Apply Exercise 2.1 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\operatorname{vol}(L)}\Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives $\Theta_L(\tau+1) = \Theta_L(\tau)$. Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

12.2 Modular forms, theta functions, and eta products

Definition 12.2 (The modular group and modular forms)**.** The modular group $SL_2(\mathbb{Z})$ acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. A *modular form of integral weight* k for $SL_2(\mathbb{Z})$ is a holomorphic function f on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion $f(\tau) = \sum_{n\geq 0} a_n q^n$ has no negative powers. It is a *cusp form* if $a_0 = 0$.

Definition 12.3 (Jacobi theta functions and the Dedekind eta function)**.** Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad\qquad \eta(\tau) = q^{1/24}(q;q)_\infty. \\ \vartheta_4(\tau) &= \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

These functions have weight 1/2 transformation laws with phases or permutations, rather than being integral-weight scalar modular forms for the full modular group. In particular,

$$\eta(\tau + 1) = e^{\pi i/12}\eta(\tau), \qquad \eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau).$$

Exercise 12.2 (Theta transformations and the Jacobi triple product)**.** Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Prove the T-transformation laws directly from the series. Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2/2} = \prod_{m\geq 1} (1 - q^m)(1 + zq^{m-1/2})(1 + z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

12.3 Eisenstein series and arithmetic applications

Definition 12.4 (Eisenstein series and the discriminant)**.** Let $\sigma_r(n) = \sum_{d|n} d^r$. The normalized Eisenstein series of weights four and six are

$$E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6(\tau) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n.$$

The modular discriminant is the weight-twelve cusp form

$$\Delta(\tau) = \eta(\tau)^{24} = q\prod_{n\geq 1}(1 - q^n)^{24} = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728}.$$

Exercise 12.3 (Eisenstein series and the E_8 shell)**.** Construct Eisenstein series by summing $(m\tau + n)^{-k}$ over nonzero lattice points for even $k > 2$, prove their modular transformation law, and obtain the displayed Fourier expansions for E_4 and E_6. For a positive-definite even unimodular lattice, first use consistency of the S and T laws to prove that its dimension d is divisible by eight, and then show that its theta series is a modular form of weight $d/2$. Construct the E_8 lattice as

$$\{x \in \mathbb{Z}^8 : \textstyle\sum_i x_i \equiv 0 \pmod 2\} \cup \{x \in (\mathbb{Z} + \frac{1}{2})^8 : \sum_i x_i \equiv 0 \pmod 2\}.$$

Enumerate its vectors of squared length two and obtain 240, matching the coefficient of q in E_4.

Exercise 12.4 (Partition asymptotics from modular inversion)**.** Put $q = e^{-t}$ and use the modular transformation of η to prove

$$P(e^{-t}) \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \qquad (t \downarrow 0).$$

Apply a saddle-point argument, or an appropriate Tauberian theorem with its hypotheses stated, to derive the Hardy–Ramanujan formula

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Interpret $\log p(n)$ as the microcanonical entropy of the bosonic modes in the first exercise of this part. Distinguish this asymptotic result from an exact coefficient formula; see Hardy–Ramanujan, *Proceedings of the London Mathematical Society* **17** (1918).

13 Automorphic Harmonic Analysis and Arithmetic Langlands

13.1 Places, local fields, and adeles

Definition 13.1 (p-adic completions)**.** For a prime p and nonzero $x \in \mathbb{Q}$, write $x = p^{v_p(x)}a/b$ with $p \nmid ab$ and define

$$|x|_p = p^{-v_p(x)}, \qquad |0|_p = 0.$$

This is a non-Archimedean absolute value:

$$|x + y|_p \le \max\{|x|_p, |y|_p\}.$$

The completion is the locally compact field $\mathbb{Q}_p$, and its compact open subring of integers is

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \le 1\}.$$

The ordinary absolute value gives the Archimedean completion $\mathbb{R}$. A *number field* is a finite extension $F/\mathbb{Q}$. It has completions F_v indexed by its Archimedean and non-Archimedean places v; at a finite place, $\mathcal{O}_v \subset F_v$ is the ring of integers and the residue field has some cardinality q_v.

Definition 13.2 (Adeles and ideles)**.** The *adele ring* of $\mathbb{Q}$ is the restricted product

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p{}' \mathbb{Q}_p = \{(x_\infty, x_2, x_3, \ldots) : x_p \in \mathbb{Z}_p \text{ for all but finitely many } p\}.$$

The restriction makes $\mathbb{A}_{\mathbb{Q}}$ a locally compact ring. Its group of units is the *idele group*

$$\mathbb{A}_{\mathbb{Q}}^{\times} = \mathbb{R}^{\times} \times \prod_{p}{}' \mathbb{Q}_p^{\times},$$

where the finite components lie in $\mathbb{Z}_p^{\times}$ for almost every p. For a number field F, define $\mathbb{A}_F$ and $\mathbb{A}_F^{\times}$ in the same way. The diagonal embedding of $F^{\times}$ allows one to define the *idele class group*

$$C_F = F^{\times} \backslash \mathbb{A}_F^{\times}.$$

Exercise 13.1 (The product formula)**.** Compute the p-adic absolute values of $x = 12/25$. For arbitrary $x \in \mathbb{Q}^{\times}$, use unique factorization to prove

$$|x|_{\infty} \prod_{p} |x|_p = 1,$$

noting that only finitely many factors differ from one. Show that the diagonal image of x is an idele, and prove that the idelic norm

$$|(x_v)|_{\mathbb{A}} = |x_{\infty}|_{\infty} \prod_{p} |x_p|_p$$

is trivial on $\mathbb{Q}^{\times}$. Explain why it therefore descends to the idele class group.

Definition 13.3 (Hecke characters and their L-functions)**.** A *Hecke character* is a continuous character

$$\chi : F^{\times} \backslash \mathbb{A}_F^{\times} \longrightarrow \mathbb{C}^{\times}$$

with the usual unitary or quasicharacter growth condition. It factors into local characters $\chi_v : F_v^{\times} \to \mathbb{C}^{\times}$. At an unramified finite place its local Euler factor is

$$L_v(s, \chi_v) = \frac{1}{1 - \chi_v(\varpi_v) q_v^{-s}},$$

where ϖ_v is a uniformizer, characterized up to a unit by $|\varpi_v|_v = q_v^{-1}$. The global L-function is the product of these factors, supplemented at ramified and Archimedean places by the appropriate local factors. For the trivial character of $\mathbb{Q}$, its finite part is the Riemann zeta function.

Definition 13.4 (Schwartz–Bruhat functions)**.** A *Schwartz–Bruhat function* on $\mathbb{A}_F$ is a finite linear combination of restricted tensor products

$$\Phi = \bigotimes_{v} \Phi_v,$$

where Φ_v is an ordinary Schwartz function at each Archimedean place, is locally constant with compact support at each finite place, and equals the indicator of $\mathcal{O}_v$ for almost every finite v.

13.2 Automorphic representations and the trace formula

Definition 13.5 (Automorphic forms and representations)**.** Let G be a reductive group over a number field F. An *automorphic form* is, roughly, a smooth function on

$$G(F) \backslash G(\mathbb{A}_F)$$

that is finite under a maximal compact subgroup and under the center of the Archimedean universal enveloping algebra, has moderate growth, and satisfies an appropriate central-character condition. It is *cuspidal* when its constant term

$$\int_{N_P(F)\backslash N_P(\mathbb{A}_F)} \varphi(ng)\, dn$$

vanishes for every proper parabolic subgroup P with unipotent radical N_P.

Right translation by $G(\mathbb{A}_F)$ acts on automorphic forms. An *automorphic representation* is an irreducible constituent π of this representation.

Exercise 13.2 (Poisson summation as an abelian trace formula)**.** For $f \in \mathcal{S}(\mathbb{R})$, use the Fourier normalization

$$\widehat{f}(m) = \int_{\mathbb{R}} f(x)e^{-2\pi i m x}\, dx$$

and define on $L^2(\mathbb{R}/\mathbb{Z})$ the integral operator with kernel

$$K_f(x,y) = \sum_{n\in\mathbb{Z}} f(x-y+n).$$

Diagonalize it in the Fourier basis $e^{2\pi i m x}$. Compute its trace once as the sum of its eigenvalues and once as $\int_0^1 K_f(x,x)\, dx$, obtaining

$$\sum_{m\in\mathbb{Z}} \widehat{f}(m) = \sum_{n\in\mathbb{Z}} f(n).$$

Identify the two sides as the spectral and geometric expansions. Explain which features of this calculation depend on commutativity and which survive for an operator defined by convolution on a nonabelian quotient.

13.3 Modular eigenforms as automorphic representations

Definition 13.6 (Classical Hecke operators)**.** Let $f(\tau) = \sum_{n\geq 1} a_n q^n$ be a cusp form of weight k for $SL_2(\mathbb{Z})$. For a prime p, define

$$(T_p f)(\tau) = p^{k-1} f(p\tau) + \frac{1}{p}\sum_{a=0}^{p-1} f\left(\frac{\tau+a}{p}\right).$$

Then

$$T_p f = \sum_{n\geq 1} \left(a_{pn} + p^{k-1} a_{n/p}\right) q^n,$$

where $a_{n/p} = 0$ if $p \nmid n$. For $n \geq 1$, define T_n on Fourier coefficients by

$$T_n f = \sum_{m\geq 1} \left(\sum_{d\mid(m,n)} d^{k-1} a_{mn/d^2}\right) q^m.$$

Definition 13.7 (Normalized Hecke eigenform)**.** A *normalized Hecke eigenform* has $a_1 = 1$ and $T_n f = a_n f$ for every n.

Exercise 13.3 (Hecke recurrences and the Euler product)**.** Derive the displayed Fourier-coefficient formula for T_p. If f is a normalized eigenform, compare coefficients to prove

$$a_{p^{r+1}} = a_p a_{p^r} - p^{k-1} a_{p^{r-1}} \qquad (r \geq 1).$$

Derive $T_m T_n = \sum_{d|(m,n)} d^{k-1} T_{mn/d^2}$ from the coefficient formula, and use it to prove $a_{mn} = a_m a_n$ for $(m, n) = 1$, and then show

$$L(f, s) = \sum_{n\geq 1} \frac{a_n}{n^s} = \prod_p \frac{1}{1 - a_p p^{-s} + p^{k-1-2s}}$$

in a right half-plane. Factor the denominator as $(1 - \alpha_p p^{-s})(1 - \beta_p p^{-s})$ and derive $\alpha_p + \beta_p = a_p$ and $\alpha_p \beta_p = p^{k-1}$.

Exercise 13.4 (The GL_2 Hecke correspondence at p)**.** Let $K_p = GL_2(\mathbb{Z}_p)$. Show that index-p sublattices of $\mathbb{Z}_p^2$ are parametrized by $\mathbb{P}^1(\mathbb{F}_p)$ and use this to obtain a decomposition of

$$K_p \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} K_p$$

into $p + 1$ one-sided cosets. Relate these representatives to the $p + 1$ terms in the formula for T_p. Explain why simultaneous Hecke diagonalization resembles a complete family of commuting observables, and why this analogy does not make Hecke symmetry generic in quantum mechanics.

13.4 Satake parameters and the dual group

Definition 13.8 (Spherical Hecke algebra)**.** Let F_v be a non-Archimedean local field, let G be split reductive over F_v, and choose a hyperspecial compact subgroup $K_v = G(\mathcal{O}_v)$. The *spherical Hecke algebra* is

$$\mathcal{H}_v(G, K_v) = C_c(K_v \backslash G(F_v)/K_v)$$

with convolution. An irreducible representation π_v with a nonzero K_v-fixed vector is *unramified.* A one-dimensional simultaneous eigenspace for $\mathcal{H}_v(G, K_v)$ defines a *spherical Hecke character.*

13.5 Galois representations and matching L-functions

Definition 13.9 (ℓ-adic representations and Frobenius)**.** Let

$$G_F = \mathrm{Gal}(\overline{F}/F)$$

be the absolute Galois group. An *ℓ-adic representation* is a continuous homomorphism

$$\rho_\ell : G_F \longrightarrow GL_n(\overline{\mathbb{Q}}_\ell).$$

At a finite place $v \nmid \ell$ where inertia acts trivially, the representation is *unramified*; a Frobenius element Frob_v is then defined up to conjugacy. Fix the Frobenius convention by declaring the local polynomial to be

$$P_v(X) = \det(1 - X\rho_\ell(\mathrm{Frob}_v))\,.$$

The Galois local factor is $P_v(q_v^{-s})^{-1}$.

Exercise 13.5 (Local zeta function of an elliptic curve)**.** Let α_p, β_p be roots of $X^2 - a_pX + p$ and define a sequence N_r by

$$N_r = p^r + 1 - \alpha_p^r - \beta_p^r,$$

use $\sum_{r\geq 1} z^r/r = -\log(1-z)$ to prove

$$\exp\left(\sum_{r\geq 1} N_r \frac{T^r}{r}\right) = \frac{1 - a_p(E)T + pT^2}{(1-T)(1-pT)}.$$

For the curve $E : y^2 = x^3 - x$ at $p = 5$, count $E(\mathbb{F}_5)$, determine $a_5 = 5 + 1 - \#E(\mathbb{F}_5)$, and compute the first three N_r. Compare the resulting degree-two factor with any weight-two Hecke eigenform having the same coefficient a_5.

13.6 The Langlands correspondence and its status

Definition 13.10 (Langlands matching and functoriality)**.** An automorphic representation π of $GL_n(\mathbb{A}_F)$ and an n-dimensional ℓ-adic representation ρ are *matched* at an unramified place v when their local polynomials agree:

$$L_v(s, \pi_v) = \det\bigl(1 - q_v^{-s}\rho(\mathrm{Frob}_v)\bigr)^{-1}.$$

They are *globally matched* when this holds at all but finitely many places. For reductive groups, an L-homomorphism

$${}^LH \longrightarrow {}^LG$$

defines *functorial transfer data* when it maps almost all unramified conjugacy classes compatibly and preserves their associated local L-factors.

14 Cyclotomic and Aperiodic Structure

14.1 Algebraic numbers, roots of unity, and cyclotomic fields

Definition 14.1 (Algebraic numbers and cyclotomic fields)**.** A complex number α is *algebraic* over $\mathbb{Q}$ if it is a root of a nonzero polynomial in $\mathbb{Q}[x]$, and an *algebraic integer* if it is a root of a monic polynomial in $\mathbb{Z}[x]$. Its *minimal polynomial* is the monic irreducible polynomial in $\mathbb{Q}[x]$ that it satisfies. A *Galois conjugate* is another root of that polynomial.

For $n \geq 1$, let $\zeta_n = e^{2\pi i/n}$. The *nth cyclotomic field* is $\mathbb{Q}(\zeta_n)$, and the *nth cyclotomic polynomial* $\Phi_n(x)$ is the minimal polynomial of ζ_n. It satisfies

$$x^n - 1 = \prod_{d|n} \Phi_d(x), \qquad [\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

Exercise 14.1 (Roots of unity in topological data)**.** Prove the cyclotomic factorization recursively and show that $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ by $\zeta_n \mapsto \zeta_n^a$. For the $SU(2)_k$ theory from Quantum Topology, put $\xi = e^{\pi i/(k+2)}$ and prove that the quantum dimensions

$$d_j = \frac{\sin((j+1)\pi/(k+2))}{\sin(\pi/(k+2))}, \qquad 0 \leq j \leq k,$$

belong to the real subfield of $\mathbb{Q}(\xi)$ and are algebraic integers. At $k = 3$, identify the nontrivial dimension $2\cos(\pi/5) = (1+\sqrt{5})/2$ and recover its polynomial $x^2 - x - 1$. Show directly that the topological twists in $SU(2)_k$ are roots of unity up to the common framing phase. Explain what this proves for this family without asserting that arbitrary algebraic numbers are physical topological data.

14.2 Diophantine structure and substitutions

Definition 14.2 (Continued fractions and Pisot numbers)**.** For irrational α, its simple continued fraction is

$$\alpha = [a_0; a_1, a_2, \ldots] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}},$$

with convergents $p_n/q_n = [a_0; \ldots, a_n]$. A real algebraic integer $\beta > 1$ is a *Pisot number* if every other Galois conjugate has absolute value less than one.

Definition 14.3 (Substitutions and the Fibonacci word)**.** A *substitution* on a finite alphabet replaces each letter by a finite word and extends by concatenation. Its incidence matrix records how many copies of each letter occur in every substituted letter. The Fibonacci substitution is

$$\sigma(a) = ab, \qquad \sigma(b) = a, \qquad M_\sigma = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its one-sided fixed word is the limit of $a, ab, aba, abaab, \ldots$.

Exercise 14.2 (Frequencies and aperiodicity of the Fibonacci chain)**.** Diagonalize M_σ, prove that its Perron–Frobenius eigenvalue is ϕ, and derive letter frequencies

$$\operatorname{freq}(a) = \frac{1}{\phi}, \qquad \operatorname{freq}(b) = \frac{1}{\phi^2}.$$

Prove that every finite subword recurs with bounded gaps. Show that the infinite word cannot be periodic because a periodic word has rational letter frequencies. If tiles of lengths ϕ and 1 replace a and b, show that substitution followed by rescaling by ϕ reproduces the tiling. Separate exact inflation symmetry from ordinary translation symmetry.

14.3 Cut-and-project order and diffraction

Definition 14.4 (Cut-and-project sets)**.** Let $\mathcal{L}$ be a lattice in $\mathbb{R}^d \times \mathbb{R}^m$, with projections $\pi_\parallel$ and $\pi_\perp$ onto physical and internal space. A *cut-and-project scheme* requires $\pi_\parallel$ to be injective on $\mathcal{L}$ and $\pi_\perp(\mathcal{L})$ to be dense. For a compact window $W \subset \mathbb{R}^m$ with nonempty interior, define the *model set*

$$\Lambda(W) = \{\pi_\parallel(x) : x \in \mathcal{L},\ \pi_\perp(x) \in W\}.$$

It is *regular* when the boundary of W has measure zero.

Exercise 14.3 (The Fibonacci chain by cut and project)**.** Let $\phi' = -1/\phi$ and embed $\mathbb{Z}^2$ by

$$(m, n) \longmapsto (m + n\phi,\ m + n\phi') \in \mathbb{R}_\parallel \times \mathbb{R}_\perp.$$

Prove the injectivity and density conditions in the definition. Choose a generic closed interval window of length $1 + 1/\phi$, with boundary disjoint from the projected lattice, and order the resulting physical points. Determine the two possible nearest-neighbor spacings and show, after an overall rescaling and a possible exchange of letters, that their sequence is a Fibonacci tiling. Track how translating the window changes the local pattern without changing its set of finite patches. Interpret the internal coordinate as an arithmetic phase invisible in physical space.

Definition 14.5 (Autocorrelation and diffraction)**.** For a uniformly discrete point set $\Lambda \subset \mathbb{R}^d$, its Dirac comb is $\omega = \sum_{x\in\Lambda} \delta_x$. When the vague limit exists, its *autocorrelation* is

$$\gamma = \lim_{R\to\infty} \frac{\omega|_{B_R} * \widetilde{\omega|_{B_R}}}{\mathrm{vol}(B_R)}, \qquad \widetilde{\mu}(f) = \overline{\mu(\widetilde{f})}, \quad \widetilde{f}(x) = \overline{f(-x)}.$$

The positive measure $\widehat{\gamma}$ is its *diffraction measure*; its atoms are Bragg peaks.

Exercise 14.4 (Sharp diffraction without periodicity)**.** For a finite sample Λ_R, show that the kinematic scattering intensity is

$$I_R(k) = \left| \sum_{x\in\Lambda_R} e^{-ik\cdot x} \right|^2$$

and relate its volume-normalized limit to $\widehat{\gamma}$. Recover the ordinary reciprocal lattice for a periodic crystal. For Fibonacci approximants obtained from the first F_n tiles, compute $I_R(k)$ on a grid of wave numbers and track the largest peaks as n increases. Express the observed peak locations as integer combinations of 1 and $1/\phi$, and verify directly that no finite reciprocal lattice contains all of them.

Compare this mechanism with the sharp noncrystallographic diffraction reported in Shechtman et al., *Physical Review Letters* **53** (1984). State what diffraction establishes—long-range coherent order and its symmetry—and what it does not establish without a structural model.

14.4 Quasicrystal spectra and gap labelling

Definition 14.6 (The Fibonacci Hamiltonian and integrated density of states)**.** Let $\alpha = 1/\phi$ and

$$v_\theta(n) = \mathbf{1}_{[1-\alpha,1)}(\theta + n\alpha \bmod 1).$$

The *Fibonacci Hamiltonian* on $\ell^2(\mathbb{Z})$ is

$$(H_\theta\psi)_n = \psi_{n+1} + \psi_{n-1} + \lambda v_\theta(n)\psi_n.$$

For the uniquely ergodic rotation of θ, its *integrated density of states* is

$$N(E) = \int_0^1 \langle \delta_0, \mathbf{1}_{(-\infty,E]}(H_\theta)\delta_0\rangle \, d\theta.$$

It is the limiting number of states below E per lattice site.

Let Ω be the closure of the shift orbit of a Fibonacci sequence in $\{0,1\}^{\mathbb{Z}}$, and let $S : \Omega \to \Omega$ be the shift. The associated observable algebra is the crossed product $\mathcal{A}_\alpha = C(\Omega) \rtimes_S \mathbb{Z}$, generated by $C(\Omega)$ and a unitary U satisfying $UfU^* = f \circ S^{-1}$. The unique invariant measure μ gives the canonical trace

$$\tau\left(\sum_n f_n U^n\right) = \int_\Omega f_0 \, d\mu.$$

The Hamiltonians above are the covariant representations of one element of this algebra.

Exercise 14.5 (Diophantine trace labels)**.** Prove that $N(E)$ is nondecreasing and is constant on every open spectral gap. Approximate α by Fibonacci convergents and interpret the finite periodic approximants as large-unit-cell crystals. Define the additive Fibonacci label group and its allowed label set by

$$\Gamma_\alpha = \mathbb{Z} + \alpha\mathbb{Z}, \qquad \mathcal{L}_\alpha = \Gamma_\alpha \cap [0,1].$$

Use the substitution recurrence to show that the measures of one-letter and two-letter cylinder sets belong to $\mathbb{Z} + \alpha\mathbb{Z}$. Prove the same for every finite disjoint union of cylinder sets. For the corresponding projections in $C(\Omega) \subset \mathcal{A}_\alpha$, compute their traces and identify them with elements of $\mathcal{L}_\alpha$. Compare these values with the plateaus of the finite-approximant integrated density of states, without asserting that every allowed label is an open spectral gap.

15 The Heisenberg Group

15.1 Central extension

Definition 15.1 (Heisenberg group)**.** For a symplectic space (E,ω), its *Heisenberg group* is

$$\mathbb{H}(E,\omega) = E \times \mathbb{R}$$

with multiplication

$$(z,s)(z',s') = (z+z', s+s' + \tfrac{1}{2}\omega(z,z')).$$

Write $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$ for the canonical symplectic form on $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$. For $E = V \oplus V^*$ and $\lambda \neq 0$, the *Schrödinger representation* on $L^2(V)$ is

$$\big(\pi_\lambda(q,p,s)\xi\big)(x) = e^{i\lambda(s+p(x-q/2))}\xi(x-q).$$

Exercise 15.1 (Central extensions and projective representations)**.** Prove that $\mathbb{H}(E,\omega)$ is a locally compact group with center $\{0\}\times\mathbb{R}$ and that

$$\pi_\lambda(0,s) = e^{i\lambda s} I.$$

Prove that projective representations of E with multiplier $e^{i\lambda\omega/2}$ correspond to representations of $\mathbb{H}(E,\omega)$ with central character $e^{i\lambda s}$.

15.2 Stone–von Neumann theorem

Exercise 15.2 (Stone–von Neumann theorem)**.** Prove that π_λ is irreducible. Prove that every strongly continuous irreducible unitary representation of $\mathbb{H}(V\oplus V^*,\omega)$ with central character $e^{i\lambda s}$, $\lambda \neq 0$, is unitarily equivalent to π_λ.

15.3 Weyl quantization

Definition 15.2 (Weyl quantization)**.** For $a \in \mathcal{S}(\mathbb{R}^{2n})$, define $\mathrm{Op}_\hbar^W(a) : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ by

$$(\mathrm{Op}_\hbar^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n}\int_{\mathbb{R}^n\times\mathbb{R}^n} e^{i(x-y)\cdot p/\hbar} a\Big(\frac{x+y}{2}, p\Big)\,\psi(y)\,dy\,dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on $\mathcal{S}(\mathbb{R}^n)$. For the coordinate functions q_j and p_j, set

$$Q_j = \mathrm{Op}_\hbar^W(q_j), \qquad P_j = \mathrm{Op}_\hbar^W(p_j).$$

The canonical Poisson bracket on $\mathbb{R}^{2n}$ is

$$\{a,b\} = \sum_{j=1}^n \left(\frac{\partial a}{\partial q_j}\frac{\partial b}{\partial p_j} - \frac{\partial a}{\partial p_j}\frac{\partial b}{\partial q_j} \right).$$

Definition 15.3 (Wigner distributions)**.** For $\psi, \varphi \in \mathcal{S}(\mathbb{R}^n)$, define

$$W_\hbar(\psi,\varphi)(q,p) = \frac{1}{(2\pi\hbar)^n} \int_{\mathbb{R}^n} e^{-ip\cdot y/\hbar}\overline{\psi(q-y/2)}\,\varphi(q+y/2)\,dy.$$

Write $W_\hbar[\psi] = W_\hbar(\psi,\psi)$.

Exercise 15.3 (Weyl calculus and the Moyal product)**.** Prove on $\mathcal{S}(\mathbb{R}^n)$ that

$$Q_j\psi(x) = x_j\psi(x), \qquad P_j\psi(x) = -i\hbar\partial_j\psi(x), \qquad [Q_j,P_k] = i\hbar\delta_{jk}I.$$

Derive the Weyl relations and recover the Schrödinger representation of the Heisenberg group. Prove

$$\mathrm{Op}_\hbar^W(a)^* = \mathrm{Op}_\hbar^W(\overline{a}), \qquad \|\mathrm{Op}_\hbar^W(a)\|_{\mathrm{HS}} = (2\pi\hbar)^{-n/2}\|a\|_2,$$

and

$$\langle \psi, \mathrm{Op}_\hbar^W(a)\varphi\rangle = \int_{\mathbb{R}^{2n}} a(q,p)W_\hbar(\psi,\varphi)(q,p)\,dq\,dp.$$

Prove that for $a,b \in \mathcal{S}(\mathbb{R}^{2n})$ there is a unique $a \star_\hbar b \in \mathcal{S}(\mathbb{R}^{2n})$ satisfying

$$\mathrm{Op}_\hbar^W(a \star_\hbar b) = \mathrm{Op}_\hbar^W(a)\,\mathrm{Op}_\hbar^W(b).$$

Derive the oscillatory-integral formula for $\star_\hbar$ and prove

$$a \star_\hbar b = ab + \frac{i\hbar}{2}\{a,b\} + O(\hbar^2), \qquad \frac{a\star_\hbar b - b\star_\hbar a}{i\hbar} = \{a,b\} + O(\hbar^2)$$

in the Schwartz-symbol topology.

Exercise 15.4 (Groenewold–van Hove obstruction)**.** Prove that Weyl quantization carries Poisson brackets to $(i\hbar)^{-1}$ times commutators exactly for affine and quadratic polynomials. Quantize q^2p^2 using

$$q^2p^2 = \frac{1}{9}\{q^3,p^3\} = \frac{1}{3}\{q^2p, qp^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \qquad \mathcal{Q}(q) = Q, \qquad \mathcal{Q}(p) = P, \qquad \mathcal{Q}(q^m) = Q^m, \qquad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f,g\}) = (i\hbar)^{-1}[\mathcal{Q}(f),\mathcal{Q}(g)]$$

on a common invariant domain.

15.4 Heisenberg Plancherel theory

Exercise 15.5 (Heisenberg Plancherel and inversion)**.** For $V = \mathbb{R}^n$ and $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$, use Haar measure $dq\,dp\,ds$ and the scalar Fourier transform $\widehat{h}(\lambda) = \int_{\mathbb{R}} h(s)e^{-i\lambda s}\,ds$, and define

$$\widehat{f}(\lambda) = \int_{\mathbb{H}_n} f(g)\pi_\lambda(g)^*\,dg.$$

Derive the Plancherel measure $(2\pi)^{-(n+1)}|\lambda|^n\,d\lambda$ and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R}\setminus\{0\}} \|\widehat{f}(\lambda)\|_{\mathrm{HS}}^2 |\lambda|^n\,d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

Part II
Deformation

16 Quantum Dynamics

Definition 16.1 (Classical and quantum evolution)**.** Let $H \in C^\infty(\mathbb{R}^{2n}, \mathbb{R})$ have bounded derivatives of order at least two, let Φ_t be its Hamiltonian flow, and let $\mathrm{Op}_\hbar^W(H)$ denote its self-adjoint realization. Define

$$U_\hbar(t) = e^{-it\,\mathrm{Op}_\hbar^W(H)/\hbar}.$$

For $a \in \mathcal{S}(\mathbb{R}^{2n})$, define

$$a_t = a \circ \Phi_t, \qquad A_\hbar(t) = U_\hbar(t)^*\,\mathrm{Op}_\hbar^W(a)U_\hbar(t).$$

Exercise 16.1 (Heisenberg equation and Egorov's theorem)**.** Prove

$$\frac{d}{dt}a_t = \{a_t, H\}, \qquad \frac{d}{dt}A_\hbar(t) = \frac{i}{\hbar}[\mathrm{Op}_\hbar^W(H), A_\hbar(t)].$$

For every $T > 0$, prove Egorov's theorem

$$\sup_{|t|\le T} \left\| A_\hbar(t) - \mathrm{Op}_\hbar^W(a \circ \Phi_t) \right\| = O_T(\hbar).$$

Prove exact equality for quadratic H. Derive Ehrenfest's equations for the expectations of Q_j and P_j.

17 Geometric Quantization

Definition 17.1 (Prequantum line bundles)**.** A *prequantum line bundle* over (M, ω) is a Hermitian line bundle $L \to M$ with a unitary connection ∇ satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega.$$

For $f \in C^\infty(M)$, its *Kostant–Souriau operator* is

$$\mathcal{Q}_{\mathrm{pre}}(f) = -i\hbar\nabla_{X_f} + f.$$

Definition 17.2 (Polarizations and metaplectic correction)**.** A *complex polarization* is an involutive Lagrangian subbundle $P \subset T_{\mathbb{C}}M$. Its canonical line is $K_P = \Lambda^{\mathrm{top}} P^*$, and a *metaplectic correction* is a line bundle δ_P with $\delta_P^{\otimes 2} \cong K_P$. The corrected quantum space is the Hilbert completion of sections of $L \otimes \delta_P$ covariantly constant along P.

Exercise 17.1 (Prequantization and polarization)**.** Prove that a prequantum line bundle exists if and only if

$$\left[\frac{\omega}{2\pi\hbar}\right] \in H^2(M;\mathbb{Z}).$$

Prove

$$[\mathcal{Q}_{\mathrm{pre}}(f), \mathcal{Q}_{\mathrm{pre}}(g)] = i\hbar\mathcal{Q}_{\mathrm{pre}}(\{f,g\}).$$

For $T^*\mathbb{R}^n$ with the vertical polarization, recover $L^2(\mathbb{R}^n)$, $Q_j\psi = x_j\psi$, and $P_j\psi = -i\hbar\partial_j\psi$. For a compact Kähler manifold with a positive prequantum line bundle, identify the corrected quantum space with $H^0(M, L \otimes K_M^{1/2})$.

18 Deformation Quantization

Definition 18.1 (Poisson manifolds)**.** A *Poisson manifold* is a smooth manifold M with a bracket on $C^\infty(M)$ that is a Lie bracket and a derivation in each argument. Equivalently, there is a bivector $\Pi \in \Gamma(\Lambda^2 TM)$ satisfying

$$\{f,g\} = \Pi(df, dg), \qquad [\Pi,\Pi]_{\mathrm{Sch}} = 0.$$

Definition 18.2 (Star products)**.** A *differential star product* on a Poisson manifold is an associative $\mathbb{C}[[\hbar]]$-bilinear product on $C^\infty(M)[[\hbar]]$ of the form

$$f \star g = fg + \sum_{r\geq 1} \hbar^r C_r(f,g),$$

where the C_r are bidifferential operators, 1 is a unit, and

$$C_1(f,g) - C_1(g,f) = i\{f,g\}.$$

Two star products are *equivalent* when they are intertwined by an operator $T = \mathrm{id} + \sum_{r\geq 1} \hbar^r T_r$ with differential T_r.

Exercise 18.1 (Fedosov and Kontsevich quantization)**.** Prove that the Moyal product is a differential star product on $\mathbb{R}^{2n}$. Prove that every symplectic manifold admits a Fedosov star product and that its equivalence class is determined by its formal characteristic class. Prove the Kontsevich formality theorem and deduce that every finite-dimensional Poisson manifold admits a deformation quantization.

19 Quantization of Symmetry

Definition 19.1 (Equivariant quantization)**.** Let a Lie group G act on a prequantized Hamiltonian manifold $(M, \omega, J, L, \nabla)$ and preserve a polarization P and metaplectic correction δ_P. An *equivariant quantization* is a lift of the G-action to $L \otimes \delta_P$ preserving its Hermitian structure, connection, and polarization. Its quantum representation is the induced unitary representation on the polarized quantum space.

Exercise 19.1 (Quantization commutes with reduction)**.** For a compact connected group acting holomorphically and Hamiltonianly on a compact prequantized Kähler manifold, assume that 0 is a regular value of J, that G acts freely on $J^{-1}(0)$, and that the prequantum and half-form data descend to the quotient. Prove that the derived quantum representation is generated by the quantized components of J and prove

$$\mathcal{Q}(M//_0G) \cong \mathcal{Q}(M)^G$$

with the metaplectic correction. Derive the multiplicity formula obtained by reduction at integral coadjoint orbits.

19.1 Poisson–Lie groups and Lie bialgebras

Definition 19.2 (Poisson–Lie groups and Lie bialgebras)**.** A *Poisson–Lie group* is a Lie group G with a Poisson structure for which multiplication $G \times G \to G$ is Poisson. A *Lie bialgebra* is a Lie algebra $\mathfrak{g}$ with a linear map

$$\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on $\mathfrak{g}^*$. It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some $r \in \mathfrak{g} \otimes \mathfrak{g}$.

Exercise 19.2 (Infinitesimal Poisson–Lie duality)**.** Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. Prove the converse for connected simply connected groups. For a coboundary Lie bialgebra, define

$$\mathrm{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Prove that the co-Jacobi identity is equivalent to $\mathrm{CYB}(r)$ being invariant under the diagonal adjoint action. Under the additional triangular hypothesis $\mathrm{CYB}(r) = 0$, recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as $\mathrm{CYB}(r) = \Omega$ for an invariant tensor Ω and distinguish the case $\Omega \neq 0$.

19.2 Drinfeld–Jimbo quantization

Definition 19.3 (Quantized enveloping algebras)**.** A *quantized enveloping algebra* of $(\mathfrak{g}, \delta)$ is a topologically free Hopf algebra $U_\hbar(\mathfrak{g})$ over $\mathbb{C}[[\hbar]]$ satisfying

$$U_\hbar(\mathfrak{g})/\hbar U_\hbar(\mathfrak{g}) \cong U(\mathfrak{g}), \qquad \delta(x) = \frac{\Delta_\hbar(x) - \Delta_\hbar^{\mathrm{op}}(x)}{\hbar} \bmod \hbar.$$

For generic q, $U_q(\mathfrak{sl}_2)$ is generated by E, F, K, K^{-1} with

$$KEK^{-1} = q^2E, \qquad KFK^{-1} = q^{-2}F, \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Exercise 19.3 (Drinfeld–Jimbo theorem)**.** Construct $U_q(\mathfrak{g})$ from the Cartan matrix of a complex semisimple Lie algebra and prove that it is a Hopf deformation of $U(\mathfrak{g})$. For $\mathfrak{sl}_2$, verify the displayed relations and coproduct and compute the semiclassical Lie bialgebra.

20 Compact Quantum Groups

Definition 20.1 (Compact quantum groups and Haar states)**.** A *compact quantum group* $\mathbb{G}$ is a unital C^*-algebra $C(\mathbb{G})$ with a unital $*$-homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where $\otimes_{\min}$ is the closure of the algebraic tensor product in $B(H \otimes K)$ for faithful representations on H and K. The coproduct satisfies

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$$

and the linear spans of $\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))$ and $\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)$ are dense in $C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$. A *Haar state* is a state h satisfying

$$(h \otimes \mathrm{id})\Delta(a) = h(a)1 = (\mathrm{id} \otimes h)\Delta(a).$$

Definition 20.2 (Corepresentations and representative functions)**.** A finite-dimensional *unitary corepresentation* on H_U is a unitary $U \in B(H_U) \otimes C(\mathbb{G})$ satisfying

$$(\mathrm{id} \otimes \Delta)(U) = U_{12}U_{13}.$$

If $U = (u_{ij})$, then $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$. Denote the linear span of the matrix coefficients of all finite-dimensional unitary corepresentations by $\mathrm{Pol}(\mathbb{G})$.

Definition 20.3 (Drinfeld–Jimbo and Woronowicz duality)**.** Let $0 < q < 1$, let G be a compact connected simply connected semisimple Lie group, and let $\mathfrak{g}_{\mathbb{C}}$ be its complexified Lie algebra. The Hopf $*$-algebra $\mathrm{Pol}(G_q)$ is spanned by the matrix coefficients $c_{\varphi,v}$ of finite-dimensional type-1 unitary $U_q(\mathfrak{g}_{\mathbb{C}})$-modules for the compact real-form involution, with pairing

$$\langle x, c_{\varphi,v} \rangle = \varphi(xv).$$

Its universal and reduced C^*-completions are denoted $C_u(G_q)$ and $C_r(G_q)$.

Exercise 20.1 (Woronowicz Peter–Weyl theorem)**.** Prove existence and uniqueness of the Haar state. Choose one representative from each equivalence class of irreducible unitary corepresentations and prove that all of their matrix coefficients are linearly independent and that $\mathrm{Pol}(\mathbb{G})$ is a dense Hopf $*$-subalgebra of $C(\mathbb{G})$. For each irreducible U^α, construct the normalized positive matrix Q_α satisfying

$$(\mathrm{id} \otimes S^2)(U^\alpha) = (Q_\alpha \otimes 1)U^\alpha(Q_\alpha^{-1} \otimes 1), \qquad \mathrm{Tr}(Q_\alpha) = \mathrm{Tr}(Q_\alpha^{-1}),$$

and derive the orthogonality relations in the Haar GNS space and the resulting Hilbert direct-sum decomposition. Prove that the displayed pairing between $U_q(\mathfrak{g}_{\mathbb{C}})$ and $\mathrm{Pol}(G_q)$ is a nondegenerate Hopf pairing.

Exercise 20.2 ($SU_q(2)$)**.** Let $C(SU_q(2))$ be generated by α, γ so that

$$u = \begin{pmatrix} \alpha & -q\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary, and define $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$. Derive the relations

$$\alpha^*\alpha + \gamma^*\gamma = 1, \qquad \alpha\alpha^* + q^2\gamma\gamma^* = 1, \qquad \gamma^*\gamma = \gamma\gamma^*,$$
$$\alpha\gamma = q\gamma\alpha, \qquad \alpha\gamma^* = q\gamma^*\alpha,$$

and prove that Δ is a coassociative $*$-homomorphism satisfying the cancellation conditions.

Definition 20.4 (Representation-category duality)**.** Let $\mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type\,1}}(U_q(\mathfrak{g}_{\mathbb{C}}))$ be the category of finite-dimensional type-1 unitary modules and intertwiners, and let $\mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q))$ be the category of finite-dimensional unitary corepresentations and intertwiners. For a module (V, π_V), define $U_V \in \mathrm{End}(V) \otimes \mathrm{Pol}(G_q)$ by

$$(\mathrm{id} \otimes \langle x, \cdot \rangle)(U_V) = \pi_V(x) \qquad (x \in U_q(\mathfrak{g}_{\mathbb{C}})).$$

Define

$$\mathcal{F} : \mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type\,1}}(U_q(\mathfrak{g}_{\mathbb{C}})) \longrightarrow \mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q)), \qquad \mathcal{F}(V) = U_V, \quad \mathcal{F}(T) = T,$$

with identity tensor constraints on the underlying spaces $V \otimes W$.

Exercise 20.3 (Matrix coefficients and tensor products)**.** For $v \in V$, $w \in W$, $\varphi \in V^*$, and $\psi \in W^*$, prove

$$c_{\varphi,v} c_{\psi,w} = c_{\varphi\otimes\psi, v\otimes w}.$$

Use the Hopf-pairing identities to prove

$$(\mathrm{id} \otimes \Delta)(U_V) = U_{V,12} U_{V,13}, \qquad U_{V\otimes W} = U_{V,13} U_{W,23}.$$

Prove that $\mathcal{F}$ is fully faithful and essentially surjective and that its tensor constraints are intertwiners. Conclude

$$\mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type\,1}}(U_q(\mathfrak{g}_{\mathbb{C}})) \simeq \mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q))$$

as monoidal $*$-categories.

Definition 20.5 (Conjugates and quantum dimension)**.** For a finite-dimensional H-module V, its *left dual* $V^\vee$ has action

$$(x\varphi)(v) = \varphi(S(x)v),$$

evaluation $\mathrm{ev}_V(\varphi \otimes v) = \varphi(v)$, and coevaluation $\mathrm{coev}_V(1) = \sum_i e_i \otimes e^i$ for a basis (e_i) and its dual basis (e^i). In a rigid C^*-tensor category, a *conjugate* of V is an object $\overline{V}$ with morphisms

$$R_V : \mathbf{1} \longrightarrow \overline{V} \otimes V, \qquad \overline{R}_V : \mathbf{1} \longrightarrow V \otimes \overline{V}$$

satisfying

$$(\overline{R}_V^* \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes R_V) = \mathrm{id}_V, \qquad (R_V^* \otimes \mathrm{id}_{\overline{V}})(\mathrm{id}_{\overline{V}} \otimes \overline{R}_V) = \mathrm{id}_{\overline{V}}.$$

The adjoints R_V^* and $\overline{R}_V^*$ are evaluation maps, and R_V and $\overline{R}_V$ are coevaluation maps. A solution is *standard* when

$$R_V^*(\mathrm{id}_{\overline{V}} \otimes T) R_V = \overline{R}_V^*(T \otimes \mathrm{id}_{\overline{V}}) \overline{R}_V \qquad (T \in \mathrm{End}(V)).$$

For a standard solution, define

$$\mathrm{tr}_q(T) = R_V^*(\mathrm{id}_{\overline{V}} \otimes T) R_V, \qquad \dim_q(V) = \mathrm{tr}_q(\mathrm{id}_V).$$

Exercise 20.4 ($SU_q(2)$ fusion and dimensions)**.** Let V_n be the irreducible type-1 $U_q(\mathfrak{sl}_2)$-module of highest weight n. Construct its conjugate equations and prove

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Prove that the normalized matrices from the Woronowicz orthogonality relations implement the standard solutions and compute

$$\dim_q(V_n) = [n+1]_q = \frac{q^{n+1} - q^{-(n+1)}}{q - q^{-1}}.$$

Prove

$$\dim V_n = n+1, \qquad \lim_{q\to 1} \dim_q(V_n) = n+1, \qquad \dim_q(V_n) > n+1$$

for $0 < q < 1$ and $n \geq 1$.

Definition 20.6 (Tannaka–Krein reconstruction)**.** Let $\mathcal{C}$ be an essentially small rigid C^*-tensor category and let $F : \mathcal{C} \to \mathrm{Hilb}_{\mathrm{fd}}$ be a faithful unitary tensor functor. Define

$$\mathcal{A}_F = \left(\bigoplus_{V \in \mathrm{Ob}\,\mathcal{C}} F(V)^* \otimes F(V) \right) \Big/ \left\langle (\psi \circ F(T)) \otimes v - \psi \otimes F(T)v \right\rangle,$$

where $T : V \to W$, $v \in F(V)$, and $\psi \in F(W)^*$. Its multiplication is induced by tensor products, and for a basis (e_i) of $F(V)$ define

$$\Delta[\varphi \otimes v]_V = \sum_i [\varphi \otimes e_i]_V \otimes [e_i^* \otimes v]_V, \qquad \varepsilon[\varphi \otimes v]_V = \varphi(v).$$

The involution and antipode are induced by the conjugate data.

Exercise 20.5 (Tannaka–Krein theorem)**.** Prove that $\mathcal{A}_F$ is a Hopf $*$-algebra and is independent of the chosen bases. For the forgetful fiber functor

$$F : \mathrm{Rep}(G_q) \longrightarrow \mathrm{Hilb}_{\mathrm{fd}},$$

prove that

$$[\varphi \otimes v]_V \longmapsto c_{\varphi,v}$$

is a Hopf $*$-isomorphism $\mathcal{A}_F \cong \mathrm{Pol}(G_q)$.

Definition 20.7 (Universal R-matrix and braiding)**.** A *universal R-matrix* for a Hopf algebra H is an invertible element $\mathcal{R}$ in a completion of $H \otimes H$ on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x), \quad (\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

For finite-dimensional modules V, W, define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

The pair $(H, \mathcal{R})$ is *quasitriangular*.

Exercise 20.6 (Hexagon identities and the Yang–Baxter equation)**.** Prove that $c_{V,W}$ is a natural module intertwiner and that the two coproduct identities for $\mathcal{R}$ give the two hexagon identities. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on $V^{\otimes 3}$,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators $c_{j,j+1}$ define braid-group representations on tensor powers of every finite-dimensional type-1 module.

Definition 20.8 (Ribbon categories)**.** Let $\mathcal{C}$ be a braided rigid $\mathbb{C}$-linear monoidal category. A *twist* is a natural isomorphism $\theta : \mathrm{id}_{\mathcal{C}} \to \mathrm{id}_{\mathcal{C}}$ satisfying

$$\theta_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}, \qquad \theta_{X \otimes Y} = c_{Y,X} c_{X,Y} (\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object X.

Exercise 20.7 (Ribbon elements)**.** For a quasitriangular Hopf algebra $(H, \mathcal{R})$, set

$$u = m(S \otimes \mathrm{id})(\mathcal{R}_{21}).$$

Let $v \in H$ be central and invertible and satisfy

$$v^2 = uS(u), \qquad S(v) = v, \qquad \varepsilon(v) = 1, \qquad \Delta(v) = (\mathcal{R}_{21}\mathcal{R})^{-1}(v \otimes v).$$

Prove that

$$\theta_V = \pi_V(v^{-1})$$

defines a ribbon twist on the category of finite-dimensional H-modules. Derive invariance of the quantum trace under framed ribbon isotopy.

21 Quantum Schur–Weyl Duality and DAHA

Definition 21.1 (Root data and affine Weyl groups)**.** Let E be a finite-dimensional Euclidean space. A finite set $R \subset E \setminus \{0\}$ is a *reduced crystallographic root system* if it spans E, is preserved by the reflections

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha, \qquad \alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle},$$

satisfies $R \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$, and has $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in R$. Define

$$W = \langle s_\alpha : \alpha \in R \rangle, \qquad Q^\vee = \mathrm{span}_{\mathbb{Z}}\{\alpha^\vee : \alpha \in R\}, \qquad W_{\mathrm{aff}} = Q^\vee \rtimes W.$$

The connected components of the complement of the affine reflection hyperplanes

$$H_{\alpha,m} = \{x \in E : \langle x, \alpha \rangle = m\}, \qquad \alpha \in R, \quad m \in \mathbb{Z},$$

are the *alcoves*.

Definition 21.2 (Finite Hecke algebras)**.** Let $(W, \{s_i\})$ be a Coxeter system, and let m_{ij} be the order of $s_i s_j$. Its *Hecke algebra* $\mathcal{H}_t(W)$ over $\mathbb{C}(t^{1/2})$ is generated by T_i with the braid relations

$$\underbrace{T_i T_j T_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{T_j T_i T_j \cdots}_{m_{ij} \text{ factors}}$$

and the quadratic relations

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

Definition 21.3 (Quantum Schur–Weyl actions)**.** Let v be generic, let $U_v(\mathfrak{gl}_N)$ act on its vector module $V = \mathbb{C}^N$, and set $t = v^2$. On $V^{\otimes r}$ define

$$\rho_r(T_i) = \mathrm{id}_V^{\otimes(i-1)} \otimes \check{R}_{V,V} \otimes \mathrm{id}_V^{\otimes(r-i-1)}, \qquad \check{R}_{V,V} = \tau(\pi_V \otimes \pi_V)(\mathcal{R}),$$

with the normalization for which $\check{R}_{V,V}$ has eigenvalues v and $-v^{-1}$.

Exercise 21.1 (Quantum Schur–Weyl duality)**.** Prove that the coproduct action of $U_v(\mathfrak{gl}_N)$ commutes with $\rho_r(T_i)$. Derive the braid and Hecke relations from the universal R-matrix and prove that ρ_r defines an action of $\mathcal{H}_{v^2}(S_r)$. For generic v and $N \geq r$, prove the double-centralizer identities

$$\rho_r(\mathcal{H}_{v^2}(S_r)) = \mathrm{End}_{U_v(\mathfrak{gl}_N)}(V^{\otimes r}), \qquad \pi_r(U_v(\mathfrak{gl}_N)) = \mathrm{End}_{\mathcal{H}_{v^2}(S_r)}(V^{\otimes r}),$$

and the decomposition

$$V^{\otimes r} \cong \bigoplus_{\substack{\lambda \vdash r \\ \ell(\lambda) \leq N}} L_\lambda^{(N)} \otimes S_t^\lambda.$$

Definition 21.4 (Affine Hecke algebras)**.** The extended affine Hecke algebra $\mathcal{H}_t^{\mathrm{aff}}(GL_r)$ is generated by $T_1, \ldots, T_{r-1}$ and commuting invertible elements $X_1, \ldots, X_r$, subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \qquad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of $\mathbb{Z}^r \rtimes S_r$.

Definition 21.5 (Affine quantum Schur–Weyl functor)**.** Let $U_v(L\mathfrak{gl}_N)$ be the quantum loop algebra and let

$$\widehat{V} = V \otimes \mathbb{C}[z^{\pm 1}]$$

be the affinization of its vector evaluation module. The tensor power $\widehat{V}^{\otimes r}$ is a $U_v(L\mathfrak{gl}_N)$–$\mathcal{H}_{v^2}^{\mathrm{aff}}(GL_r)$ bimodule, with the right action determined by the normalized spectral R-matrices and the loop operators. Define

$$\mathsf{SW}_{N,r}^{\mathrm{aff}}(M) = \widehat{V}^{\otimes r} \otimes_{\mathcal{H}_{v^2}^{\mathrm{aff}}(GL_r)} M.$$

Exercise 21.2 (Affine quantum Schur–Weyl duality)**.** Use the spectral Yang–Baxter equation to prove that the two actions on $\widehat{V}^{\otimes r}$ commute and satisfy the affine braid relations. Use the two eigenvalues of the vector R-matrix to derive the affine Hecke quadratic relation. For $N > r$, prove that $\mathsf{SW}_{N,r}^{\mathrm{aff}}$ is an equivalence from finite-dimensional $\mathcal{H}_{v^2}^{\mathrm{aff}}(GL_r)$-modules to the full subcategory of finite-dimensional polynomial $U_v(L\mathfrak{gl}_N)$-modules generated by subquotients of tensor products of r vector evaluation modules.

Definition 21.6 (Double-affine braid group)**.** Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and let

$$\mathrm{Conf}_r(T^2) = \{(z_1, \ldots, z_r) \in (T^2)^r : z_i \neq z_j \text{ for } i \neq j\}.$$

The *elliptic braid group* is

$$B_r^{\mathrm{ell}} = \pi_1(\mathrm{Conf}_r(T^2)/S_r).$$

Its standard generators are adjacent braids T_i and loops X_j, Y_j moving the jth point around the two fundamental cycles of T^2. The central extension in which the total X–Y commutator is denoted q is the *double-affine braid group* of type GL_r.

Exercise 21.3 (From affine to double-affine braids)**.** Derive a presentation of B_r^{ell} from the motions T_i, X_j, Y_j. Prove that the T_i satisfy the finite braid relations, that the X_j commute, that the Y_j commute, and that conjugation by T_i permutes the adjacent loop generators. Compute the central commutator of the two total torus motions. Cutting T^2 along one cycle, prove that the remaining generators form an affine braid group.

Definition 21.7 (Double affine Hecke algebras)**.** Let $B^{\mathrm{da}}(R)$ be the double-affine braid group of a root datum R, with braid generators T_i and central parameter q. For nonzero parameters q and t_i, define

$$\mathsf{H}_{q,\mathbf{t}}(R) = \mathbb{C}(q,\mathbf{t})[B^{\mathrm{da}}(R)] \Big/ \left\langle \mathsf{q} - q, (T_i - t_i^{1/2})(T_i + t_i^{-1/2}) \right\rangle.$$

This is the *double affine Hecke algebra* of R.

Definition 21.8 (Rank-one DAHA and its spherical representation)**.** The universal DAHA $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ is generated by $t_i^{\pm 1}$ for $0 \le i \le 3$, subject to

$$t_i t_i^{-1} = 1, \qquad t_i + t_i^{-1} \text{ central}, \qquad t_0 t_1 t_2 t_3 = q^{-1}.$$

For $k_i \neq 0$, the specialization

$$t_i + t_i^{-1} = k_i + k_i^{-1}$$

is the four-parameter rank-one DAHA. After the standard scalar normalization, let T be its finite Hecke generator with $(T - k)(T + k^{-1}) = 0$, and define

$$e = \frac{T + k^{-1}}{k + k^{-1}}.$$

Its basic polynomial representation is on $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$ by reflection–q-difference operators, and

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}].$$

The action of $e\widehat{\mathsf{H}}_q e$ on $e\mathcal{L}$ is the *spherical rank-one polynomial representation.*

Definition 21.9 (Ordinary hypergeometric series)**.** For $m \in \mathbb{Z}_{\ge 0}$, define

$$(a)_0 = 1, \qquad (a)_m = a(a+1)\cdots(a+m-1) \quad (m \ge 1).$$

Where the quotient is defined, $(a)_m = \Gamma(a+m)/\Gamma(a)$; the product gives its continuation at nonpositive integral a. If no denominator parameter b_j is a nonpositive integer, the *generalized hypergeometric series* is

$${}_rF_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer. Otherwise it converges for every z when $r \le s$, for $|z| < 1$ when $r = s+1$, and only at $z = 0$ when $r > s+1$. The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z)\frac{d^2}{dz^2} + \big(c - (a+b+1)z\big)\frac{d}{dz} - ab.$$

Exercise 21.4 (Gauss hypergeometric theorem)**.** Prove

$$ {}_1F_0\left(\begin{matrix} a \\ - \end{matrix}; z\right) = (1-z)^{-a} $$

and prove that ${}_2F_1(a,b;c;z)$ is the solution of $\mathcal{L}_{a,b,c}f = 0$ analytic at $z=0$ with $f(0)=1$. Derive Euler's integral and transformation

$$ {}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\, du, $$

$$ {}_2F_1(a,b;c;z) = (1-z)^{c-a-b}{}_2F_1(c-a,c-b;c;z), $$

and Gauss's summation

$$ {}_2F_1(a,b;c;1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}. $$

State the parameter conditions for each identity and extend them by analytic continuation.

Definition 21.10 (Jacobi and Wilson polynomials)**.** For $\alpha, \beta > -1$, the *Jacobi polynomial* is

$$ P_n^{(\alpha,\beta)}(x) = \frac{(\alpha+1)_n}{n!}{}_2F_1\left(\begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2}\right). $$

For positive a, b, c, d, the *Wilson polynomial* is

$$ \begin{aligned} W_n(x^2; a, b, c, d) = (a+b)_n(a+c)_n(a+d)_n& \\ \times {}_4F_3\left(\begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1\right)&. \end{aligned} $$

Exercise 21.5 (Rational rank-one spectral theory)**.** Prove

$$ \left((1-x^2)\frac{d^2}{dx^2} + (\beta-\alpha-(\alpha+\beta+2)x)\frac{d}{dx}\right) P_n^{(\alpha,\beta)} = -n(n+\alpha+\beta+1)P_n^{(\alpha,\beta)}. $$

Derive Jacobi orthogonality on $[-1,1]$ with weight $(1-x)^\alpha(1+x)^\beta$. Prove the Wilson difference equation and orthogonality on $[0,\infty)$ with weight

$$ w_W(x) = \left| \frac{\Gamma(a+ix)\Gamma(b+ix)\Gamma(c+ix)\Gamma(d+ix)}{\Gamma(2ix)} \right|^2. $$

Derive the three-term recurrences of both families.

Definition 21.11 (Basic hypergeometric series)**.** For $m \in \mathbb{N}$, define

$$ (a;q)_m = \prod_{j=0}^{m-1}(1-aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1-aq^j), $$

and

$$ (a_1,\ldots,a_r;q)_m = \prod_{i=1}^{r}(a_i;q)_m, \qquad (a_1,\ldots,a_r;q)_\infty = \prod_{i=1}^{r}(a_i;q)_\infty. $$

The *basic hypergeometric series* is

$$ {}_r\phi_s\left(\begin{matrix} a_1,\ldots,a_r \\ b_1,\ldots,b_s \end{matrix}; q, z\right) = \sum_{m=0}^{\infty} \frac{(a_1,\ldots,a_r;q)_m}{(q,b_1,\ldots,b_s;q)_m}\left((-1)^m q^{m(m-1)/2}\right)^{1+s-r} z^m. $$

It is *terminating* when one numerator parameter is q^{-n}.

Exercise 21.6 (Basic hypergeometric summations)**.** Prove the q-binomial theorem

$$ {}_1\phi_0\left(\begin{matrix} a \\ - \end{matrix}; q, z\right) = \frac{(az;q)_\infty}{(z;q)_\infty}, \qquad |z| < 1, $$

the q-Gauss summation

$$ {}_2\phi_1\left(\begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab}\right) = \frac{(c/a, c/b;q)_\infty}{(c, c/(ab);q)_\infty}, \qquad \left|\frac{c}{ab}\right| < 1, $$

and the terminating q-Pfaff–Saalschütz summation

$$ {}_3\phi_2\left(\begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q\right) = \frac{(c/a, c/b;q)_n}{(c, c/(ab);q)_n}. $$

Definition 21.12 (Askey–Wilson operator and polynomials)**.** For $0 < q < 1$ and nonzero a, b, c, d, define

$$ A(z) = \frac{(1-az)(1-bz)(1-cz)(1-dz)}{(1-z^2)(1-qz^2)} $$

and, on $f(z) = f(z^{-1})$,

$$ (\mathcal{D}_{AW} f)(z) = A(z)\big(f(qz) - f(z)\big) + A(z^{-1})\big(f(q^{-1}z) - f(z)\big). $$

For $x = (z + z^{-1})/2$, the *Askey–Wilson polynomial* is

$$ \begin{aligned} p_n(x; a, b, c, d \mid q) &= a^{-n}(ab, ac, ad;q)_n \\ &\qquad \times {}_4\phi_3\left(\begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q\right). \end{aligned} $$

Its weight on the unit circle is

$$ w_{AW}(z) = \frac{(z^2, z^{-2};q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z;q)_\infty}. $$

Exercise 21.7 (Rank-one DAHA and the Askey–Wilson theorem)**.** Construct the basic representation of $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ from the involutions

$$ (s_1 f)(z) = f(z^{-1}), \qquad (s_0 f)(z) = f(qz^{-1}) $$

and first-order reflection–q-difference operators. Verify the four Hecke relations and the product relation. Prove that the spherical subalgebra is generated in its polynomial representation by multiplication by $z + z^{-1}$ and $\mathcal{D}_{AW}$. Prove

$$ \mathcal{D}_{AW} p_n = (q^{-n} - 1)(1 - abcdq^{n-1})p_n. $$

For $a, b, c, d \in (-1, 1)$, prove that $\mathcal{D}_{AW}$ is symmetric for

$$ \langle f, g\rangle_{AW} = \int_{|z|=1} f(z)\overline{g(z)}w_{AW}(z)\frac{dz}{2\pi i z}, $$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

Definition 21.13 (Type-A polynomial representation)**.** Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \ldots, x_r^{\pm 1}],$$

let X_j act by multiplication by x_j, and let s_i exchange x_i and x_{i+1}. Define

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \qquad 1 \le i < r,$$

and

$$(\pi f)(x_1, \ldots, x_r) = f(x_2, \ldots, x_r, qx_1).$$

For generic independent parameters q, t, the operator algebra generated by $X_j^{\pm 1}$, T_i, and $\pi^{\pm 1}$ is the polynomial realization of the GL_r *double affine Hecke algebra* $\mathsf{H}_{q,t}(GL_r)$. The subalgebra generated by T_i and $\pi^{\pm 1}$ is an extended affine Hecke algebra; its translation elements form a commuting family $Y_1, \ldots, Y_r$ of q-difference-reflection operators.

Exercise 21.8 (Polynomial representation and the DAHA quotient)**.** Prove that the displayed operators preserve $\mathcal{P}_r$ and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \le i < r), \qquad \pi X_r \pi^{-1} = qX_1,$$

and identify the two affine-Hecke subalgebras generated respectively by T_i, X_j and by T_i, Y_j. Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to q.

Definition 21.14 (Macdonald operators and polynomials)**.** Let

$$(T_{q,x_i} f)(x_1, \ldots, x_r) = f(x_1, \ldots, qx_i, \ldots, x_r).$$

The first type-A *Macdonald operator* is

$$D_{q,t} = \sum_{i=1}^{r} \left(\prod_{j \ne i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

For generic q, t, the *Macdonald polynomial* $P_\lambda(x; q, t)$ is the symmetric polynomial triangular in the monomial-symmetric basis and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^{r} q^{\lambda_i} t^{r-i} \right) P_\lambda, \qquad \lambda_1 \ge \cdots \ge \lambda_r \ge 0.$$

For $0 < q, t < 1$, define

$$\Delta_{q,t}(x) = \prod_{i \ne j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^{r} \frac{dx_j}{2\pi i x_j}.$$

Exercise 21.9 (DAHA Fourier transform and Macdonald spectral theory)**.** Construct the commuting Y-operators and prove that their S_r-invariant Laurent polynomials preserve $\mathcal{P}_r^{S_r}$. Identify the first elementary symmetric polynomial in the Y_j with $D_{q,t}$ up to normalization. Prove existence, uniqueness, orthogonality, and joint diagonalization of the Macdonald polynomials. Construct the DAHA Fourier transform exchanging the X- and Y-polynomial representations and derive its projective $SL_2(\mathbb{Z})$ relations.

Exercise 21.10 (Ordinary and basic hypergeometric degeneration)**.** Set $q = e^{-\varepsilon}$. Prove

$$\lim_{\varepsilon\to 0^+} \frac{(q^a;q)_m}{(1-q)^m} = (a)_m, \qquad \lim_{q\to 1^-} (1-q)^{1-a} \frac{(q;q)_\infty}{(q^a;q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q\to 1^-} {}_r\phi_{r-1}\left(\begin{matrix} q^{a_1},\dots,q^{a_r} \\ q^{b_1},\dots,q^{b_{r-1}} \end{matrix}; q, z\right) = {}_rF_{r-1}\left(\begin{matrix} a_1,\dots,a_r \\ b_1,\dots,b_{r-1} \end{matrix}; z\right).$$

Derive Gauss's summation from the q-Gauss summation. Prove

$$\lim_{q\to 1^-} (1-q)^{-3n} p_n\left(\frac{q^{ix}+q^{-ix}}{2}; q^a, q^b, q^c, q^d \mid q\right) = W_n(x^2; a, b, c, d).$$

Definition 21.15 (Root-system basic hypergeometric weights)**.** Write $f(z^{\pm m}) = f(z^m)f(z^{-m})$ and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon,\delta=\pm 1} f(z_i^\epsilon z_j^\delta).$$

For $|q|, |t|, |a_s| < 1$, define

$$\begin{aligned}\Delta_{BC_r}(z) = &\prod_{i=1}^r \frac{(z_i^{\pm 2};q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1};q)_\infty} \\ &\times \prod_{1\le i<j\le r} \frac{(z_i^{\pm 1} z_j^{\pm 1};q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1};q)_\infty}.\end{aligned}$$

The associated *Gustafson integral* uses the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

Exercise 21.11 (Root systems and Gustafson integrals)**.** Prove that Δ_{BC_r} is invariant under permutations and independent inversions of the z_i. Identify its factors with the roots $\pm e_i$, $\pm 2e_i$, and $\pm e_i \pm e_j$. For $r = 1$, recover the Askey–Wilson weight and its order-two Weyl normalization. Set $t = q^g$ with $g \in \mathbb{N}$, reduce the quotients of q-products to finite products, and compute the limit $q \to 1^-$ with $z_i = e^{i\theta_i}$.

Definition 21.16 (Elliptic shifted factorials and gamma function)**.** For $|p|, |q| < 1$, define

$$\theta(z;p) = (z;p)_\infty (p/z;p)_\infty, \qquad (a;q,p)_m = \prod_{j=0}^{m-1} \theta(aq^j;p),$$

$$(a_1,\dots,a_s;q,p)_m = \prod_{i=1}^s (a_i;q,p)_m,$$

and

$$\Gamma(z;p,q) = \prod_{j,k\ge 0} \frac{1-p^{j+1}q^{k+1}/z}{1-p^j q^k z}.$$

Definition 21.17 (Hypergeometric deformation hierarchy)**.** For a series $\sum_m c_m$, set $h_m = c_{m+1}/c_m$. It is *ordinary hypergeometric* when h_m is rational in m, and *basic hypergeometric* when h_m is rational in q^m. It is *elliptic hypergeometric* when, after writing $x = q^m$, its term ratio $h(x)$ is multiplicatively p-periodic:

$$h(px) = h(x).$$

Its parameters are *balanced* when the products of the numerator and denominator parameters required by this ellipticity agree. The deformation and degeneration orders are

$$\text{ordinary/additive} \longrightarrow \text{basic/multiplicative} \longrightarrow \text{elliptic},$$

$$\text{elliptic} \xrightarrow{p\to 0} \text{basic} \xrightarrow{q\to 1} \text{ordinary}.$$

Exercise 21.12 (Elliptic functional equations and degenerations)**.** Prove

$$\theta(pz;p) = -z^{-1}\theta(z;p), \qquad \Gamma(qz;p,q) = \theta(z;p)\Gamma(z;p,q),$$

$$\Gamma(pz;p,q) = \theta(z;q)\Gamma(z;p,q), \qquad \Gamma(z;p,q)\Gamma(pq/z;p,q) = 1.$$

Prove

$$\theta(z;0) = 1 - z, \qquad (a;q,0)_m = (a;q)_m, \qquad \Gamma(z;0,q) = \frac{1}{(z;q)_\infty}.$$

Definition 21.18 (Frenkel–Turaev series and elliptic beta integral)**.** For $a^2q^{n+1} = bcde$, define

$$\begin{aligned}{}_{10}V_9(a;b,c,d,e,q^{-n};q,p) = \sum_{m=0}^{n} &\frac{\theta(aq^{2m};p)}{\theta(a;p)}\\ &\times \frac{(a,b,c,d,e,q^{-n};q,p)_m}{(q,aq/b,aq/c,aq/d,aq/e,aq^{n+1};q,p)_m} q^m.\end{aligned}$$

For parameters $t_1,\ldots,t_6$ satisfying

$$|t_j| < 1, \qquad \prod_{j=1}^{6} t_j = pq,$$

the *elliptic beta integral* is

$$\frac{(p;p)_\infty(q;q)_\infty}{4\pi i}\int_{|z|=1} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)}\frac{dz}{z}.$$

Exercise 21.13 (Frenkel–Turaev and elliptic beta evaluations)**.** Prove the Frenkel–Turaev summation

$${}_{10}V_9(a;b,c,d,e,q^{-n};q,p) = \frac{(aq,aq/(bc),aq/(bd),aq/(cd);q,p)_n}{(aq/b,aq/c,aq/d,aq/(bcd);q,p)_n}.$$

Prove the elliptic beta evaluation

$$\frac{(p;p)_\infty(q;q)_\infty}{4\pi i}\int_{|z|=1} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)}\frac{dz}{z} = \prod_{1\le j<k\le 6}\Gamma(t_jt_k;p,q).$$

Take $p \to 0$ in both identities and identify the resulting terminating very-well-poised ${}_8\phi_7$ summation and basic hypergeometric beta integral. Obtain the Askey–Wilson integral by specialization.

Definition 21.19 (Elliptic Ruijsenaars operator)**.** For type A_{r-1}, define

$$D^{\mathrm{ell}}_{p,q,t} = \sum_{i=1}^{r} \left(\prod_{j \neq i} \frac{\theta(tx_i/x_j;p)}{\theta(x_i/x_j;p)} \right) T_{q,x_i}.$$

Exercise 21.14 (Differential, basic, and elliptic operator hierarchy)**.** Construct the commuting higher elliptic Ruijsenaars operators. Prove

$$D^{\mathrm{ell}}_{p,q,t} \longrightarrow D_{q,t} \qquad (p \to 0).$$

Set $q = e^{\varepsilon}$ and $t = e^{g\varepsilon}$ and derive the Heckman–Opdam differential operator from the second-order term as $\varepsilon \to 0$. Take the rational scaling limit of its hyperbolic coefficients and obtain the rational Calogero–Moser operator. Organize the eigenfunction limits as

elliptic Ruijsenaars functions $\longrightarrow$ Macdonald polynomials $\longrightarrow$ Heckman–Opdam functions.

Exercise 21.15 (The Calogero–Sutherland limit)**.** Write $x_i = e^{z_i}$, set $q = e^{\epsilon}$ and $t = e^{g\epsilon}$, and expand $D_{q,t}$ through order ϵ^2. After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i<j} \coth\left(\frac{z_i - z_j}{2}\right)(\partial_{z_i} - \partial_{z_j}).$$

On the circle, put $z_i = i\theta_i$ and conjugate by

$$\Psi_0(\theta) = \prod_{i<j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = -\sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for $r = 2$, degree two. Identify their Sutherland eigenvalues.

Exercise 21.16 (Selberg measure and the Jacobi beta ensemble)**.** For coupling $g > 0$, set $\beta = 2g$. For $a, b > 0$, take the BC_N Calogero–Sutherland ground state in Jacobi coordinates $x_i \in (0,1)$,

$$\Psi_0(x) = C \prod_{i=1}^{N} x_i^{(a-1)/2}(1 - x_i)^{(b-1)/2} \prod_{i<j} |x_i - x_j|^{\beta/2}.$$

Prove

$$|\Psi_0(x)|^2 = \frac{1}{Z^{\mathrm{Jac}}_{N,\beta}(a,b)} \prod_{i=1}^{N} x_i^{a-1}(1-x_i)^{b-1} \prod_{i<j} |x_i - x_j|^{\beta},$$

where the Selberg integral gives

$$Z^{\mathrm{Jac}}_{N,\beta}(a,b) = \prod_{j=0}^{N-1} \frac{\Gamma(a + j\beta/2)\Gamma(b + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(a + b + (N + j - 1)\beta/2)\Gamma(1 + \beta/2)}.$$

Interpret $|\Psi_0|^2 dx_1 \cdots dx_N$ as the joint law of N random eigenvalues. For the type-A circular ground state, prove instead

$$|\Psi_0(\theta)|^2 \propto \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^{\beta}.$$

22 Root-of-Unity Categories and Topological Field Theory

Definition 22.1 (Root-of-unity semisimplification)**.** Fix $k \in \mathbb{N}$ and

$$q = e^{\pi i/(k+2)}.$$

Let $\mathcal{T}_k$ be the ribbon category of finite-dimensional type-1 tilting modules for the Lusztig integral form of $U_q(\mathfrak{sl}_2)$ specialized at q. A morphism $f : X \to Y$ is *negligible* when

$$\operatorname{tr}_q(gf) = 0 \qquad \text{for every } g : Y \to X.$$

Let $\mathcal{N}_k$ be the negligible morphisms and define

$$\mathcal{C}_k = \operatorname{Kar}(\mathcal{T}_k/\mathcal{N}_k),$$

where Kar denotes idempotent completion.

Exercise 22.1 (Semisimplified $SU(2)_k$ fusion)**.** Prove that $\mathcal{N}_k$ is a tensor ideal preserved by duals, braiding, and twist. Prove that $\mathcal{C}_k$ is semisimple with simple objects $V_0, \dots, V_k$ and that

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \le c \le \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod 2}} V_c.$$

Compute

$$\dim_q(V_a) = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

Definition 22.2 (Modular tensor categories)**.** A *fusion category* is a semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit, finite-dimensional morphism spaces, and finitely many simple-object classes. A *modular tensor category* is a ribbon fusion category for which the matrix

$$\widetilde{S}_{ij} = \operatorname{tr}_q(c_{X_j,X_i} c_{X_i,X_j})$$

on the simple objects $(X_i)_{i \in I}$ is invertible. Define

$$d_i = \dim_q(X_i), \qquad \mathcal{D} = \left(\sum_{i \in I} d_i^2\right)^{1/2}, \qquad S = \mathcal{D}^{-1}\widetilde{S}, \qquad T_{ij} = \delta_{ij}\theta_i,$$

where $\theta_{X_i} = \theta_i \operatorname{id}_{X_i}$. It is *unitary* when it is a C^*-tensor category and its braiding and twist are unitary.

Exercise 22.2 (Modularity of $SU(2)_k$)**.** Prove that $\mathcal{C}_k$ is a unitary modular tensor category. For $0 \le a, b \le k$, compute

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\!\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \qquad \theta_a = \exp\!\left(\frac{\pi i\, a(a+2)}{2(k+2)}\right).$$

Prove directly that S is unitary and invertible.

Definition 22.3 (Reshetikhin–Turaev TQFT)**.** Let $\mathrm{Bord}_{3,2}^{\mathrm{ext}}(\mathcal{C})$ be the symmetric monoidal category whose objects are closed oriented surfaces with $\mathcal{C}$-labeled marked points and Lagrangian subspaces of first homology, and whose morphisms are oriented three-dimensional bordisms with colored framed ribbon graphs and integral anomaly weights. A *Reshetikhin–Turaev TQFT* is a symmetric monoidal functor

$$Z_{\mathcal{C}}^{\mathrm{RT}} : \mathrm{Bord}_{3,2}^{\mathrm{ext}}(\mathcal{C}) \longrightarrow \mathrm{Vect}_{\mathbb{C}}^{\mathrm{fd}}$$

obtained from surgery colored by the Kirby element

$$\Omega = \sum_{i \in I} d_i X_i.$$

For $\mathcal{C} = \mathcal{C}_k$, write Z_k^{RT}.

Exercise 22.3 (Reshetikhin–Turaev theorem)**.** Prove invariance of the surgery construction under the two Kirby moves, including the anomaly-weight correction. Prove compatibility with gluing and disjoint union and deduce that $Z_{\mathcal{C}}^{\mathrm{RT}}$ is a symmetric monoidal functor. Prove that a colored framed link in S^3 is evaluated by the ribbon-graph functor of $\mathcal{C}$.

Definition 22.4 (Modular functors and sewing)**.** A *modular functor* assigns a finite-dimensional vector space

$$\mathcal{V}(\Sigma; i_1, \ldots, i_n)$$

to every oriented surface with parametrized boundary circles and labeled marked points, projective linear maps to mapping classes, and tensor products to disjoint unions. For two boundary circles with opposite orientations, its *sewing map* is an isomorphism

$$\mathcal{V}(\Sigma_{\mathrm{sew}}; \boldsymbol{i}) \cong \bigoplus_{a \in I} \mathcal{V}(\Sigma; \boldsymbol{i}, a, a^{\vee}).$$

Exercise 22.4 (RT modular functor)**.** Prove that the surface state spaces of $Z_{\mathcal{C}}^{\mathrm{RT}}$ define a modular functor and satisfy the sewing isomorphism. Prove

$$\mathcal{V}(S^2; i, j, k) \cong \mathrm{Hom}(\mathbf{1}, X_i \otimes X_j \otimes X_k)$$

and derive the pair-of-pants decomposition and its independence under the pentagon and hexagon moves.

Exercise 22.5 (Torus state space from the Atiyah axioms)**.** For each simple object X_i, let M_i be a solid torus whose core is colored by X_i, and set

$$e_i = Z_{\mathcal{C}}^{\mathrm{RT}}(M_i)(1) \in Z_{\mathcal{C}}^{\mathrm{RT}}(T^2).$$

Use Atiyah functoriality, duality, multiplicativity, and gluing, together with the sewing isomorphism, to prove

$$Z_{\mathcal{C}}^{\mathrm{RT}}(T^2) \cong \bigoplus_{i \in I} \mathbb{C} e_i.$$

For an orientation-preserving diffeomorphism $f : T^2 \to T^2$, let M_f be its mapping cylinder. Prove that

$$\rho([f]) = Z_{\mathcal{C}}^{\mathrm{RT}}(M_f)$$

defines a projective mapping-class-group action. Prove

$$\mathrm{MCG}^{+}(T^2) \cong SL_2(\mathbb{Z})$$

with generators

$$\mathsf{s} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{t} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Evaluate the two mapping cylinders by gluing solid tori and prove, up to the anomaly scalar,

$$\rho(\mathsf{s}) = S, \qquad \rho(\mathsf{t}) = T.$$

Exercise 22.6 (Modular relations and the Verlinde formula)**.** For a unitary modular tensor category, let

$$N_{ij}^{\ \ k} = \dim \operatorname{Hom}(X_k, X_i \otimes X_j), \qquad C_{ij} = \delta_{i,j^\vee}.$$

Use sewing and the $SL_2(\mathbb{Z})$ action to prove

$$S^2 = C, \qquad (ST)^3 = \xi C$$

for the anomaly scalar ξ. Prove the Verlinde formula

$$N_{ij}^{\ \ k} = \sum_{a \in I} \frac{S_{ia} S_{ja} \overline{S_{ka}}}{S_{0a}}$$

and the closed-surface dimension formula

$$\dim Z_{\mathcal{C}}^{\mathrm{RT}}(\Sigma_g) = \sum_{a \in I} S_{0a}^{2-2g}.$$

For $\mathcal{C}_k$, recover the truncated $SU(2)_k$ fusion rule from the displayed sine matrix.

23 Index Theory

23.1 Homotopy, homology, and generalized cohomology

Definition 23.1 (Homotopy and fundamental group)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $H : X \times [0,1] \to Y$ with endpoints f and g. The *fundamental group* $\pi_1(X, x_0)$ is the group of endpoint-fixed homotopy classes of loops based at x_0, with concatenation.

Definition 23.2 (Homology and cohomology)**.** Fix a commutative ring R with identity. The *singular chain group* $C_n(X; R)$ is the free R-module on continuous maps $\Delta^n \to X$, with alternating face boundary ∂. The *homology* is $H_n(X; R) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. The *cochain complex* is $C^n(X; R) = \operatorname{Hom}_R(C_n(X; R), R)$ with coboundary $\delta\phi = \phi\partial$, and its cohomology is $H^n(X; R)$.

Definition 23.3 (Cofibers, finite CW complexes, and cohomology theories)**.** A based inclusion $A \hookrightarrow X$ is a *cofibration* if every homotopy on A that extends at its initial time to X extends over the whole homotopy. Its *cofiber* is X/A. More generally, the cofiber of a based map $f : A \to X$ is the pushout $X \cup_f CA$, where CA is the reduced cone; the reduced suspension is $\Sigma A = CA/A$.

A *finite CW complex* is obtained from the empty space by finitely many pushouts

$$\begin{array}{ccc} \coprod_\alpha S^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_\alpha D^n & \longrightarrow & X^n, \end{array}$$

where X^n is the nth skeleton. A *reduced cohomology theory* on based finite CW complexes is a graded contravariant functor $\widetilde{E}^*$ that is homotopy invariant, vanishes on a point, takes finite wedges to direct sums, has natural suspension isomorphisms $\widetilde{E}^{q+1}(\Sigma X) \cong \widetilde{E}^q(X)$, and makes

$$\widetilde{E}^q(X/A) \longrightarrow \widetilde{E}^q(X) \longrightarrow \widetilde{E}^q(A)$$

exact for every cofibration $A \hookrightarrow X$.

Definition 23.4 (Products and pairings)**.** The *Kronecker pairing* is $H^n(X;R) \times H_n(X;R) \to R$, $([\phi],[c]) \mapsto \phi(c)$. The *cup product* is the product on cohomology induced by the front–back face formula on cochains.

Exercise 23.1 (Homology, cohomology, and degree)**.** Prove $\partial^2 = 0$, $\delta^2 = 0$, and homotopy invariance. For a CW pair (X,A), define $C_*(X,A;R) = C_*(X;R)/C_*(A;R)$, construct barycentric subdivision, and use it to prove excision and $H^q(X,A;R) \cong \widetilde{H}^q(X/A;R)$. Construct the Puppe sequence of a cofibration and derive the long exact sequence of a pair and the Mayer–Vietoris sequence. Deduce that reduced singular cohomology is a reduced cohomology theory.

Prove that $X^{n-1} \hookrightarrow X^n$ is a cofibration and X^n/X^{n-1} is a finite wedge of n-spheres. By induction over the skeleta, prove that a natural transformation between reduced cohomology theories that is an isomorphism on S^0 in every degree is an isomorphism on every finite CW complex; prove the required five-term diagram chase. Compute the homology and cohomology rings of spheres and tori. For a continuous map $f : S^n \to S^n$, define its degree by $f_*[S^n] = \deg(f)[S^n]$ and prove multiplicativity and agreement with winding number for $n = 1$.

23.2 Bundles, holonomy, and characteristic classes

Exercise 23.2 (Aharonov–Bohm holonomy)**.** Using the connection, curvature, and holonomy of Definitions 9.7 and 9.8, consider on $X = \mathbb{R}^2 \setminus \{0\}$ the $U(1)$ gauge potential $A = (\Phi/2\pi)d\theta$. Verify that $F = dA = 0$ on X but $\oint_\gamma A = \Phi$ for a loop of winding number one. Show that a particle of charge q acquires the gauge-invariant phase $\exp(iq\Phi/\hbar)$, derive the relative phase of two paths enclosing the excluded flux, and obtain the flux period $h/|q|$. Prove that A cannot be removed by a single-valued gauge transformation unless its holonomy is trivial. Compare the predicted fringe displacement with the shielded-flux electron-holography experiment of Tonomura et al., *Physical Review Letters* **56** (1986). Explain why the result establishes the physical significance of holonomy, not of a gauge-dependent potential by itself, even though curvature vanishes along the electron paths.

Definition 23.5 (Grassmannians and universal bundles)**.** For a finite-dimensional Hermitian space V, the *Grassmannian* $\mathrm{Gr}_n(V)$ is the space of n-dimensional subspaces of V. Its *tautological bundle* has fiber W over $W \in \mathrm{Gr}_n(V)$. Isometric inclusions of Hermitian spaces induce compatible inclusions of their Grassmannians and tautological bundles, and

$$BU(n) = \varinjlim_N \mathrm{Gr}_n(\mathbb{C}^N), \qquad \gamma_n = \varinjlim_N \gamma_{n,N}.$$

Exercise 23.3 (Universal Chern classes)**.** Identify $BU(1)$ with $\mathbb{P}(\mathbb{C}^\infty)$ and use its even-dimensional cell decomposition to compute $H^*(BU(1);\mathbb{Z}) = \mathbb{Z}[x]$, where x evaluates as -1 on the projective line carrying the tautological line bundle. Let

$$s : BU(1)^n \longrightarrow BU(n)$$

classify the direct sum of the n tautological lines, and set $x_i = \mathrm{pr}_i^* x$. Construct the Schubert cell decompositions of the finite Grassmannians and their flag bundles. Use the cellular filtration and the splitting map s to prove that s^* is injective with image

$$\mathbb{Z}[x_1, \dots, x_n]^{\mathfrak{S}_n} = \mathbb{Z}[e_1(x), \dots, e_n(x)].$$

Define $c_j(\gamma_n) \in H^{2j}(BU(n); \mathbb{Z})$ by $s^* c_j(\gamma_n) = e_j(x_1, \dots, x_n)$. For the block-sum map $BU(n) \times BU(m) \to BU(n+m)$, prove

$$c(\gamma_n \oplus \gamma_m) = c(\gamma_n)c(\gamma_m), \qquad c = 1 + c_1 + c_2 + \cdots .$$

Definition 23.6 (Chern classes and character)**.** If a rank-n complex vector bundle $E \to X$ is classified by $f : X \to BU(n)$, its *Chern classes* are $c_j(E) = f^* c_j(\gamma_n) \in H^{2j}(X; \mathbb{Z})$. If the formal roots of $1 + c_1(E) + \cdots + c_n(E)$ are $y_1, \dots, y_n$, the *Chern character* is the degreewise class

$$\mathrm{ch}(E) = \sum_{i=1}^{n} e^{y_i} \in \prod_{k \geq 0} H^{2k}(X; \mathbb{Q}),$$

where each symmetric polynomial in the y_i is expressed in the Chern classes by the Newton identities.

Exercise 23.4 (Curvature representatives)**.** For a Hermitian bundle with unitary connection, prove that

$$\det\Big(1 + \frac{i}{2\pi} F_\nabla\Big) \quad \text{and} \quad \mathrm{Tr}\exp\Big(\frac{i}{2\pi} F_\nabla\Big)$$

are closed forms whose cohomology classes are independent of the connection. Identify their de Rham classes with the images of the Chern classes and Chern character, and prove their Whitney-sum formulas.

23.3 Berry geometry and topological K-theory

Definition 23.7 (Berry geometry)**.** For a smooth family of one-dimensional spectral projections P_x on a Hilbert space, the ranges form a line bundle. The *Berry connection* is $\nabla = P \circ d$ and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral $\langle c_1(L), [\Sigma] \rangle$ is the *Chern number* over an oriented closed surface Σ.

Exercise 23.5 (Berry phase and quantized response)**.** Compute the Berry curvature of a two-level Hamiltonian $H(n) = n \cdot \sigma$ over S^2 for both the lower and upper eigenline bundles. With the standard orientation on S^2, compute their opposite Chern numbers and state the sign convention used. Identify the two eigenlines with classifying maps $f_\pm : S^2 \to \mathbb{P}(\mathbb{C}^2)$ and, for the Fubini–Study normalization $\omega_{\mathrm{FS}}([u], [v]) = -\operatorname{Im}\langle u, v \rangle$, prove

$$F^{\pm}_{\mathrm{Berry}} = -2i f_\pm^* \omega_{\mathrm{FS}}.$$

For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as e^2/h times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus. For an adiabatic closed parameter loop, compute the Berry holonomy, show that it is independent of traversal rate, and identify the corresponding interference-fringe shift.

Definition 23.8 (Grothendieck completion and stable equivalence)**.** The *Grothendieck group* of a commutative monoid M is the universal abelian group receiving a monoid map from M. Vector bundles E, F are *stably equivalent* if $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ for some m, n. The group $K^0(X)$ is the Grothendieck group of complex vector bundles on compact X. For based X, let $\widetilde{K}^0(X)$ be the kernel of restriction to the basepoint and, for every $j \geq 0$, define

$$\widetilde{K}^j(X) = \widetilde{K}^0(\Sigma^j X).$$

For unbased X, write $X_+ = X \sqcup \{*\}$ and set $K^j(X) = \widetilde{K}^j(X_+)$ for $j \geq 0$. The Chern character extends additively to a homomorphism

$$\mathrm{ch} : K^0(X) \longrightarrow \prod_{k \geq 0} H^{2k}(X; \mathbb{Q}).$$

Definition 23.9 (Relative and compactly supported K-theory)**.** For a compact pair (X, A) and a locally compact space Y, define

$$K^j(X, A) = \widetilde{K}^j(X/A), \qquad K^j_c(Y) = \widetilde{K}^j(Y^+),$$

where Y^+ is the one-point compactification. A triple (F^0, F^1, σ) of vector bundles on Y, with $\sigma : F^0 \to F^1$ invertible off a compact set, represents a class in $K^0_c(Y)$ up to homotopy and stabilization.

Definition 23.10 (The K-theory Thom class)**.** Let $\pi : E \to X$ be a rank-r Hermitian complex vector bundle. Exterior multiplication by v and its adjoint give the odd endomorphism

$$c(v) = \varepsilon(v) - \iota(v) : \Lambda^* E_{\pi(v)} \longrightarrow \Lambda^* E_{\pi(v)}, \qquad c(v)^2 = -\|v\|^2.$$

Writing $c^+(v)$ for its even-to-odd part, the *K-theory Thom class* is

$$u_E = [\pi^* \Lambda^{\mathrm{even}} E, \pi^* \Lambda^{\mathrm{odd}} E, c^+] \in K^0_c(E).$$

Exercise 23.6 (Thom isomorphism and Bott periodicity)**.** Prove from the bundle and triple models the cofibration exact sequence

$$\cdots \longrightarrow \widetilde{K}^j(X/A) \longrightarrow \widetilde{K}^j(X) \longrightarrow \widetilde{K}^j(A) \longrightarrow \widetilde{K}^{j+1}(X/A) \longrightarrow \cdots$$

and derive Mayer–Vietoris. For $X = \mathrm{pt}$, prove from the displayed Thom triple that

$$K^0_c(\mathbb{C}) \cong \mathbb{Z}, \qquad K^1_c(\mathbb{C}) = 0,$$

with $u_{\mathbb{C}}$ as generator. Apply these results and cellular induction to prove the trivial-line result over finite CW complexes. Iterate it for trivial bundles and then use Mayer–Vietoris over bundle trivializations to prove the complex Thom isomorphism

$$K^j(X) \longrightarrow K^j_c(E), \qquad a \longmapsto \pi^* a\, u_E, \qquad j = 0, 1,$$

for finite CW complexes X. For the trivial line over a compact based finite CW complex, identify its relative Thom space with $S^2 \wedge X$ and write $\beta = u_{\mathbb{C}} \in \widetilde{K}^0(S^2)$. Deduce complex Bott periodicity:

$$\widetilde{K}^j(X) \xrightarrow{\boxtimes \beta} \widetilde{K}^j(S^2 \wedge X)$$

is an isomorphism.

Exercise 23.7 (K-theory of tori and class-A phases)**.** Derive from the cofibration for $X \times S^1$ and Bott periodicity the split formula $K^j(X \times S^1) \cong K^j(X) \oplus K^{j-1}(X)$, with indices taken modulo two, and compute the complex K-groups of tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the strong and weak class-A invariants on Brillouin tori in dimensions one through three.

Definition 23.11 (Operator K-theory)**.** For a unital C^*-algebra A, $K_0(A)$ is the Grothendieck group of stable homotopy classes of projections in matrix algebras over A, where orthogonal sum gives the underlying monoid, and $K_1(A)$ is the stable homotopy group of unitaries in matrix algebras over A.

23.4 Fredholm operators and elliptic index theory

Definition 23.12 (Compact and Fredholm operators)**.** An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator $T : H_0 \to H_1$ is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is $\operatorname{ind} T = \dim\ker T - \dim\operatorname{coker} T$. The *Calkin algebra* is $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$.

Exercise 23.8 (Fredholm parametrices and the Calkin algebra)**.** For $T : H_0 \to H_1$, prove that T is Fredholm exactly when there is a bounded $S : H_1 \to H_0$ such that $ST - \mathrm{id}_{H_0}$ and $TS - \mathrm{id}_{H_1}$ are compact. When $H_0 = H_1 = H$, identify this condition with invertibility of the image of T in $\mathcal{Q}(H)$. Prove homotopy invariance, compact-perturbation invariance, and additivity of the index.

Definition 23.13 (Elliptic symbols)**.** For vector bundles E, F over a compact smooth manifold, the *principal symbol* of an order-m differential operator $D : \Gamma(E) \to \Gamma(F)$ is the homogeneous bundle map $\sigma_D : \pi^*E \to \pi^*F$ on T^*M. The operator is *elliptic* if $\sigma_D(x, \xi)$ is invertible for every $\xi \neq 0$. For an integer $s \geq 0$, $H^s(E)$ is the completion of smooth sections in the norm formed from their covariant derivatives through order s; negative orders are defined by duality, and different auxiliary choices give equivalent norms. When its Sobolev extension

$$D : H^s(E) \longrightarrow H^{s-m}(F)$$

is Fredholm, its Fredholm index is the *analytic index*.

Exercise 23.9 (Elliptic regularity and analytic index)**.** Construct a pseudodifferential parametrix for an elliptic operator on a compact manifold. Prove elliptic regularity and prove that every Sobolev extension displayed above is Fredholm with index independent of s. Construct from the symbol a compactly supported class in $K^0_c(T^*M)$ and prove that homotopies through elliptic symbols preserve the analytic index.

Exercise 23.10 (Symbol winding and Toeplitz index)**.** Let $H^2(S^1)$ be the closed span of $1, z, z^2, \ldots$ in $L^2(S^1)$ and let P_+ be the orthogonal projection onto it. For $m \in \mathbb{Z}$, compute the kernel and cokernel of $T_{z^m} = P_+ M_{z^m} P_+$ and prove

$$\operatorname{ind}(T_{z^m}) = -m.$$

Identify m with the winding number of the symbol z^m. Use compact- perturbation invariance and multiplication of monomials to show that this index is stable under compact perturbations of the infinite-dimensional Toeplitz operator. If P_N projects onto $\operatorname{span}\{1, z, \ldots, z^N\}$, compute the kernel and cokernel of $P_N T_{z^m} P_N$ for $N \geq |m|$ and show that its finite-dimensional index is zero: the artificial upper cutoff creates a defect that cancels the one-sided Toeplitz defect. Explain why square finite-rank spectral truncations therefore do not preserve the Fredholm index. Compare this compression with Exercise 35.2.

23.5 Dirac operators, anomalies, and topological phases

Definition 23.14 (Spin structures and Dirac operators)**.** A *spin structure* on an oriented Riemannian n-manifold is a lift of its oriented orthonormal frame bundle through $\mathrm{Spin}(n) \to SO(n)$. Its complex spin representation defines the *spinor bundle* S, with Clifford action c and spin connection. The *Dirac operator* is $D = c \circ \nabla^S$. In even dimensions the chirality operator splits $S = S^+ \oplus S^-$. In formal curvature roots,

$$\widehat{A}(TM) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

Definition 23.15 (Chirality and instanton number)**.** Let M be an oriented Riemannian spin four-manifold with the spinor splitting above. For a Hermitian vector bundle E with unitary connection A, the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

For an instanton or anti-instanton from Exercise 9.8, its *instanton number* is

$$k = -\frac{1}{8\pi^2} \int_M \mathrm{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

With the Chern–Weil convention $\mathrm{ch}(E) = \mathrm{Tr}\exp(iF_A/2\pi)$ used above, $k = \int_M \mathrm{ch}_2(E)$. Thus for an $SU(r)$ bundle, $\mathrm{ch}_2(E) = -c_2(E)$; conventions that call $\int_M c_2(E)$ the instanton charge use the opposite sign.

Exercise 23.11 (Index theory and invertible free-fermion phases)**.** Let $H(k)$ be a continuous family of finite-dimensional self-adjoint free-fermion Hamiltonians over a compact momentum space X, with a uniform gap at zero. Flatten its spectrum to

$$Q(k) = \mathrm{sgn}\, H(k) = H(k)|H(k)|^{-1}, \qquad P(k) = \frac{1 - Q(k)}{2},$$

and prove that the occupied spaces $E_k = \mathrm{im}\, P(k)$ form a vector bundle $E \to X$. Show that its stable class $[E] \in K^0(X)$ is invariant under gap-preserving deformations.

Prove that stacking Hamiltonians gives direct sum and addition in $K^0(X)$. After passing to reduced K-theory, show that spectral reversal replaces E by $E^\perp$ and gives the additive inverse because $E \oplus E^\perp$ is trivial. Explain why an invertible phase can have protected boundary modes or symmetry defects but no nontrivial intrinsic bulk fusion sectors. Contrast this group-valued classification with the sector-valued noninvertible phases in the next section.

Exercise 23.12 (The free-fermion periodic table of topological insulators and superconductors)**.** Let $H(k)$ be a spectrally flattened Bloch or Bogoliubov–de Gennes Hamiltonian on the compactified momentum space S^d. Time-reversal, particle-hole, and chiral symmetries act by

$$\mathsf{T}H(k)\mathsf{T}^{-1} = H(-k), \qquad \mathsf{C}H(k)\mathsf{C}^{-1} = -H(-k), \qquad \mathsf{S}H(k)\mathsf{S}^{-1} = -H(k),$$

where T and C are antiunitary, with square $+1$ or -1 when present, and S is unitary, with its phase chosen so that $\mathsf{S}^2 = 1$. Organize the complex classes $\mathrm{A}, \mathrm{AIII}$ and the real classes in the order

s	0	1	2	3	4	5	6	7
class	AI	BDI	D	DIII	AII	CII	C	CI
T^2	$+1$	$+1$	0	-1	-1	-1	0	$+1$
C^2	0	$+1$	$+1$	$+1$	0	-1	-1	-1

and verify that when both antiunitary symmetries occur their product gives a chiral symmetry, up to an irrelevant phase.

For this exercise, define the complex strong-classification entries in dimension d by

$$\pi_0(C_{-d}) \quad \text{for A}, \qquad \pi_0(C_{1-d}) \quad \text{for AIII},$$

and define the entry in real class s by

$$\pi_0(R_{s-d}) \cong KO^{d-s}(\mathrm{pt}).$$

Indices are periodic, and their coefficient groups are

$$\pi_0(C_q) = \begin{cases} \mathbb{Z} & q \equiv 0 \pmod 2, \\ 0 & q \equiv 1 \pmod 2, \end{cases} \qquad \begin{array}{c|cccccccc} q \bmod 8 & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline \pi_0(R_q) & \mathbb{Z} & \mathbb{Z}/2 & \mathbb{Z}/2 & 0 & \mathbb{Z} & 0 & 0 & 0. \end{array}$$

Use these data, rather than a memorized chart, to fill every entry of the periodic table:

class	T^2	C^2	S	$d = 0$	1	2	3	4	5	6	7
A	0	0	0								
AIII	0	0	1								
AI	+1	0	0								
BDI	+1	+1	1								
D	0	+1	0								
DIII	−1	+1	1								
AII	−1	0	0								
CII	−1	−1	1								
C	0	−1	0								
CI	+1	−1	1								

Recover in particular the integer quantum Hall phase in class A, the one-dimensional Majorana chain in class D, the two-dimensional quantum spin Hall phase in class AII, and the three-dimensional class-DIII topological superconductor. Explain why an abstract $\mathbb{Z}$ entry coming from R_4 is often printed as $2\mathbb{Z}$ when its generator maps to twice the corresponding complex invariant; in dimension zero this is the even complex dimension forced by Kramers degeneracy. Finally, explain why this is only the table of *strong* phases: a lattice Brillouin torus can also carry weak invariants. Relate the symmetry-free case to the earlier decomposition of $K^*(\mathbb{T}^d)$, and state why antiunitary symmetries require real, equivariant, or twisted K-theory, following Freed–Moore, *Annales Henri Poincaré* **14** (2013). State why this stable noninteracting classification need not remain unchanged in the presence of strong interactions.

23.6 Spectral triples, ultraviolet cutoffs, and operator systems

Definition 23.16 (Spectral triples and spectral distance)**.** A *spectral triple* (A, H, D) consists of a unital $*$-algebra A represented faithfully by bounded operators on a Hilbert space H and a self-adjoint operator D with dense domain such that $(1 + D^2)^{-1/2}$ is compact, and each commutator $[D, a]$, initially defined on $\mathrm{Dom}(D)$, extends to a bounded operator. Its Lipschitz seminorm is

$$L_D(a) = \|[D, a]\|.$$

On the state space of the C^*-closure of A, the associated extended *spectral distance* is

$$d_D(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : a = a^*,\ L_D(a) \leq 1\}.$$

The value may be infinite if the commutator fails to distinguish enough observables.

Exercise 23.13 (Recovering Riemannian distance)**.** Let M be a compact connected Riemannian spin manifold, let $A = C^\infty(M)$ act by multiplication on $H = L^2(M, S)$, and let D be its Dirac operator. Prove

$$[D, f] = c(df), \qquad \|[D, f]\| = \sup_{x\in M} \|df_x\|.$$

Using the characterization of geodesic distance by 1-Lipschitz functions, show for the evaluation states δ_x, δ_y that

$$d_D(\delta_x, \delta_y) = d_M(x, y).$$

Thus the metric can be recovered from how the Dirac operator fails to commute with the observable algebra.

Definition 23.17 (Spectral truncation)**.** For $\Lambda > 0$, let

$$P_\Lambda = \mathbf{1}_{[-\Lambda,\Lambda]}(D), \qquad H_\Lambda = P_\Lambda H, \qquad D_\Lambda = P_\Lambda D P_\Lambda.$$

The compressed observables are

$$S_\Lambda = P_\Lambda A P_\Lambda = \{P_\Lambda a P_\Lambda : a \in A\} \subseteq B(H_\Lambda).$$

A state on S_Λ is a positive linear functional ω with $\omega(P_\Lambda) = 1$. The truncated commutator seminorm and spectral distance are

$$L_\Lambda(x) = \|[D_\Lambda, x]\|, \qquad d_\Lambda(\omega, \eta) = \sup_{\substack{x=x^*\\ L_\Lambda(x)\le 1}} |\omega(x) - \eta(x)|.$$

Exercise 23.14 (Metric geometry of the truncated circle)**.** Prove that compactness of $(1 + D^2)^{-1/2}$ makes P_Λ finite-rank and that S_Λ is an operator system with identity P_Λ, inherited adjoint, and inherited matrix-order cones.

Return to the Toeplitz operator system S_N from Exercise 35.2, restrict the algebra to $A = C^\infty(S^1)$, and set

$$D = -i\frac{d}{d\theta}, \qquad D_N = P_N D P_N.$$

Verify

$$[D_N, P_N M_f P_N]_{jk} = (j - k)\widehat{f}(j - k)$$

and deduce that the kernel of L_N on S_N consists only of scalars.

Show that rotations commute with D_N and hence preserve L_N and the spectral distance on the state space of S_N. For

$$v_{N,\theta} = (2N + 1)^{-1/2} \sum_{k=-N}^{N} e^{-ik\theta} e_k,$$

identify the expectation $\langle v_{N,\theta}, P_N M_f P_N v_{N,\theta}\rangle$ with convolution by a Fejér kernel and show that it tends to $f(\theta)$ as $N \to \infty$. Interpret these vector states as points blurred by finite spectral resolution; see Connes–van Suijlekom, *Spectral Truncations in Noncommutative Geometry and Operator Systems*, Communications in Mathematical Physics **383** (2021).

24 Gauge Duality and the Geometric Langlands Program

24.1 Supersymmetry and electric–magnetic duality

Definition 24.1 ($\mathcal{N}=4$ super Yang–Mills)**.** Let G be a compact Lie group and let $P \to M$ be a principal G-bundle over an oriented Riemannian four-manifold. The bosonic fields of $\mathcal{N}=4$ super Yang–Mills theory are a connection A and six adjoint-valued scalar fields $X_1, \dots, X_6$. With an invariant positive inner product on $\mathfrak{g}$, the bosonic action has the form

$$\frac{1}{g^2}\int_M \left(\frac{1}{2}|F_A|^2 + \frac{1}{2}\sum_I |d_A X_I|^2 + \frac{1}{2}\sum_{I<J} |[X_I, X_J]|^2\right) \mathrm{vol} - i\theta k(A),$$

supplemented by fermions and Yukawa couplings fixed by supersymmetry. The six scalars transform under the R-symmetry group $\mathrm{Spin}(6)_R \cong SU(4)_R$, and the theory has sixteen real supercharges. Its gauge coupling and theta angle combine into

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}.$$

The potential is minimized when the X_I commute, so scalar expectation values can break G to a maximal torus and make electric and magnetic charges visible simultaneously.

Definition 24.2 (Langlands dual group)**.** Let $T \subset G$ be a maximal torus. Its character and cocharacter lattices are

$$X^*(T) = \mathrm{Hom}(T, U(1)), \qquad X_*(T) = \mathrm{Hom}(U(1), T).$$

The *Langlands dual group* ${}^L G$ is the compact form whose complexified root datum is dual to that of G: characters and cocharacters are exchanged, and roots are exchanged with coroots. Thus

$$X^*({}^L T) = X_*(T), \qquad X_*({}^L T) = X^*(T).$$

Ignoring global-form refinements, $U(1)$ is self-dual, ${}^L SU(n) = PSU(n)$, and the root systems B_n and C_n are exchanged. The complex reductive group used in geometric Langlands is ${}^L G_{\mathbb{C}}$.

Definition 24.3 (Electric–magnetic duality transformation)**.** The basic *S-transformation* on gauge-theory parameters is

$$(G, \tau) \longmapsto ({}^L G, -1/\tau).$$

It exchanges character and cocharacter lattices, and hence electric and magnetic charge labels. For a non-simply-laced root system define the transformation by $\tau \mapsto -1/(n_G \tau)$, where n_G is the ratio of squared long- and short-root lengths.

Exercise 24.1 (Charge lattices and the dual group)**.** For a torus $T = U(1)^r$, prove that a representation restricts to characters $\lambda : T \to U(1)$ with $\lambda \in X^*(T)$. A magnetic monopole defines a transition map around its equator, hence a cocharacter $\mu : U(1) \to T$ with $\mu \in X_*(T)$. Show that composition gives an integer pairing

$$\langle \lambda, \mu \rangle \in \mathbb{Z},$$

which is the Dirac quantization condition. Verify directly that exchanging $X^*(T)$ and $X_*(T)$ exchanges electric and magnetic labels. For $G = SU(2)$, compare the weight and coweight lattices and explain why the global dual group is adjoint rather than simply connected.

24.2 The geometric Langlands twist

Definition 24.4 (Cohomological twist data)**.** In Euclidean signature the rotation group is locally $\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-$. A *topological twist* replaces it by a diagonal combination with a $\mathrm{Spin}(4)$ subgroup of $\mathrm{Spin}(6)_R$. The *cohomological twist data* include scalar odd operators Q_ℓ, Q_r whose projective linear combinations

$$Q_t = Q_\ell + tQ_r, \qquad t \in \mathbb{P}^1,$$

satisfy $Q_t^2 = 0$ on gauge-invariant quantities. The associated observables are classes

$$H_{Q_t} = \ker Q_t / \operatorname{im} Q_t.$$

Definition 24.5 (Canonical parameter)**.** For $(\tau, t) \in \mathbb{H} \times \mathbb{P}^1$, define the *canonical parameter*

$$\Psi = \frac{\tau + \bar\tau}{2} + \frac{\tau - \bar\tau}{2} \frac{t - t^{-1}}{t + t^{-1}} = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2} \frac{t - t^{-1}}{t + t^{-1}}.$$

Under the S-transformation, define the twist-parameter change by $t \mapsto \pm t\tau/|\tau|$; the sign is immaterial because t and $-t$ define equivalent twists. This rule and $\tau \mapsto -1/(n_G\tau)$ give $\Psi \mapsto -1/(n_G\Psi)$, where n_G is the ratio of the squared lengths of long and short roots and equals one in the simply laced case.

Definition 24.6 (Kapustin–Witten equations)**.** For a connection A, an adjoint-valued one-form ϕ, and $t \in \mathbb{P}^1$, the *Kapustin–Witten equations* are

$$\begin{aligned} (F_A - \phi \wedge \phi + t\, d_A\phi)^+ &= 0, \\ (F_A - \phi \wedge \phi - t^{-1} d_A\phi)^- &= 0, \\ d_A^*\phi &= 0. \end{aligned} \tag{KW}$$

At $t = 0$ and $t = \infty$, multiply the appropriate equation by t or t^{-1} before taking the limit.

Exercise 24.2 (Q-cohomology and the Kapustin–Witten parameter)**.** Let $Q^2 = 0$. Prove that adding QV to a Q-closed element does not change its class in $\ker Q / \operatorname{im} Q$, and determine why the same statement is undefined outside $\ker Q$. Next substitute $t = i$ into (KW) and verify directly that its first two lines combine into

$$F_{A+i\phi} = F_A - \phi \wedge \phi + i\, d_A\phi = 0.$$

Finally, compute Ψ at $t = \pm 1$ and at $t = \pm i$, and verify algebraically that S-duality acts by $\Psi \mapsto -1/(n_G\Psi)$.

24.3 Dimensional reduction and Hitchin moduli space

Definition 24.7 (Hitchin equations and Higgs bundles)**.** For a connection A on a principal G-bundle over C and $\phi \in \Omega^1(C, \operatorname{ad} P)$, the *Hitchin equations* are

$$F_A - \phi \wedge \phi = 0, \qquad d_A\phi = 0, \qquad d_A^*\phi = 0. \tag{H}$$

Writing the $(1,0)$ part of ϕ as φ, these become, up to a conventional normalization of φ,

$$\bar\partial_A \varphi = 0, \qquad F_A + [\varphi, \varphi^\dagger] = 0.$$

Thus (E, φ) is a $G_{\mathbb{C}}$-*Higgs bundle*: E is a holomorphic principal $G_{\mathbb{C}}$-bundle and

$$\varphi \in H^0(C, K_C \otimes \operatorname{ad} E).$$

The quotient of solutions to (H) by compact gauge transformations is the *Hitchin moduli space* $\mathcal{M}_H(G, C)$. Equivalently, after the appropriate stability condition is imposed, it is the moduli space of polystable $G_{\mathbb{C}}$-Higgs bundles.

Exercise 24.3 (Hitchin equations from four-dimensional instantons)**.** On $\mathbb{R}^4$ take a connection invariant under translations in x^3, x^4 and write $X = A_3$, $Y = A_4$. Decompose the anti-self-duality equations into

$$F_{12} + [X, Y] = 0, \qquad D_1 X - D_2 Y = 0, \qquad D_1 Y + D_2 X = 0,$$

up to the sign determined by the chosen orientation. With $z = x^1 + ix^2$ and a suitable choice of $\varphi = (X + iY)\, dz$, show that the last two equations give $\bar{\partial}_A \varphi = 0$ and the first gives the real Hitchin equation. This calculation is not the full supersymmetric compactification, but it exhibits the same dimensional-reduction mechanism using the instantons already studied in Exercise 9.8.

24.4 From gauge theory to branes

Definition 24.8 (Sigma-model reduction)**.** A two-dimensional sigma model on Σ with target X has fields $\Phi : \Sigma \to X$. In the adiabatic limit of the twisted gauge theory on $\Sigma \times C$, its low-energy configurations are slowly varying solutions of (H), hence maps

$$\Phi : \Sigma \longrightarrow \mathcal{M}_H(G, C).$$

The twist determines which topological sigma model survives. At $\Psi = \infty$ it is the B-model in complex structure J; at $\Psi = 0$ it is the A-model with symplectic form ω_K.

Definition 24.9 (Mirror-dual Hitchin targets)**.** The Hitchin targets for G and ${}^L G$ are *mirror-dual at the canonical parameters* when there is an equivalence

$$B_J(\mathcal{M}_H(G, C)) \simeq A_K(\mathcal{M}_H({}^L G, C)), \tag{M}$$

where the subscripts record the complex or symplectic structure.

24.5 Line operators, Hecke modifications, and eigensheaves

Definition 24.10 (Wilson and 't Hooft lines)**.** A Wilson line measures the holonomy of a connection in a representation R:

$$W_R(\gamma) = \operatorname{Tr}_R \operatorname{Hol}_\gamma(\mathcal{A}).$$

At $\Psi = \infty$, the topological line uses the flat complex connection $\mathcal{A} = A + i\phi$ and is labeled by a representation of G. A *'t Hooft line* is defined instead by prescribing a magnetic singularity transverse to the line. Its charge is a cocharacter of G, equivalently a weight of ${}^L G$. The S transformation exchanges these two kinds of probes. When a line lies along a curve in Σ at a point $x \in C$, it acts on branes in the reduced sigma model.

Definition 24.11 (Hecke modification)**.** Let $x \in C$. A *Hecke modification* of a holomorphic principal $G_{\mathbb{C}}$-bundle E at x is another bundle E' together with an isomorphism

$$E|_{C\setminus\{x\}} \cong E'|_{C\setminus\{x\}}.$$

Its relative position at x is labeled by a dominant coweight of G. A *Hecke functor* H_V is the correspondence on bundles and sheaves obtained by summing modifications with dominant coweight labeled by the representation V of ${}^LG$.

Definition 24.12 (D-modules and Hecke eigensheaves)**.** Let $\mathrm{Bun}_G(C)$ denote the moduli stack of holomorphic $G_{\mathbb{C}}$-bundles on C; a stack is needed because bundles can have automorphisms. A D-module on it is, schematically, a sheaf carrying a compatible action of differential operators. Let $\mathcal{E}$ be a flat ${}^LG_{\mathbb{C}}$-bundle on C. A *Hecke eigensheaf* with eigenvalue $\mathcal{E}$ is a D-module $\mathcal{F}_{\mathcal{E}}$ equipped, for every representation V of ${}^LG$, with compatible isomorphisms

$$\mathsf{H}_V(\mathcal{F}_{\mathcal{E}}) \cong V_{\mathcal{E}} \boxtimes \mathcal{F}_{\mathcal{E}}, \tag{HE}$$

where $V_{\mathcal{E}}$ is the flat vector bundle obtained from $\mathcal{E}$ through V. Compatibility with tensor products of representations is part of the definition.

Definition 24.13 (The Kapustin–Witten dictionary)**.** The *Kapustin–Witten dictionary* is the chain of labels

$$\begin{gathered}\text{flat } {}^LG_{\mathbb{C}}\text{-connection } \mathcal{E}\\ \updownarrow\\ \text{point } B\text{-brane on } \mathcal{M}_H({}^LG, C)\\ \updownarrow\\ \text{Hecke eigenbrane on } \mathcal{M}_H(G, C)\\ \updownarrow\\ \text{Hecke eigensheaf } \mathcal{F}_{\mathcal{E}} \text{ on } \mathrm{Bun}_G(C).\end{gathered}$$

The line-operator labels on the two middle entries are exchanged by character–cocharacter duality.

Exercise 24.4 (From an eigenbrane to the Hecke condition)**.** Suppose topological line operators $L_V(x)$ obey the representation-ring law

$$L_V(x)L_W(x) \cong L_{V\otimes W}(x)$$

and a brane $\mathcal{B}$ obeys $L_V(x)\mathcal{B} \cong \mathcal{B} \otimes E_{V,x}$. Show that the spaces $E_{V,x}$ must satisfy $E_{V\otimes W,x} \cong E_{V,x} \otimes E_{W,x}$. Explain why topological motion of x makes E_V a local system on C. After Wilson–'t Hooft label exchange, rewrite this statement as (HE).

25 Integrable Models and Root-System Special Functions

25.1 Yang–Baxter optical tests

Exercise 25.1 (A direct optical Yang–Baxter test)**.** Consider the unitary polarization transformations

$$A(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \qquad B(\theta) = \begin{pmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{pmatrix}.$$

Multiply the matrices and prove that, away from singular boundary cases,

$$A(\theta_1)B(\theta_2)A(\theta_3) = B(\theta_3)A(\theta_2)B(\theta_1)$$

exactly when

$$\tan\theta_2 = \frac{\sin(\theta_1+\theta_3)}{\cos(\theta_1-\theta_3)}.$$

For $\theta_1 = 56^\circ$ and $\theta_3 = 23^\circ$, obtain $\theta_2 \approx 49.49^\circ$. For an arbitrary input qubit $|\psi\rangle$, show that the output-state overlap has absolute value one when the relation holds and is generally smaller away from it. Compare these predictions with the wave-plate experiment of Zheng et al., *Journal of the Optical Society of America B* **30** (2013). State clearly that this experiment tests a two-dimensional Temperley–Lieb reduction of the Yang–Baxter equation, not the full representation theory of a quantum group or DAHA.

25.2 Quantum spherical functions and qKZ transport

Definition 25.1 (Quantum spherical functions)**.** Let $H = U_q(\mathfrak{g})$ and let $B \subseteq H$ be a right coideal subalgebra, so $\Delta(B) \subseteq B \otimes H$. It plays the role of the stabilizer algebra of a quantum homogeneous space. If an H-module V contains a vector v satisfying

$$bv = \varepsilon(b)v \qquad (b \in B),$$

then the matrix coefficient

$$\varphi_V(x) = \langle v, \pi_V(x)v\rangle$$

is a *quantum zonal spherical function.* It belongs to the coordinate Hopf dual of H and is invariant under the left and right quantum stabilizer actions.

Exercise 25.2 (Flatness, qKZ transport, and affine Hecke monodromy)**.** Let a vector-valued function satisfy the adjacent exchange rule

$$\Psi(\ldots, z_i, z_{i+1}, \ldots) = \check{R}_i(z_i/z_{i+1})\Psi(\ldots, z_{i+1}, z_i, \ldots).$$

Show that consistency of the two routes reversing three adjacent variables is exactly the spectral Yang–Baxter equation. Add a periodic rule that returns z_1 at the end as qz_1, and explain why translations enlarge the finite braid action to an affine one. If the R-matrices satisfy a Hecke quadratic, prove that the exchange monodromy factors through an affine Hecke algebra. Write the resulting reflection and translation operators in the same generators used in the polynomial representation defined above.

25.3 Elliptic star–triangle and Yang–Baxter identities

Definition 25.2 (Star–triangle data)**.** A family of edge weights $W_u(x, y)$, one-spin weights $S(x)$, and scalar normalizations $R(u, v, w)$ is *star–triangle data* when

$$\int S(x)W_u(a, x)W_v(b, x)W_w(c, x)\,dx = R(u, v, w)W_{u'}(b, c)W_{v'}(a, c)W_{w'}(a, b)$$

for its parameter constraint and contour. It is *elliptic* when its weights are finite products and ratios of $\Gamma(\,cdot\,; p, q)$.

Exercise 25.3 (Elliptic beta integral and star–triangle relation)**.** Parametrize the six variables in the elliptic beta integral by three spectral parameters and three external spins. Derive an elliptic star–triangle identity from its evaluation. Apply the identity twice to a three-row lattice strip and prove equality of the two local reorderings that yields commuting transfer matrices. Take $p \to 0$ and then $q \to 1$ and derive the corresponding trigonometric and rational star–triangle relations. Compare this integral construction with the $6j$-symbol/recoupling construction of the Yang–Baxter equation.

Part III

Counting

26 Linear Spaces, Tensor Products, and Duality

Definition 26.1 (Linear spaces and maps)**.** Let k be $\mathbb{R}$ or $\mathbb{C}$. A *linear space* over k is an abelian group V with scalar multiplication satisfying

$$a(v+w) = av + aw, \quad (a+b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map $T : V \to W$ is *linear* when $T(av + bw) = aT(v) + bT(w)$. Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For $T : V \to V$, a scalar λ is an *eigenvalue* when $\ker(T - \lambda I) \neq 0$; a nonzero vector in this kernel is an *eigenvector*.

Definition 26.2 (Dual spaces and tensor products)**.** The *dual space* of V is $V^\vee = \operatorname{Hom}_k(V, k)$. Its canonical evaluation pairing is

$$\mathrm{ev}_V : V^\vee \times V \longrightarrow k, \qquad (\varphi, v) \longmapsto \varphi(v).$$

The algebraic transpose of $T : V \to W$ is $T^\vee : W^\vee \to V^\vee$, defined by $T^\vee(\varphi) = \varphi \circ T$.

The *tensor product* $V \otimes W$ is characterized by a bilinear map $(v, w) \mapsto v \otimes w$ with the following universal property: every bilinear map $b : V \times W \to X$ factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\widetilde{b}} X.$$

The evaluation pairing induces a linear map $\mathrm{ev}_V : V^\vee \otimes V \to k$.

Definition 26.3 (Dualizability and finite dimensionality)**.** A linear space V is *dualizable* when there is a coevaluation map

$$\mathrm{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the two snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^\vee}) \circ (\mathrm{id}_{V^\vee} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^\vee}.$$

A linear space is *finite-dimensional* if it is dualizable.

Definition 26.4 (Categorical trace and dimension)**.** Let V be dualizable, let $s_{V,V^\vee} : V \otimes V^\vee \to V^\vee \otimes V$ exchange the factors, and let $T \in \operatorname{End}(V)$. The *categorical trace* is the scalar

$$\operatorname{tr}(T) = \mathrm{ev}_V \circ s_{V,V^\vee} \circ (T \otimes \mathrm{id}_{V^\vee}) \circ \mathrm{coev}_V, \qquad \dim V = \operatorname{tr}(\mathrm{id}_V).$$

Exercise 26.1 (Dualizability recovers ordinary finite dimension)**.** Construct $V \otimes W$ as a quotient of the free linear space on $V \times W$ and prove its universal property. If V has basis $e_1, \ldots, e_n$ with dual basis $e^1, \ldots, e^n$, show that

$$\mathrm{coev}_V(1) = \sum_{j=1}^{n} e_j \otimes e^j$$

is independent of the chosen basis and satisfies the snake identities. Conversely, write any proposed coevaluation as $\mathrm{coev}_V(1) = \sum_{j=1}^{r} v_j \otimes \varphi_j$ and use a snake identity to prove that the v_j span V. Conclude that dualizability is equivalent to ordinary finite dimensionality. Finally prove

$$\mathrm{tr}(T) = \sum_{j=1}^{n} e^j(Te_j), \qquad \mathrm{tr}(ST) = \mathrm{tr}(TS), \qquad \dim V = n.$$

27 Hilbert Spaces

Definition 27.1 (Hilbert spaces and orthogonality)**.** An *inner product* on a complex linear space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies $\langle x, y\rangle = \overline{\langle y, x\rangle}$. Its norm is $\|x\| = \sqrt{\langle x, x\rangle}$. A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace $M \subseteq H$, define

$$M^{\perp} = \{x : \langle m, x\rangle = 0 \text{ for every } m \in M\}.$$

Definition 27.2 (Bounded operators and adjoints)**.** A linear map $T : H \to K$ between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \le 1} \|Tx\| < \infty.$$

An *adjoint* of T is an operator $T^* : K \to H$ satisfying

$$\langle x, Ty\rangle = \langle T^*x, y\rangle.$$

An operator is self-adjoint when $T = T^*$, unitary when $T^*T = TT^* = I$, and normal when $T^*T = TT^*$. An *orthogonal projection* is an operator P satisfying $P = P^* = P^2$.

Exercise 27.1 (Hilbert-space foundations)**.** Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \qquad (\operatorname{im} T)^{\perp} = \ker T^*.$$

For every closed subspace $M \subseteq H$, prove $H = M \oplus M^{\perp}$ and construct the orthogonal projection $P_M : H \to M$. Show that an orthonormal family (e_j) is complete exactly when $\|x\|^2 = \sum_j |\langle e_j, x\rangle|^2$ for every $x \in H$.

28 Measure and Probability

Definition 28.1 (Measure spaces and Haar measure)**.** A *σ-algebra* $\mathcal{F}$ on a set X contains $\varnothing$ and is closed under complements and countable unions. A *measure* is a countably additive map

$\mu : \mathcal{F} \to [0,\infty]$ with $\mu(\varnothing) = 0$, and $(X, \mathcal{F}, \mu)$ is a *measure space*. A map is *measurable* when inverse images of measurable sets are measurable. For $1 \le p < \infty$, define

$$L^p(X,\mu) = \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} /\text{equality a.e.}$$

with its usual norm, and let $L^\infty(X,\mu)$ consist of essentially bounded measurable functions.

A *topological group* is a group with a Hausdorff topology for which multiplication and inversion are continuous. It is *locally compact* when every point has a compact neighborhood. A *Radon measure* is a Borel measure finite on compact sets, outer regular on Borel sets, and inner regular on open sets. A *left Haar measure* on a locally compact group is a nonzero left-invariant Radon measure, so $\mu(gE) = \mu(E)$. Its *modular function* is determined by

$$\mu(Eg) = \Delta_G(g)^{-1}\mu(E).$$

Write $C_b(G)$, $C_0(G)$, and $C_c(G)$ for the continuous functions that are bounded, vanish at infinity, and have compact support, respectively.

Definition 28.2 (Probability spaces and convergence)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map $X : \Omega \to Y$, its *law* is $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$, and, when it is scalar-valued and integrable, its *expectation* is its integral. A random vector $X = (X_1, \dots, X_N)$ has *joint density* p with respect to a measure μ when

$$\mathbb{P}(X \in B) = \int_B p \, d\mu.$$

The Gaussian law $N(m, v)$ on $\mathbb{R}$ has density

$$\frac{1}{\sqrt{2\pi v}} \exp\left[-\frac{(x-m)^2}{2v} \right].$$

A sequence of random variables in a metric space (Y, d) *converges in probability* to X when

$$\mathbb{P}(d(X_N, X) > \varepsilon) \longrightarrow 0 \qquad (\varepsilon > 0).$$

It *converges in distribution*, written $X_N \Longrightarrow X$, when

$$\int f \, d\mathbb{P}_{X_N} \longrightarrow \int f \, d\mathbb{P}_X$$

for every bounded continuous f.

29 C^*-Algebras and Functional Calculus

Definition 29.1 (C^*-algebras, spectrum, and positivity)**.** A *C^*-algebra* is a complete normed complex algebra A with a conjugate-linear involution satisfying

$$(ab)^* = b^* a^*, \qquad \|ab\| \le \|a\| \|b\|, \qquad \|a^* a\| = \|a\|^2.$$

For a unital C^*-algebra, define

$$\operatorname{spec}_A(a) = \{ \lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible} \}.$$

An element is *positive*, written $a \ge 0$, when $a = b^* b$ for some $b \in A$. It is *self-adjoint* when $a = a^*$ and *normal* when $a^* a = a a^*$.

Definition 29.2 (Gelfand spectrum)**.** For a commutative C^*-algebra A, let $\widehat{A}$ be the space of nonzero homomorphisms $\chi : A \to \mathbb{C}$ with its weak-$*$ topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\widehat{A}), \qquad \Gamma(a)(\chi) = \chi(a).$$

Exercise 29.1 (Spectrum, Gelfand duality, and functional calculus)**.** Prove that $\operatorname{spec}_A(a)$ is nonempty and compact and that

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}.$$

For normal a, prove $r(a) = \|a\|$. Prove that positive elements have unique positive square roots and that $a \geq 0$ if and only if $a = a^*$ and $\operatorname{spec}_A(a) \subseteq [0, \infty)$. Prove that the Gelfand transform is an isometric $*$-isomorphism. For every normal element a of a unital C^*-algebra, prove that there is a unique unital isometric $*$-homomorphism

$$C(\operatorname{spec}(a)) \longrightarrow C^*(1, a), \qquad f \longmapsto f(a),$$

sending the coordinate function to a, and prove the continuous spectral mapping theorem.

30 Operator Spaces and Operator Tensor Products

Definition 30.1 (Operator spaces and completely bounded maps)**.** An *operator space* is a closed linear subspace $E \subseteq B(H)$ with the matrix norms inherited from

$$M_n(E) \subseteq B(H^{\oplus n}).$$

For a linear map $u : E \to F$, define

$$\|u\|_{cb} = \sup_{n\geq 1} \| \operatorname{id}_{M_n} \otimes u\|.$$

The map is *completely bounded* when $\|u\|_{cb} < \infty$ and a *complete isometry* when every $\operatorname{id}_{M_n} \otimes u$ is isometric. The dual operator-space structure on E^* is determined by

$$M_n(E^*) = \mathrm{CB}(E, M_n)$$

isometrically.

Definition 30.2 (Row and column spaces)**.** For a Hilbert space H, its *column space* and conjugate *row space* are

$$H_c = B(\mathbb{C}, H), \qquad \overline{H}_r = B(H, \mathbb{C})$$

with their inherited operator-space structures. For operator spaces E and F, the *Haagerup norm* is

$$\|u\|_h = \inf \left\{ \|[x_1 \ \cdots \ x_m]\|_{M_{1,m}(E)} \|[y_1 \ \cdots \ y_m]^t\|_{M_{m,1}(F)} : u = \sum_{j=1}^m x_j \otimes y_j \right\}.$$

The completion is denoted $E \otimes_h F$. The *operator-space projective norm* is

$$\|u\|_\wedge = \inf\{\|\alpha\| \, \|x\| \, \|y\| \, \|\beta\| : u = \alpha(x \otimes y)\beta\},$$

where $x \in M_p(E)$, $y \in M_q(F)$, $\alpha \in M_{1,pq}$, and $\beta \in M_{pq,1}$. Its completion is denoted $E \mathbin{\widehat{\otimes}} F$.

Definition 30.3 (Trace and Hilbert–Schmidt classes)**.** For $x, y \in H$, let

$$\theta_{x,y}(\xi) = \langle y, \xi \rangle x.$$

The algebra $\mathcal{K}(H)$ of *compact operators* is the norm closure of the span of the operators $\theta_{x,y}$. The *trace class* $S_1(H)$ consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j,y_j}, \qquad \sum_{j=1}^{\infty} \|x_j\| \, \|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\mathrm{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j \rangle.$$

The *Hilbert–Schmidt class* $S_2(H)$ consists of operators satisfying

$$\|A\|_2^2 = \sum_j \|Ae_j\|^2 < \infty$$

for an orthonormal basis (e_j) of H.

Definition 30.4 (Ruan's axioms)**.** Matrix norms on a complex linear space E satisfy *Ruan's axioms* when

$$\|x \oplus y\|_{M_{n+m}(E)} = \max\{\|x\|_{M_n(E)}, \|y\|_{M_m(E)}\}$$

and

$$\|\alpha x \beta\|_{M_{r,s}(E)} \le \|\alpha\| \, \|x\|_{M_n(E)} \, \|\beta\|$$

for scalar matrices of compatible sizes.

Exercise 30.1 (Row–column tensor duality)**.** Prove that the definitions of $S_1(H)$, Tr, and $S_2(H)$ are independent of all displayed choices. Prove the complete isometries

$$H_c \otimes_h \overline{H}_r \cong \mathcal{K}(H), \qquad \overline{H}_r \mathbin{\widehat{\otimes}} H_c \cong \overline{H}_r \otimes_h H_c \cong S_1(H), \qquad S_1(H)^* \cong B(H),$$

where $\overline{y} \otimes x$ corresponds to $\theta_{x,y}$ and the last pairing is $(A, T) \mapsto \mathrm{Tr}(AT)$. Prove that $\langle A, B \rangle_{S_2} = \mathrm{Tr}(A^* B)$ makes $S_2(H)$ a Hilbert space.

Exercise 30.2 (Ruan's representation theorem)**.** Prove that matrix norms arise from a complete isometry $E \to B(H)$ if and only if they satisfy Ruan's axioms. Prove $M_n(E^*) = \mathrm{CB}(E, M_n)$ completely isometrically.

31 Spectral Theorems from Harmonic Analysis

Definition 31.1 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X is a map E from measurable subsets of X to orthogonal projections on H satisfying

$$E(\varnothing) = 0, \qquad E(X) = I, \qquad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral $\int_X f \, dE$ for bounded measurable f is defined by uniform approximation from simple functions. For a topological space X, the measure is *regular* when every scalar measure $B \mapsto \langle x, E(B)y \rangle$ is a regular Borel measure.

Exercise 31.1 (Representations of $C_0(X)$)**.** Let X be locally compact Hausdorff. A representation $\pi : C_0(X) \to B(H)$ is *nondegenerate* when $\overline{\pi(C_0(X))H} = H$. Prove that every nondegenerate $*$-representation has a unique regular projection-valued measure E_π satisfying

$$\pi(f) = \int_X f(x)\, dE_\pi(x), \qquad f \in C_0(X).$$

Exercise 31.2 (Bounded normal spectral theorem)**.** For a normal operator $T \in B(H)$, apply the preceding representation theorem to the continuous functional calculus $C(\mathrm{spec}(T)) \to B(H)$ and prove that there is a unique projection-valued measure E_T on $\mathrm{spec}(T)$ satisfying

$$T = \int_{\mathrm{spec}(T)} z\, dE_T(z), \qquad f(T) = \int_{\mathrm{spec}(T)} f(z)\, dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space is unitarily diagonalizable.

Exercise 31.3 (Compact spectral theorem)**.** Prove that every trace-class operator is compact. For a compact normal operator K, prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_j \lambda_j P_j$$

in operator norm for pairwise orthogonal finite-rank spectral projections P_j. For $K \geq 0$ in $S_1(H)$, prove

$$K = \sum_j \lambda_j \theta_{e_j, e_j}, \qquad \lambda_j \geq 0, \qquad \sum_j \lambda_j = \operatorname{Tr} K.$$

Definition 31.2 (Unbounded self-adjoint operators)**.** An unbounded operator is a linear map $A : \mathcal{D}(A) \to H$ on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax\rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For $y \in \mathcal{D}(A^*)$, the vector A^*y is determined by $\langle A^*y, x\rangle = \langle y, Ax\rangle$ for $x \in \mathcal{D}(A)$. The operator is *self-adjoint* when $A = A^*$, including equality of domains, and *closed* when its graph is closed in $H \oplus H$.

Exercise 31.4 (Unbounded spectral theorem and Stone's theorem)**.** For a self-adjoint operator A, prove that $A \pm iI$ are bijective from $\mathcal{D}(A)$ to H with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with $\ker(I - C_A) = 0$. Apply the bounded normal spectral theorem to C_A and the inverse Cayley map to prove that there is a unique projection-valued measure E_A on $\mathbb{R}$ satisfying

$$A = \int_{\mathbb{R}} \lambda\, dE_A(\lambda), \qquad \mathcal{D}(A) = \left\{x : \int_{\mathbb{R}} \lambda^2\, d\langle x, E_A(\lambda)x\rangle < \infty\right\}.$$

Use E_A to prove that e^{-itA} is a strongly continuous unitary group and that every strongly continuous one-parameter unitary group has a unique self-adjoint generator.

Exercise 31.5 (Spectral lines and time evolution)**.** If $A\psi_n = \lambda_n\psi_n$ and an observable B has matrix elements $B_{mn} = \langle \psi_m, B\psi_n\rangle$, prove

$$\langle \psi_m, e^{itA/\hbar} B e^{-itA/\hbar} \psi_n\rangle = e^{it(\lambda_m - \lambda_n)/\hbar} B_{mn}.$$

Extend the calculation from pure point spectrum to the projection-valued spectral measure.

32 Ergodic Theory

32.1 Probability, conditioning, and entropy

Definition 32.1 (Conditional expectation)**.** For $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, a *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is a $\mathcal{G}$-measurable integrable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P} \qquad (G \in \mathcal{G}).$$

Definition 32.2 (Shannon entropy)**.** For a discrete probability vector p, its *Shannon entropy* is

$$H(p) = -\sum_i p_i \log p_i.$$

The corresponding physical entropy is $S(p) = k_B H(p)$. For probability vectors p, q with $p_i = 0$ whenever $q_i = 0$, their *relative entropy* is

$$D(p\|q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

For a joint distribution p_{XY}, the *mutual information* is $I(X : Y) = D(p_{XY}\|p_X p_Y)$. For a phase-space probability density ρ relative to a fixed dimensionless normalization of Liouville measure μ_L (equivalently, to a chosen phase-space cell), its *statistical entropy* is $S(\rho) = -k_B \int \rho \log \rho\, d\mu_L$; for densities ρ, σ, set $D(\rho\|\sigma) = \int \rho \log(\rho/\sigma)\, d\mu_L$ when defined.

32.2 Time–bandwidth concentration and Gaussian channels

Definition 32.3 (Time and band limiting)**.** On $L^2(\mathbb{R})$, define the time-limiting and band-limiting projections by

$$(Q_T f)(x) = \mathbf{1}_{[-T,T]}(x) f(x), \qquad B_W = \mathcal{F}^{-1}\mathbf{1}_{[-W,W]}\mathcal{F},$$

where $T, W > 0$. The *time–band concentration operator* is

$$K_{T,W} = Q_T B_W Q_T.$$

After identifying $Q_T L^2(\mathbb{R})$ with $L^2([-T,T])$, its normalized eigenfunctions are the time-limited representatives of the *prolate spheroidal wave functions*, also called *Slepian functions*. The dimensionless parameter $c = TW$ is the time–bandwidth product.

Exercise 32.1 (Optimal simultaneous concentration)**.** Use Plancherel's theorem to prove that Q_T and B_W are orthogonal projections. Starting from the Fourier-transform convention in Harmonic Analysis, derive

$$(K_{T,W} f)(x) = \int_{-T}^{T} \frac{\sin W(x-y)}{\pi(x-y)} f(y)\, dy, \qquad |x| \le T,$$

where the diagonal value of the kernel is W/π. Prove that $K_{T,W}$ is a positive compact self-adjoint contraction. If its positive eigenvalues are $\lambda_0 \ge \lambda_1 \ge \cdots > 0$, use the compact spectral theorem to prove the Rayleigh–Ritz formula

$$\lambda_0 = \sup_{\|f\|=1} \langle f, K_{T,W} f\rangle,$$

and hence show

$$\lambda_0 = \sup_{\substack{f\neq 0 \\ \operatorname{supp} f\subseteq[-T,T]}} \frac{\displaystyle\int_{-W}^{W} |\widehat{f}(\xi)|^2\, d\xi}{\displaystyle\int_{\mathbb{R}} |f(x)|^2\, dx}.$$

Show likewise that $B_W Q_T B_W$ has the same nonzero eigenvalues and that, for a band-limited eigenfunction g_n, λ_n is the fraction of its energy lying in $[-T,T]$. Thus successive eigenfunctions solve the optimal concentration problem subject to orthogonality to the preceding solutions.

Exercise 32.2 (Slepian modes and Gaussian channel capacity)**.** Let

$$A_{T,W} = B_W Q_T : Q_T L^2(\mathbb{R}) \longrightarrow B_W L^2(\mathbb{R}).$$

Show that $A_{T,W}^* A_{T,W} = K_{T,W}$. If f_n is a normalized λ_n-eigenfunction of $K_{T,W}$, prove that $g_n = \lambda_n^{-1/2} A_{T,W} f_n$ is normalized, that the g_n are mutually orthogonal, and that

$$A_{T,W} f_n = \sqrt{\lambda_n}\, g_n.$$

Thus the numbers $\sqrt{\lambda_n}$ are the singular gains of the time–bandwidth channel.

Restrict to the first N real modes and add independent Gaussian noise:

$$Y_n = \sqrt{\lambda_n}\, X_n + Z_n, \qquad Z_n \sim \mathcal{N}(0, N_0), \qquad 0 \le n < N.$$

For a real random vector with density p, write $h(p) = -\int p \log p$. Prove nonnegativity of relative entropy for such densities from $\log u \le u - 1$, and use it to prove that a Gaussian maximizes h among densities with fixed covariance. Taking channel capacity to be the supremum of $I(X:Y)$, show that under the power constraint $\sum_n \mathbb{E}[X_n^2] \le P$ it is, in nats per channel use,

$$C_N(P) = \max_{\substack{p_n\ge 0 \\ \sum_n p_n \le P}} \frac{1}{2} \sum_{n=0}^{N-1} \log\left(1 + \frac{\lambda_n p_n}{N_0}\right).$$

Show that independent centered Gaussian inputs attain the maximum and use Lagrange multipliers to obtain the water-filling allocation

$$p_n = \left(\nu - \frac{N_0}{\lambda_n}\right)_+, \qquad \sum_n p_n = P.$$

Explain why $2TW/\pi = \sum_n \lambda_n$ estimates the number of useful modal channels, while capacity depends on the full eigenvalue distribution, the noise level, and the power constraint.

32.3 Ergodic dynamics and equilibrium states

Definition 32.4 (Ergodic dynamics and dynamical entropy)**.** A measurable transformation T of a probability space is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$. Its *Koopman operator* is the isometry $U_T f = f \circ T$ on $L^2(\mu)$. The system is *ergodic* if every invariant measurable set has measure zero or one, and *mixing* if

$$\mu(T^{-n}A \cap B) \longrightarrow \mu(A)\mu(B)$$

for all measurable A, B. An *equilibrium state* is an invariant probability measure for the dynamics.

For a finite measurable partition $\mathcal{P}$, set

$$H_\mu(\mathcal{P}) = -\sum_{A\in\mathcal{P}} \mu(A)\log\mu(A), \qquad h_\mu(T,\mathcal{P}) = \lim_{n\to\infty}\frac{1}{n}H_\mu\left(\bigvee_{j=0}^{n-1} T^{-j}\mathcal{P}\right).$$

The *Kolmogorov–Sinai entropy* is $h_\mu(T) = \sup_{\mathcal{P}} h_\mu(T,\mathcal{P})$.

Exercise 32.3 (Ergodic averages, mixing, and entropy rate)**.** Prove that the fixed vectors of U_T are precisely the invariant L^2-observables, and that ergodicity is equivalent to all such observables being constant almost everywhere. Prove the von Neumann mean ergodic theorem

$$\frac{1}{N}\sum_{n=0}^{N-1} U_T^n f \longrightarrow P_{\mathrm{inv}} f \quad \text{in } L^2(\mu),$$

where P_{inv} is orthogonal projection onto the invariant observables. Prove that mixing implies ergodicity and that it gives decay of correlations $\int (f\circ T^n) g\, d\mu - \int f\, d\mu \int g\, d\mu \to 0$ for bounded observables.

For every finite partition $\mathcal{P}$, prove by subadditivity that the limit defining $h_\mu(T,\mathcal{P})$ exists. For the two-sided Bernoulli shift on $\{1,\dots,r\}^{\mathbb{Z}}$ with product probabilities $p_1,\dots,p_r$, prove mixing first for cylinder sets and then for all measurable sets. Show that the time-zero coordinate partition satisfies

$$h_\mu(T,\mathcal{P}) = -\sum_{i=1}^{r} p_i \log p_i = H(p).$$

Interpret this equality as the information produced per observed symbol.

Definition 32.5 (Gibbs states)**.** For a classical Hamiltonian H on a phase space with Liouville measure μ_L, the *canonical Gibbs state* at inverse temperature β is

$$d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H}\, d\mu_L, \qquad Z(\beta) = \int e^{-\beta H}\, d\mu_L,$$

provided $0 < Z(\beta) < \infty$. It is stationary under the Hamiltonian dynamics generated by H.

Exercise 32.4 (Entropy and equilibrium)**.** For every density ρ with finite mean energy, prove

$$D(\rho\|\rho_\beta) = -\frac{S(\rho)}{k_B} + \beta\langle H\rangle_\rho + \log Z(\beta).$$

Deduce that the Gibbs ensemble uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Prove $\langle H\rangle = -\partial_\beta \log Z$ and $\mathrm{Var}(H) = \partial_\beta^2 \log Z$. Prove invariance of Gibbs measure under Hamiltonian flow. For a finite measure-preserving system, apply the L^2 mean ergodic result proved above to an observable and identify the long-time mean-square limit in the ergodic case.

Exercise 32.5 (Ideal-gas thermodynamics)**.** Compute the classical canonical partition function of N indistinguishable noninteracting particles in a box, including the phase-space normalization $h^{3N}N!$. Derive the ideal-gas law, internal energy, and entropy, and the one-particle Maxwell–Boltzmann speed distribution. Derive its temperature scaling and compare both predictions with dilute-gas measurements in the classical regime.

For a wall of area A, express the pressure measured during time τ as the time-averaged normal momentum transfer

$$P_\tau = \frac{1}{A\tau} \sum_{\substack{\text{collisions with the wall} \\ 0<t_j\le\tau}} 2m|v_{j,\perp}|.$$

Average the collision flux using the Maxwell–Boltzmann distribution and recover the same pressure $P = Nk_BT/V$. Identify the stationarity and ergodic assumptions that relate the long-time reading of one gas sample to the ensemble calculation. Explain why agreement of pressure, velocity statistics, and relaxation rates for selected observables and finite times tests consequences of those assumptions but does not prove that the full microscopic dynamics is mathematically ergodic.

33 Operator Algebras

33.1 Representations and GNS construction

Definition 33.1 (Representations, states, and cyclic representations)**.** A *representation* of a C^*-algebra A is a $*$-homomorphism $\pi : A \to B(H)$. A *state* is a positive linear functional of norm one. A representation is *cyclic* when there is a vector $\Omega \in H$ such that $\overline{\pi(A)\Omega} = H$. For a state φ, define

$$N_\varphi = \{a \in A : \varphi(a^*a) = 0\}, \qquad \langle [a], [b] \rangle_\varphi = \varphi(a^*b)$$

on A/N_φ.

Exercise 33.1 (GNS theorem)**.** Prove that N_φ is a left ideal, that the displayed form is an inner product, and that left multiplication extends to a cyclic representation $(\pi_\varphi, H_\varphi, \Omega_\varphi)$ satisfying

$$\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle.$$

Prove uniqueness up to cyclic unitary equivalence. For a discrete group, apply the construction to the coefficient-at-the-identity state on $C_r^*(G)$ and recover the left regular representation on $\ell^2(G)$.

33.2 Noncommutative probability

Definition 33.2 (Noncommutative probability)**.** A *noncommutative probability space* is a unital $*$-algebra A together with a linear functional $\varphi : A \to \mathbb{C}$ satisfying $\varphi(a^*a) \ge 0$ and $\varphi(1) = 1$. Its random variables are elements of A, and the joint $*$-distribution of $X_1, \dots, X_r$ is the functional

$$p \longmapsto \varphi\big(p(X_1, \dots, X_r, X_1^*, \dots, X_r^*)\big)$$

on noncommutative $*$-polynomials. For a self-adjoint variable X, its moments are $\varphi(X^n)$. A *tracial probability space* also satisfies $\varphi(ab) = \varphi(ba)$. Random Hermitian matrices form such spaces after combining expectation with the normalized trace.

33.3 Quantum states and dynamics

Definition 33.3 (Quantum states and observables)**.** A *density operator* on H is a positive element $\rho \in S_1(H)$ with $\operatorname{Tr} \rho = 1$. An *observable* is a self-adjoint operator A with spectral measure E_A. The *Born probability* of an event B is $\operatorname{Tr}(\rho E_A(B))$. A closed-system *unitary evolution* is $\rho(t) = U_t\rho(0)U_t^*$, where $(U_t)_{t\in\mathbb{R}}$ is a strongly continuous one-parameter unitary group. For unbounded observables and generators, spectral measures, expectations, and exponentials use the earlier unbounded spectral calculus and its natural domains.

Exercise 33.2 (Quantized harmonic oscillator)**.** For $H = P^2/(2m) + m\omega^2 Q^2/2$, define creation and annihilation operators on the Schwartz core $\mathcal{S}(\mathbb{R})$ and prove there that $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Construct the eigenstates with $E_n = \hbar\omega(n + 1/2)$ and use the spectral-lines exercise to obtain the frequency spacing ω. Prove that their span is a core for the self-adjoint closure of H. For a two-level internal transition of frequency Ω_0 coupled linearly to Q, compute the oscillator matrix elements, obtain motional sidebands at $\Omega_0 \pm \omega$, and compare their spacing with trapped-ion spectra.

Exercise 33.3 (Liouville and von Neumann equations)**.** Prove that the Born rule defines a probability measure and that its expected value is $\mathrm{Tr}(\rho A)$ when defined.

Let $\mu_t = (\phi_t)_*\mu_0$ have smooth density p_t relative to the Liouville measure μ_L of a time-independent Hamiltonian system. Using Exercise 10.4, prove

$$\partial_t p_t + \{p_t, H\} = 0, \qquad \frac{d}{dt}\int_M f\, d\mu_t = \int_M \{f, H\}\, d\mu_t.$$

For $U_t = e^{-itH/\hbar}$, set $\rho_t = U_t \rho_0 U_t^*$. Prove

$$\dot{\rho}_t = -\frac{i}{\hbar}[H, \rho_t].$$

Prove

$$\int_M f\, d\mu_t = \int_M (f \circ \phi_t)\, d\mu_0, \qquad \mathrm{Tr}(\rho_t A) = \mathrm{Tr}(\rho_0 U_t^* A U_t),$$

and prove conservation of expected energy in both theories.

Exercise 33.4 (Blackbody radiation from quantized Fourier modes)**.** For electromagnetic radiation in a large conducting cavity, use its Fourier modes and two polarizations to derive the mode density $g(\omega) = \omega^2/(\pi^2 c^3)$ per unit volume. Show that classical Gibbs equipartition assigns mean energy $k_B T$ to each mode and produces the ultraviolet divergence. Using the oscillator levels derived in Exercise 33.2, compute the quantum Gibbs partition function and the thermal mean energy $\hbar\omega/(e^{\beta\hbar\omega} - 1)$. Deduce Planck's law

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1},$$

the Stefan–Boltzmann T^4 law, and Wien scaling $\omega_{\max} \propto T$. Compare the low- and high-frequency limits with the cavity spectra discussed by Planck, *Annalen der Physik* **309** (1901), and identify precisely where quantization removes the classical divergence.

33.4 Fermionic statistics and the CAR algebra

Definition 33.4 (CAR algebra and fermionic Fock space)**.** For a complex Hilbert space K, the *canonical anticommutation relations algebra* $\mathrm{CAR}(K)$ is the unital C^*-algebra generated by conjugate-linear symbols $a(f)$ satisfying

$$\{a(f), a(g)\} = 0, \qquad \{a(f), a(g)^*\} = \langle f, g\rangle 1.$$

Let $\Lambda_2^n K$ be the Hilbert-space completion of the nth antisymmetric tensor power of K, with $\Lambda_2^0 K = \mathbb{C}$. The *antisymmetric Fock space* is the Hilbert direct sum

$$\mathcal{F}_-(K) = \bigoplus_{n\geq 0} \Lambda_2^n K.$$

Its vacuum is $\Omega = 1 \in \Lambda_2^0 K$; creation by f is exterior multiplication, and annihilation by f is its Hilbert-space adjoint.

Exercise 33.5 (CAR and GNS predict fermionic antibunching)**.** Prove that creation and annihilation on $\mathcal{F}_-(K)$ satisfy the CAR and that $\omega_0(x) = \langle \Omega, x\Omega \rangle$ is a state. Construct the cyclic map $[x] \mapsto x\Omega$ from its GNS pre-Hilbert space and prove that its completion is the Fock representation. Let ω_C be a gauge-invariant quasifree state with one-particle density $0 \le C \le 1$, and use its determinantal Wick rule to prove, for orthonormal detector modes f, g, that

$$\omega_C(N_f N_g) = \omega_C(N_f)\omega_C(N_g) - |\langle f, Cg \rangle|^2, \qquad N_f = a(f)^* a(f).$$

Deduce $N_f^2 = N_f$ and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

33.5 Von Neumann algebras and noncommutative conditioning

Exercise 33.6 (Macroscopic magnetization and von Neumann closure)**.** Let $\mathcal{A}_N = M_2(\mathbb{C})^{\otimes N}$, include $\mathcal{A}_N$ in $\mathcal{A}_{N+1}$ by $a \mapsto a \otimes 1$, and form the quasilocal spin algebra

$$\mathcal{A} = \overline{\bigcup_{N \ge 1} \mathcal{A}_N}^{\|\cdot\|}.$$

For $Z = \mathrm{diag}(1, -1)$, let Z_j act as Z at site j and set

$$M_N = \frac{1}{N} \sum_{j=1}^{N} Z_j.$$

Prove $\|M_{2N} - M_N\| = 1$, so (M_N) has no norm limit in $\mathcal{A}$.

Let $|+\rangle, |-\rangle$ be the Z-eigenvectors and define product states on local elementary tensors by

$$\omega_\pm(a_1 \otimes \cdots \otimes a_N) = \prod_{j=1}^{N} \langle \pm | a_j | \pm \rangle.$$

Prove that these extend to states on $\mathcal{A}$. In their GNS triples $(H_\pm, \pi_\pm, \Omega_\pm)$, prove

$$\pi_+(M_N) \longrightarrow I_{H_+}, \qquad \pi_-(M_N) \longrightarrow -I_{H_-}$$

in the strong operator topology. First prove the convergence on vectors $\pi_\pm(a)\Omega_\pm$ with $a \in \mathcal{A}_k$, using

$$\|[M_N, a]\| \le \frac{2k}{N} \|a\|,$$

and then extend it to all vectors using $\|M_N\| \le 1$.

On $H_+ \oplus H_-$, let $\pi = \pi_+ \oplus \pi_-$ and $\Omega = 2^{-1/2}(\Omega_+ \oplus \Omega_-)$. Prove that Ω represents the mixed state $\omega = (\omega_+ + \omega_-)/2$ and is cyclic. Show that

$$\pi(M_N) \longrightarrow C = I_{H_+} \oplus (-I_{H_-})$$

strongly, that $C \in \pi(\mathcal{A})'' \cap \pi(\mathcal{A})'$, and that the spectral projections $(I \pm C)/2$ select the two magnetized phases. Thus the C^*-algebra describes the common local experiments, while its von Neumann closure in a chosen state also contains the macroscopic phase observable. Interpret this as an idealized zero-temperature ferromagnet; in an extremal low-temperature phase of an interacting ferromagnet, the corresponding limit is $\pm m_\beta I$ under the appropriate ergodic or clustering hypotheses.

Definition 33.5 (Noncommutative conditional expectation)**.** For a von Neumann subalgebra $N \subseteq M$, a *conditional expectation* is a normal positive unital projection $E : M \to N$ satisfying

$$E(n_1 a n_2) = n_1 E(a) n_2 \qquad (a \in M,\ n_1, n_2 \in N).$$

It is *φ-preserving* for a normal state φ when $\varphi \circ E = \varphi$. When $M = L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ and $N = L^\infty(\Omega, \mathcal{G}, \mathbb{P})$, this recovers classical conditional expectation onto $\mathcal{G}$.

Exercise 33.7 (Conditioning on subalgebras)**.** Show that classical conditional expectation is positive, unital, contractive, and satisfies the bimodule identity above. For a normal state ψ on N, prove that

$$\mathrm{id}_M \,\overline{\otimes} \psi : M \,\overline{\otimes}\, N \longrightarrow M$$

is a normal conditional expectation onto $M \otimes 1$ after the canonical identification, and determine which product states it preserves. If M is a finite von Neumann algebra with normalized trace τ and $N \subseteq M$, show that the orthogonal projection $L^2(M, \tau) \to L^2(N, \tau)$ restricts to the unique τ-preserving conditional expectation $M \to N$. Derive the tower property for nested von Neumann subalgebras.

33.6 Factor types, modular theory, and relative entropy

Definition 33.6 (Traces and factor types)**.** A *finite trace* on a von Neumann algebra is a normal positive functional τ satisfying $\tau(ab) = \tau(ba)$. A possibly unbounded *normal tracial weight* is an additive, positively homogeneous map $\tau : M_+ \to [0, \infty]$ satisfying $\tau(x^*x) = \tau(xx^*)$ and $\tau(x_\lambda) \uparrow \tau(x)$ whenever $x_\lambda \uparrow x$. It is *semifinite* if every nonzero $x \in M_+$ majorizes a nonzero $y \in M_+$ with $\tau(y) < \infty$.

Projections $p, q \in M$ are *Murray–von Neumann equivalent*, written $p \sim q$, if some $v \in M$ satisfies $v^*v = p$ and $vv^* = q$. A projection p is *finite* if $q \le p$ and $q \sim p$ imply $q = p$, and it is *minimal* if its only subprojections are 0 and p.

A *factor* is a von Neumann algebra with center $\mathbb{C}1$. A factor is *type I* if it has a nonzero minimal projection, equivalently if it is isomorphic to some $B(K)$; it is *type II* if it has nonzero finite projections but no minimal projections; and it is *type III* if it has no nonzero finite projection. A factor is *hyperfinite* if it is the weak closure of an increasing union of finite-dimensional $*$-subalgebras.

Exercise 33.8 (Projection dimension and factors)**.** For a finite factor with normalized trace τ, prove additivity and unitary invariance of τ on projections. Relate the possible projection values to matrix factors and to diffuse finite factors. Prove that $B(K)$ is type I and that its standard trace is semifinite but is not a finite functional when K is infinite-dimensional. Prove that a type-III factor has no nonzero normal tracial state and no trace-density representation of its normal states. Explain why finite-dimensional quantum systems see only type-I factors, whereas the classification permits genuinely different infinite systems.

Definition 33.7 (Cyclic and separating vectors)**.** For $M \subseteq B(H)$, a vector Ω is *cyclic* if $\overline{M\Omega} = H$ and *separating* if $a\Omega = 0$ implies $a = 0$. For such a pair (M, Ω), define the densely defined antilinear operator $S_0(a\Omega) = a^*\Omega$.

Definition 33.8 (KMS states)**.** With the present sign convention, a state φ is *β-KMS* for a flow α_t if, for analytic a, b,

$$\varphi(a\alpha_{t+i\beta}(b)) = \varphi(\alpha_t(b)a).$$

Exercise 33.9 (Finite modular flow and Gibbs equilibrium)**.** Let $H = H^*$ on a finite-dimensional Hilbert space and $\rho = Z^{-1}e^{-\beta H}$. On $B(H)$ define $\alpha_t(a) = e^{itH}ae^{-itH}$. Prove directly, first on matrix units in an energy eigenbasis and then by linearity, that

$$\mathrm{Tr}\big(\rho a \alpha_{t+i\beta}(b)\big) = \mathrm{Tr}\big(\rho \alpha_t(b) a\big).$$

Show that the modular flow of ρ is $a \mapsto \rho^{it} a \rho^{-it} = \alpha_{-\beta t}(a)$ and that $-\log\rho = \beta H + \log Z$. Deduce that the same spectral data determine energy gaps, equilibrium weights, and modular frequencies.

Definition 33.9 (Relative entropy of operator-algebraic states)**.** Let φ and ψ be faithful normal states on M, represented by vectors Ω_φ and Ω_ψ in standard form. Their relative Tomita operator is the closure of

$$S_{\varphi|\psi,0}(a\Omega_\psi) = a^*\Omega_\varphi,$$

and their *relative modular operator* is

$$\Delta_{\varphi|\psi} = S^*_{\varphi|\psi} S_{\varphi|\psi}.$$

The *Araki relative entropy* is the extended nonnegative number

$$D_M(\varphi\|\psi) = \langle \Omega_\varphi, \log\Delta_{\varphi|\psi}\, \Omega_\varphi\rangle.$$

For $M = B(H)$ and faithful density operators ρ, σ, the standard Hilbert–Schmidt representation gives $\Delta_{\rho|\sigma}(X) = \rho X \sigma^{-1}$ and hence

$$D(\rho\|\sigma) = \mathrm{Tr}(\rho(\log\rho - \log\sigma))\,.$$

34 Quantum Information Theory

34.1 Measurements, entropy, and tomography

Definition 34.1 (Measurements)**.** A *positive operator-valued measure* is a map E from measurable events to positive operators with $E(X) = 1$ that is countably additive in the weak operator topology. In state ρ, its outcome probability is $\mathrm{Tr}(\rho E(B))$.

Exercise 34.1 (POVMs and dilation)**.** Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions, realizing every POVM's outcome probabilities by a projective measurement on a larger Hilbert space. Show that state discrimination can require a nonprojective POVM.

Definition 34.2 (Von Neumann entropy)**.** For a density operator with discrete spectral resolution $\rho = \sum_{\lambda>0} \lambda P_\lambda$, its *von Neumann entropy* is

$$S(\rho) = -k_B\,\mathrm{Tr}(\rho\log\rho) = -k_B \sum_{\lambda>0} (\mathrm{rank}\, P_\lambda)\lambda\log\lambda,$$

with $0\log 0 = 0$; in infinite dimension the value may be $+\infty$. It is k_B times the Shannon entropy of the outcomes of a projective measurement in an eigenbasis of ρ, and it vanishes exactly for pure states.

Definition 34.3 (Quadratures, Wigner distributions, and homodyne data)**.** For one canonical degree of freedom, choose $q_0, p_0 > 0$ with $q_0 p_0 = \hbar$ and set

$$X = Q/q_0, \qquad Y = P/p_0, \qquad X_\theta = X\cos\theta + Y\sin\theta,$$

so that $[X, Y] = i$. For a density operator ρ, its *Weyl characteristic function* is

$$\chi_\rho(u, v) = \mathrm{Tr}\Big(\rho e^{i(uX+vY)}\Big),$$

and its *Wigner distribution* is the inverse Fourier transform

$$W_\rho(x, y) = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \chi_\rho(u, v)e^{-i(ux+vy)}\, du\, dv,$$

interpreted as a tempered distribution when necessary. If E_θ is the spectral measure of X_θ, the *homodyne quadrature distribution* is

$$p_\rho(r \mid \theta)\, dr = \mathrm{Tr}(\rho E_\theta(dr)).$$

In balanced optical homodyne detection, the local-oscillator phase selects θ and repeated measurements estimate this distribution.

Exercise 34.2 (Optical homodyne tomography)**.** Use Exercise 15.5 and the spectral theorem to prove the Fourier-slice identity

$$\widetilde{p}_\rho(k \mid \theta) = \int_{\mathbb{R}} p_\rho(r \mid \theta)e^{ikr}\, dr = \chi_\rho(k\cos\theta, k\sin\theta).$$

Deduce, in the distributional sense, that the measured quadrature density is the Radon transform of the Wigner distribution:

$$p_\rho(r \mid \theta) = \int_{\mathbb{R}^2} W_\rho(x, y)\delta(r - x\cos\theta - y\sin\theta)\, dx\, dy.$$

Prove that the data for $0 \le \theta < \pi$ determine χ_ρ, hence W_ρ and ρ, and derive the filtered-backprojection formula

$$W_\rho(x, y) = \frac{1}{(2\pi)^2}\int_0^\pi \int_{\mathbb{R}} |k|\, \widetilde{p}_\rho(k \mid \theta)e^{-ik(x\cos\theta + y\sin\theta)}\, dk\, d\theta.$$

Show that every $p_\rho(\,\cdot \mid \theta)$ is a genuine probability density although W_ρ may be negative. Compute the data and reconstruction for vacuum, coherent, squeezed, and one-photon states. Explain why the factor $|k|$ amplifies high-frequency noise and why finite-angle experimental data require regularization or constrained density-matrix estimation. Compare the predicted reconstruction of vacuum and squeezed states with Smithey et al., *Physical Review Letters* **70** (1993).

Definition 34.4 (Quantum relative entropy)**.** For finite-dimensional density operators with $\operatorname{supp}\rho \le \operatorname{supp}\sigma$, their *quantum relative entropy* is

$$D(\rho\|\sigma) = \mathrm{Tr}(\rho(\log\rho - \log\sigma)),$$

and it is $+\infty$ otherwise. It reduces to classical relative entropy when ρ and σ commute.

Exercise 34.3 (Entropy and quantum Gibbs states)**.** Let K be n-dimensional, let $P_1, \dots, P_n$ be mutually orthogonal rank-one projections with sum 1_K, and let $\mathcal{D} = \operatorname{span}\{P_1, \dots, P_n\} \cong \mathbb{C}^n$. Identify a state on $\mathcal{D}$ with a probability vector $p = (p_i)$ and the density operator $\rho_p = \sum_i p_i P_i$. Prove

$$S(\rho_p) = -k_B \sum_i p_i \log p_i,$$

so von Neumann entropy restricts to classical entropy on a commutative observable algebra. Prove more generally that $S(U\rho U^*) = S(\rho)$ for every unitary U.

Let $H = H^* \in B(K)$ have at least two distinct eigenvalues, and let the prescribed energy E lie strictly between its smallest and largest eigenvalues. Set

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)}, \qquad Z(\beta) = \operatorname{Tr}(e^{-\beta H}).$$

Use the spectral resolution of H to reduce the partition function to a finite classical Gibbs distribution. Referring to Exercise 32.4 and its variance identity, prove that there is a unique $\beta \in \mathbb{R}$ for which $\operatorname{Tr}(\rho_\beta H) = E$. For an arbitrary density operator ρ, now prove the genuinely noncommutative identity

$$D(\rho \| \rho_\beta) = -\frac{S(\rho)}{k_B} + \beta \operatorname{Tr}(\rho H) + \log Z(\beta).$$

Use nonnegativity to obtain the quantum variational principle from the classical one: ρ_β uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Finally, restrict ρ_β to the commutative algebra generated by the spectral projections of H and recover the classical Gibbs probabilities, including their energy-level degeneracies. If g_i is the degeneracy of E_i and p_i is the total population of that energy level, derive

$$\frac{p_i}{p_j} = \frac{g_i}{g_j} e^{-\beta(E_i - E_j)}.$$

Explain why the degeneracy factor is absent for the ratio of populations of individual eigenstates, and compare both formulas with measured level-population ratios.

34.2 Composite systems and entanglement

Definition 34.5 (Composite and reduced states)**.** The Hilbert space of a composite system is $H_A \otimes H_B$, and its normal states are the density operators on this tensor product. A density operator is *separable* if it lies in the trace-norm closed convex hull of product density operators and *entangled* otherwise. The *reduced state* $\rho_A = \operatorname{Tr}_B \rho$ is determined by $\operatorname{Tr}(\rho_A A) = \operatorname{Tr}(\rho(A \otimes \mathrm{id}))$ for all bounded A.

Definition 34.6 (Classical and tensor independence)**.** Families of σ-algebras are *independent* when the probability of every finite intersection of events, one from each family, is the product of their probabilities. Equivalently, bounded random variables measurable with respect to distinct families have factorizing mixed expectations.

Commuting unital subalgebras $M_1, M_2 \subseteq M$ are independent in the state φ when $\varphi(ab) = \varphi(a)\varphi(b)$ for $a \in M_1$ and $b \in M_2$. They are *tensor independent* when the algebra they generate, together with the state, is a spatial tensor product with a product state.

Exercise 34.4 (Wishart matrices and typical entanglement). Let G be a $d_A \times d_B$ matrix of independent standard complex Gaussian entries and set

$$W = GG^*, \qquad \rho_A = \frac{W}{\operatorname{Tr} W}.$$

Show that ρ_A has the same distribution as the reduced state of a Haar-random unit vector in $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$. Using Gaussian or Haar integration, prove

$$\mathbb{E}\operatorname{Tr}(\rho_A^2) = \frac{d_A + d_B}{d_A d_B + 1}.$$

Deduce that when $d_B \gg d_A$, a typical reduced state is close to maximally mixed in Hilbert–Schmidt norm. Interpret this as a random-matrix explanation of why a typical pure state of a large composite system is highly entangled.

Definition 34.7 (Entanglement entropy and mutual information). For a pure state $\rho = |\psi\rangle\langle\psi|$ on $H_A \otimes H_B$, its *entanglement entropy* is $S_A(\psi) = S(\rho_A)$. For an arbitrary bipartite state, its *quantum mutual information* is

$$I(A:B)_\rho = D(\rho \| \rho_A \otimes \rho_B).$$

Exercise 34.5 (Schmidt projections and entanglement). Let H_A and H_B be finite-dimensional. For a pure vector $\psi \in H_A \otimes H_B$, apply the projection form of the singular-value decomposition to the contraction map $C_\psi : H_A^* \to H_B$. If Q_s and R_s are its domain and range projections, decompose ψ into mutually orthogonal spectral blocks supported on $(Q_s H_A^*)^* \otimes R_s H_B$. Explain why a rank-one refinement within a block of multiplicity greater than one is not canonical. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectral values with multiplicities $\operatorname{rank} Q_s = \operatorname{rank} R_s$. Derive

$$S_A(\psi) = -k_B \sum_{s>0} (\operatorname{rank} Q_s) s^2 \log s^2$$

and prove that it vanishes exactly for separable pure states. Compute the joint and reduced states, local outcome frequencies, and entanglement entropy of a Bell state, and compare them with entangled-pair tomography. For every bipartite density operator ρ, prove

$$I(A:B)_\rho = \frac{S(\rho_A) + S(\rho_B) - S(\rho)}{k_B} \geq 0,$$

with equality exactly for product states.

Exercise 34.6 (Bell correlations and CHSH). For a polarization-entangled two-photon state, derive the CHSH bound 2 for local hidden-variable models. Choose four polarization angles for which the quantum correlations attain $2\sqrt{2}$, and derive the predicted coincidence frequencies. As experimental input, use the adjusted local-realistic p-value 2.3×10^{-7} reported by Shalm et al., *Physical Review Letters* **115** (2015). Determine at the 5%, 1%, and five-standard-deviation significance levels, using the one-sided Gaussian five-sigma threshold 2.87×10^{-7}, whether the local-realistic null is excluded. State the tested freedom-of-choice and spacelike-separation assumptions and explain why the result does not exclude arbitrary superdeterministic or signaling models.

34.3 Quantum channels and data processing

Definition 34.8 (Positive and completely positive maps)**.** A linear map $\Phi : A \to B$ between C^*-algebras is *positive* if it preserves positive elements and *completely positive* if every $\mathrm{id}_{M_n} \otimes \Phi$ is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a normal unital completely positive map on observables in the adjoint picture.

Exercise 34.7 (Stinespring and Kraus representations)**.** Prove that a unital completely positive map $\Phi : A \to B(H)$ has a representation $\Phi(a) = V^*\pi(a)V$. In finite dimensions, when $\Phi = \mathcal{N}^*$ is the Heisenberg adjoint of a Schrödinger-picture channel $\mathcal{N}$, derive the paired Kraus formulas

$$\mathcal{N}(\rho) = \sum_i K_i \rho K_i^*, \qquad \Phi(a) = \sum_i K_i^* a K_i, \qquad \sum_i K_i^* K_i = 1.$$

Prove that every channel is unitary evolution on a larger system followed by discarding an environment. Use the bimodule identity to prove that every conditional expectation introduced earlier is completely positive. For a finite POVM $(E_x)_x$, prove that

$$\mathcal{M}(\rho) = \sum_x \mathrm{Tr}(\rho E_x)|x\rangle\langle x|$$

is a quantum-to-classical channel, compute its adjoint, and identify its Stinespring dilation with the Naimark dilation constructed earlier.

Exercise 34.8 (Relative entropy and data processing)**.** Prove the log-sum inequality and deduce $D(p\|q) \geq 0$, with equality exactly when $p = q$. For a stochastic map $K(y \mid x)$, prove

$$D(Kp\|Kq) \leq D(p\|q).$$

Interpret a random variable, finite-resolution detector, or forgotten label as a stochastic map and explain why none can increase distinguishability. Prove $I(X : Y) = S(p_X)/k_B + S(p_Y)/k_B - S(p_{XY})/k_B$, and show that it vanishes exactly when X and Y are independent.

Exercise 34.9 (Amplitude damping reproduces spontaneous emission)**.** For a qubit with ground state $|0\rangle$, excited state $|1\rangle$, and environment vacuum $|0\rangle_E$, let

$$\begin{aligned} &V_t|0\rangle = |0\rangle|0\rangle_E, \\ &V_t|1\rangle = \sqrt{\eta_t}\,|1\rangle|0\rangle_E + \sqrt{1-\eta_t}\,|0\rangle|1\rangle_E, \qquad \eta_t = e^{-t/T_1}. \end{aligned}$$

In the Schrödinger picture, trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|, \qquad \mathcal{N}_t(\rho) = K_0\rho K_0^* + K_1 \rho K_1^*.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive $\rho_{11}(t) = e^{-t/T_1}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-t/(2T_1)}\rho_{01}(0)$. Compare the predicted decay and the environmental dependence of T_1 with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

34.4 Quantum error correction and operator systems

Definition 34.9 (Quantum codes and correctable errors)**.** Let $V : K \to H$ be an isometry between finite-dimensional Hilbert spaces. Its image $C = \operatorname{im} V$, with projection $P = VV^*$, is a *quantum* *code*; K is the logical system and H the physical system. Write $\mathcal{T}(H)$ for the trace-class operators on H. A channel $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ is *correctable on* C if there is a recovery channel $\mathcal{R}$ such that

$$(\mathcal{R} \circ \mathcal{N})(V\rho V^*) = V\rho V^*$$

for every state ρ on K. A *stabilizer code* is the common $+1$ spectral subspace of commuting self-adjoint unitaries. When $H = \bigotimes_{j=1}^n H_j$, the *distance* of C is the least number of tensor factors supporting an operator E for which PEP is not a scalar multiple of P, or $+\infty$ if there is no such operator.

Exercise 34.10 (Knill–Laflamme correction and decoupling)**.** Let $\mathcal{N}(\rho) = \sum_a E_a\rho E_a^*$ and let P project onto a code C. Prove that $\mathcal{N}$ is correctable on C exactly when there is a positive semidefinite matrix (λ_{ab}) such that

$$PE_a^*E_bP = \lambda_{ab}P \qquad \text{for every } a, b.$$

Show that this condition is independent of the chosen Kraus representation, and construct a recovery from the polar decomposition of the combined error map

$$W : C \otimes \ell^2(\{a\}) \longrightarrow H', \qquad W(\psi \otimes \delta_a) = E_a\psi,$$

together with the spectral projections of (λ_{ab}). Deduce that a code of distance d corrects every error supported on at most $\lfloor (d-1)/2 \rfloor$ tensor factors.

For an encoding $V : K \to H_A \otimes H_B$, prove that erasure of B is correctable exactly when the complementary channel $\rho \mapsto \mathrm{Tr}_A(V\rho V^*)$ is constant on logical states. Interpret this as the equivalence between recovery from A and absence of encoded information in B; see Knill–Laflamme, *Physical Review A* **55** (1997).

Definition 34.10 (Operator systems)**.** An *operator system* is a closed self-adjoint linear subspace $S \subseteq B(H)$ containing the identity. Its matrix order is the family of cones

$$M_n(S)_+ = M_n(S) \cap B(H^{\oplus n})_+.$$

Completely positive maps are precisely the maps that preserve these cones at every matrix level.

Definition 34.11 (The confusability operator system of a channel)**.** Let H, H' be finite-dimensional and let $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ be a channel with Kraus operators $E_a : H \to H'$. Its *confusability operator system* is

$$S_{\mathcal{N}} = \mathrm{span}\{E_a^*E_b : a, b\} \subseteq B(H).$$

It is self-adjoint, and it contains the identity because $\sum_a E_a^*E_a = 1_H$.

Definition 34.12 (Protected observable algebras)**.** Let P project onto a code $C \subseteq H$. A unital $*$-subalgebra $\mathcal{A} \subseteq PB(H)P$, whose identity is P, is *protected* or *correctable* for $\mathcal{N}$ if there is a recovery channel $\mathcal{R} : \mathcal{T}(H') \to \mathcal{T}(H)$ such that

$$P(\mathcal{R} \circ \mathcal{N})^*(A)P = A \qquad (A \in \mathcal{A}).$$

Write $\mathcal{A}' = \{X \in PB(H)P : [X, A] = 0 \text{ for all } A \in \mathcal{A}\}$ for the commutant in the code corner.

Exercise 34.11 (Exact operator-algebra quantum error correction)**.** Prove that $\mathcal{A}$ is correctable for $\mathcal{N}$ if and only if

$$[PE_a^*E_bP, A] = 0 \qquad (a, b,\ A \in \mathcal{A}),$$

or, equivalently,

$$PS_{\mathcal{N}}P \subseteq \mathcal{A}'.$$

Derive necessity by applying a Stinespring dilation to the recovery identity, and prove sufficiency by decomposing the finite-dimensional algebra and constructing a recovery on each summand. For $\mathcal{A} = PB(H)P \cong B(C)$, use $\mathcal{A}' = \mathbb{C}P$ to recover the ordinary Knill–Laflamme equations $PE_a^*E_bP = \lambda_{ab}P$. Show that a commutative algebra protects a classical label, while a decomposition

$$\mathcal{A} \cong \bigoplus_{\alpha}(B(H_\alpha^A) \otimes 1_{H_\alpha^B})$$

protects hybrid information: the central label α is classical, the H_α^A factor is quantum, and H_α^B is an unprotected gauge subsystem. Compare with Bény–Kempf–Kribs, *Quantum Error Correction of Observables*, Physical Review A **76** (2007).

35 Operator-Space Applications

35.1 Channel discrimination

Definition 35.1 (Ancilla-assisted channel discrimination)**.** A *one-use discrimination strategy* for channels $\Phi_1, \dots, \Phi_m$ prepares a state on $H \otimes K$, applies $\Phi_j \otimes \mathrm{id}_K$ to the unknown channel input, and measures the output with a POVM. The strategy is *ancilla assisted* when $\dim K > 1$. It is *unambiguous* when every conclusive outcome identifies the channel without error. If channel Φ_j is selected with prior probability η_j, its *inconclusive probability* P_I is the prior-weighted average probability of an inconclusive outcome.

Exercise 35.1 (An ancilla improves measurement-channel discrimination)**.** For $0 < \theta < \pi/4$, define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let $\mathcal{M}, \mathcal{N}$ be the two projective measurement channels with rank-one projection families $\{P_\phi, P_{\phi^\perp}\}$ and $\{P_\psi, P_{\psi^\perp}\}$. Assume that the two channels are selected with equal prior probabilities. Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of $|\phi\rangle$ and $|\psi\rangle$ and prove

$$P_I^{\mathrm{anc}} = \cos(2\theta), \qquad P_I^{\mathrm{sep}} = \frac{1 + \cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for $0 < \theta < \pi/4$ and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).

35.2 Equivariant compression and Toeplitz operator systems

Definition 35.2 (Equivariant compression)**.** Let a compact group G act on a Hilbert space H by a unitary representation U, and let a unital C^*-algebra $A \subseteq B(H)$ be invariant under conjugation by U_g. If a finite-rank orthogonal projection P commutes with every U_g, define

$$S_P = PAP = \{PaP : a \in A\} \subseteq B(PH).$$

The restricted representation $U_g^P = U_g|_{PH}$ acts on S_P by conjugation. We call S_P an *equivariant compression* of A.

Exercise 35.2 (Rotation-covariant Toeplitz compression)**.** Let $A = C(S^1)$ act by multiplication on $H = L^2(S^1)$, with normalized Fourier basis $e_k(\theta) = (2\pi)^{-1/2}e^{ik\theta}$. Let P_N project onto $H_N = \operatorname{span}\{e_k : -N \le k \le N\}$ and set $S_N = P_N A P_N$. Prove

$$\langle e_j, P_N M_f P_N e_k\rangle = \widehat{f}(j-k), \qquad -N \le j,k \le N,$$

and deduce that S_N is exactly the operator system of $(2N+1)$-dimensional Toeplitz matrices.

For $z(\theta) = e^{i\theta}$, put $T_N = P_N M_z P_N$. Show that T_N is a truncated shift and that $T_N T_N^*$ has nonconstant main diagonal. Conclude that S_N is not an algebra, although the C^*-algebra generated by S_N is the full matrix algebra $B(H_N)$.

Finally, let rotations act by $(U_\phi\xi)(\theta) = \xi(\theta - \phi)$. Prove that P_N commutes with U_ϕ and that conjugation by $U_\phi|_{H_N}$ preserves S_N. Verify directly that the cutoff retains rotation covariance although its compressed observables do not form an algebra.

35.3 Extension, tensor norms, and bilinear forms

Exercise 35.3 (Finite-dimensional Arveson–Wittstock extension)**.** Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing C^*-algebra. Let $E \subseteq A$ be an operator space in a finite-dimensional C^*-algebra and let $u : E \to B(H)$ be completely bounded. Deduce, by the 2×2 operator-system construction, that u extends to a map $\widetilde{u} : A \to B(H)$ satisfying $\|\widetilde{u}\|_{cb} = \|u\|_{cb}$.

Definition 35.3 (Operator-space tensor products)**.** The *minimal operator-space tensor norm* on $E \otimes F$ is inherited from concrete inclusions in $B(H \otimes K)$. The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \, \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form $u : E \times F \to \mathbb{C}$ is *jointly completely bounded* when the maps

$$u_n([x_{ij}], [y_{kl}]) = [u(x_{ij}, y_{kl})]_{(i,k),(j,l)}$$

from $M_n(E) \times M_n(F)$ to M_{n^2} are uniformly bounded; the supremum of their norms is $\|u\|_{jcb}$.

Exercise 35.4 (Tensor norms linearize bilinear maps)**.** Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

35.4 Nonlocal games as operator-space tensors

Definition 35.4 (Bell functionals and nonlocal games)**.** A *Bell functional* is an array $M = (M_{xy}^{ab})$ evaluated on conditional distributions by $\langle M, p\rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a,b \mid x,y)$. A two-player *nonlocal game* $G = (\pi, V)$ consists of a question distribution $\pi(x,y)$ and a payoff $0 \le V(a,b \mid x,y) \le 1$. Its classical and entangled values $\omega(G)$ and $\omega^*(G)$ are the optimal expected payoffs using, respectively, shared randomness and local POVMs $\{E_a^x\}_a, \{F_b^y\}_b$ on a shared finite-dimensional state ρ, with $p(a,b \mid x,y) = \operatorname{Tr}(\rho(E_a^x \otimes F_b^y))$. An *XOR game* has binary answers and scores only their parity, equivalently a correlation $\sum_{x,y} M_{xy}\mathbb{E}[a_x b_y]$ for signs a_x, b_y.

Exercise 35.5 (Magic-square pseudotelepathy)**.** The referee chooses $(x, y) \in \{1, 2, 3\}^2$ uniformly. Alice returns a row $a \in \{\pm 1\}^3$ with product $+1$, Bob returns a column $b \in \{\pm 1\}^3$ with product -1, and they win when $a_y = b_x$. Prove by multiplying all row and column constraints that no deterministic classical strategy wins every question, construct one losing on only one question, and hence show $\omega(G_{\mathrm{MS}}) = 8/9$. Using two Bell pairs and a Mermin–Peres square of commuting two-qubit Pauli observables, construct a strategy with $\omega^*(G_{\mathrm{MS}}) = 1$.

Compare the ideal query-by-query predictions with the hyperentangled-photon experiment of Xu et al., *Physical Review Letters* **129** (2022). Explain separately the locality and fair-sampling assumptions in that experiment. Show why a score above 8/9 is a device-independent witness under the stated assumptions.

Definition 35.5 (The tensor of a game)**.** Set

$$E_{X,A} = \ell_1^X(\ell_\infty^A), \qquad \|z\|_{E_{X,A}} = \sum_x \max_a |z_{x,a}|,$$

with its standard dual operator-space structure. The tensor associated to $G = (\pi, V)$ is

$$\widehat{G} = \sum_{x,y,a,b} \pi(x, y) V(a, b \mid x, y)(e_x \otimes e_a) \otimes (e_y \otimes e_b) \in E_{X,A} \otimes E_{Y,B}.$$

Exercise 35.6 (Game values as Banach and operator-space norms)**.** Identify deterministic strategies with extreme points of the relevant unit balls and prove

$$\omega(G) = \|\widehat{G}\|_{E_{X,A} \otimes_\varepsilon E_{Y,B}}, \qquad \omega^*(G) = \|\widehat{G}\|_{E_{X,A} \otimes_{\min} E_{Y,B}}.$$

Equivalently, show that these are the bounded and completely bounded norms of the bilinear payoff form on the dual mixed-norm spaces. Locate in the proof the use of positivity of a game tensor, and explain why the same simple formula need not hold for an arbitrary signed Bell functional. Use Palazuelos–Vidick, *Survey on Nonlocal Games and Operator Space Theory*, *Journal of Mathematical Physics* **57** (2016) as a guide.

36 Random Matrix Theory

Definition 36.1 (Scalar determinants, Vandermonde determinant, and Dyson classes)**.** For $A = (a_{ij}) \in M_N(\mathbb{C})$, define

$$\det A = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) \prod_{i=1}^N a_{i,\sigma(i)}.$$

For $\lambda = (\lambda_1, \ldots, \lambda_N)$, define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N = \prod_{i<j} (\lambda_j - \lambda_i).$$

Let $\mathbb{F} = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$ and set

$$\beta = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}.$$

The Gaussian orthogonal, unitary, and symplectic ensembles, denoted GOE, GUE, and GSE, consist of the self-adjoint matrices $H = H^*$ over $\mathbb{R}, \mathbb{C}$, and $\mathbb{H}$, respectively, with density

$$d\mathbb{P}_{N,\beta}(H) = C_{N,\beta} \exp\Big[-\frac{\beta N}{4\sigma^2} \operatorname{Tr}(H^2)\Big]\, dH.$$

Here H^* is conjugate transpose and dH is Lebesgue measure on the independent real coordinates.

Exercise 36.1 (Eigenvalue–eigenvector Jacobian)**.** Diagonalize $H = U\,\mathrm{diag}(\lambda)U^{-1}$ in each Dyson class. Use the invariant metric $\mathrm{Tr}(dH^2)$ to prove

$$dH = C\,|\Delta(\lambda)|^{\beta}\prod_{i=1}^{N} d\lambda_i\,d\mu(U),$$

with the stabilizer and permutation factors included in C. Integrate over the eigenvectors and derive

$$p_{N,\beta}(\lambda) = \frac{1}{Z_{N,\beta}}\exp\left[-\frac{\beta N}{4\sigma^2}\sum_{i=1}^{N}\lambda_i^2\right]\prod_{i<j}|\lambda_i-\lambda_j|^{\beta}.$$

Definition 36.2 (Beta ensembles and Coulomb gases)**.** For an interval $I \subseteq \mathbb{R}$, a confining potential V, and $\beta > 0$, the *beta ensemble* is

$$d\mathbb{P}_{N,\beta,V}(\lambda) = \frac{1}{Z_{N,\beta,V}}\prod_{i<j}|\lambda_i-\lambda_j|^{\beta}\prod_{i=1}^{N}e^{-\beta N V(\lambda_i)/2}\,d\lambda_i.$$

Its *Coulomb-gas Hamiltonian* is

$$\mathcal{H}_N(\lambda) = \frac{N}{2}\sum_{i=1}^{N}V(\lambda_i) - \sum_{i<j}\log|\lambda_i-\lambda_j|, \qquad d\mathbb{P}_{N,\beta,V} = Z^{-1}e^{-\beta\mathcal{H}_N}\,d\lambda.$$

The Jacobi beta ensemble is the Selberg law on $[0,1]^N$, and the circular beta ensemble is

$$\frac{1}{Z^{\mathrm{circ}}_{N,\beta}}\prod_{i<j}|e^{i\theta_i}-e^{i\theta_j}|^{\beta}\prod_{i=1}^{N}\frac{d\theta_i}{2\pi}.$$

Exercise 36.2 (Logarithmic equilibrium and the semicircle law)**.** For the empirical measure $L_N = N^{-1}\sum_i \delta_{\lambda_i}$, derive the mean-field energy

$$\mathcal{I}_V(\mu) = \int V(x)\,d\mu(x) - \iint \log|x-y|\,d\mu(x)d\mu(y).$$

Prove that its minimizer satisfies

$$V(x) - 2\int \log|x-y|\,d\mu_V(y) = \ell$$

on its support. For $V(x) = x^2/(2\sigma^2)$, solve the singular integral equation and obtain

$$\rho_{\mathrm{sc}}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2-x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x).$$

Definition 36.3 (Wigner matrices and empirical spectral measures)**.** A *Wigner matrix* H_N is an $N\times N$ Hermitian random matrix whose upper-triangular entries are independent and centered, with $\mathbb{E}|H_{ij}|^2 = \sigma^2/N$ off the diagonal and, for every fixed m,

$$\sup_{N,i,j}\mathbb{E}|\sqrt{N}H_{ij}|^m < \infty.$$

Its *empirical spectral measure* is

$$L_{H_N} = \frac{1}{N}\sum_{j=1}^{N}\delta_{\lambda_j(H_N)}, \qquad \int x^k\,dL_{H_N}(x) = \frac{1}{N}\mathrm{Tr}(H_N^k).$$

Exercise 36.3 (Wigner semicircle law)**.** Expand $N^{-1}\mathbb{E}\,\mathrm{Tr}(H_N^k)$ as a sum over closed index walks. Prove that only noncrossing pair-matched walks survive as $N \to \infty$ and that their number is the Catalan number. Prove that the variance tends to zero and conclude that L_{H_N} converges in probability, in moments, to $\rho_{\mathrm{sc}}(x)dx$.

Definition 36.4 (Unfolded spectra and level repulsion)**.** For a smooth approximation $\overline{N}(E)$ to the cumulative level count, define

$$x_j = \overline{N}(E_j), \qquad s_j = x_{j+1} - x_j.$$

The variables x_j are the *unfolded levels*. Their local mean spacing is one. Exact symmetry sectors are unfolded separately. The factor $|\Delta(\lambda)|^\beta$ defines *level repulsion* of order β.

Exercise 36.4 (Wigner surmises and repulsion exponents)**.** For 2×2 GOE and GUE matrices, separate the trace from the traceless coordinates and compute the radial distribution of the eigenvalue gap. After rescaling to unit mean spacing, derive

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \qquad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Use the beta-ensemble density to prove

$$P_\beta(s) \sim c_\beta s^\beta \qquad (s \downarrow 0),$$

and compare it with the Poisson spacing law e^{-s}.

Definition 36.5 (Global, bulk, and edge scales)**.** Order the eigenvalues as $\lambda_1 \le \cdots \le \lambda_N$. Define the classical location γ_j by

$$\frac{j}{N} = \int_{-2\sigma}^{\gamma_j} \rho_{\mathrm{sc}}(x)\, dx.$$

At $E \in (-2\sigma, 2\sigma)$, the bulk coordinate is

$$u = N\rho_{\mathrm{sc}}(E)(\lambda - E),$$

and at the upper soft edge it is

$$\xi = \frac{N^{2/3}}{\sigma}(\lambda - 2\sigma).$$

Exercise 36.5 (Bulk and edge scales)**.** Use the expected count $N\rho_{\mathrm{sc}}(E)dE$ to derive the bulk spacing scale $[N\rho_{\mathrm{sc}}(E)]^{-1}$. Prove

$$\rho_{\mathrm{sc}}(2\sigma - s) \sim \frac{1}{\pi\sigma^{3/2}}\sqrt{s} \qquad (s \downarrow 0),$$

and derive the edge scale $s \asymp \sigma N^{-2/3}$.

Definition 36.6 (Airy kernel and the Tracy–Widom law)**.** Let Ai be the solution of $y'' = xy$ decaying at $+\infty$. Define

$$K_{\mathrm{Ai}}(x, y) = \frac{\mathrm{Ai}(x)\,\mathrm{Ai}'(y) - \mathrm{Ai}'(x)\,\mathrm{Ai}(y)}{x - y}$$

by continuity on the diagonal. The GUE Tracy–Widom distribution is

$$F_2(s) = \det(I - K_{\mathrm{Ai}})_{L^2(s,\infty)}.$$

36.1 Correlation functions and point processes

Definition 36.7 (Point processes and correlation functions)**.** Let (X, μ) be a locally compact measure space. A *simple point process* is a random locally finite counting measure

$$\Xi = \sum_i \delta_{x_i}$$

with distinct points almost surely. Its k-point correlation function ρ_k is defined by

$$\mathbb{E} \sum_{i_1,\ldots,i_k}^{\neq} f(x_{i_1}, \ldots, x_{i_k}) = \int_{X^k} f(x_1, \ldots, x_k)\rho_k(x_1, \ldots, x_k) \prod_{j=1}^{k} d\mu(x_j).$$

For a symmetric joint density p_N of exactly N points,

$$\rho_k(x_1, \ldots, x_k) = \frac{N!}{(N-k)!} \int_{X^{N-k}} p_N(x_1, \ldots, x_N) \prod_{j=k+1}^{N} d\mu(x_j).$$

Exercise 36.6 (Factorial moments and gap probabilities)**.** For a relatively compact measurable set B and $N_B = \Xi(B)$, prove, whenever either side is finite,

$$\mathbb{E}(N_B)_k = \int_{B^k} \rho_k \, d\mu^{\otimes k}, \qquad (N_B)_k = N_B(N_B - 1) \cdots (N_B - k + 1).$$

If $\mathbb{E}[2^{N_B}] < \infty$, use inclusion–exclusion and justify the exchange of expectation and summation to prove

$$\mathbb{P}(N_B = 0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{B^k} \rho_k \, d\mu^{\otimes k}.$$

36.2 Determinantal/Pfaffian processes and Fredholm determinants

Definition 36.8 (Determinantal and Pfaffian point processes)**.** For an antisymmetric matrix $A = (a_{ij}) \in M_{2m}(\mathbb{C})$, define

$$\mathrm{Pf}(A) = \frac{1}{2^m m!} \sum_{\sigma \in S_{2m}} \mathrm{sgn}(\sigma) \prod_{j=1}^{m} a_{\sigma(2j-1),\sigma(2j)}.$$

A point process is *determinantal* with kernel K when

$$\rho_k(x_1, \ldots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^{k}.$$

It is *Pfaffian* with antisymmetric 2×2 matrix kernel $\mathbb{K}$ when

$$\rho_k(x_1, \ldots, x_k) = \mathrm{Pf}[\mathbb{K}(x_i, x_j)]_{i,j=1}^{k}.$$

For a trace-class integral operator K_B on $L^2(B, \mu)$, its *Fredholm determinant* is

$$\det(I + zK_B) = \sum_{k=0}^{\infty} \frac{z^k}{k!} \int_{B^k} \det[K(x_i, x_j)]_{i,j=1}^{k} \, d\mu^{\otimes k}.$$

The Fredholm Pfaffian is defined by the analogous series with Pfaffians of the matrix kernel. A *positive contraction* is an operator K satisfying $0 \leq K \leq I$.

Exercise 36.7 (Counting statistics and Fredholm formulas)**.** For a determinantal process, prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z-1)K_B), \qquad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Derive the corresponding Fredholm-Pfaffian formula for a Pfaffian process. If K_B is a positive trace-class contraction with eigenvalues (λ_j), prove

$$\det(I + zK_B) = \prod_j (1 + z\lambda_j)$$

and that N_B is distributed as a sum of independent Bernoulli variables with parameters λ_j.

36.3 Orthogonal-polynomial ensembles

Definition 36.9 (Orthogonal-polynomial ensembles)**.** Let $w(x)d\mu(x)$ be a measure with finite moments and let p_j be its monic orthogonal polynomials,

$$\int p_j(x)p_k(x)w(x)d\mu(x) = h_j\delta_{jk}.$$

The associated *orthogonal-polynomial ensemble* has density

$$p_N(x_1, \dots, x_N) = \frac{1}{Z_N}\Delta(x)^2 \prod_{i=1}^{N} w(x_i), \qquad Z_N = N! \prod_{j=0}^{N-1} h_j.$$

Its correlation kernel is

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{p_j(x)p_j(y)}{h_j}.$$

Exercise 36.8 (Andréief identity and determinantal correlations)**.** Prove Andréief's identity

$$\int_{X^N} \det[f_i(x_j)] \det[g_i(x_j)] \prod_{j=1}^{N} d\mu(x_j) = N! \det\left[\int_X f_i(x)g_j(x)d\mu(x)\right].$$

Use the Vandermonde determinant to prove

$$p_N(x_1, \dots, x_N) = \frac{1}{N!} \det[K_N(x_i, x_j)]_{i,j=1}^{N}$$

and

$$\rho_k(x_1, \dots, x_k) = \det[K_N(x_i, x_j)]_{i,j=1}^{k}.$$

Exercise 36.9 (Skew-orthogonal ensembles)**.** For an antisymmetric bilinear form $\langle \cdot, \cdot \rangle_s$, construct skew-orthogonal polynomials satisfying

$$\langle p_{2j}, p_{2k+1} \rangle_s = r_j\delta_{jk}, \qquad \langle p_{2j}, p_{2k} \rangle_s = \langle p_{2j+1}, p_{2k+1} \rangle_s = 0.$$

Prove de Bruijn's Pfaffian integration identity and use it to derive the matrix correlation kernels for the GOE and GSE.

36.4 Christoffel–Darboux and integrable kernels

Definition 36.10 (Christoffel–Darboux and integrable kernels)**.** Let $q_j = p_j/\sqrt{h_j}$ and let κ_j be the leading coefficient of q_j. The Christoffel–Darboux kernel is

$$K_N(x,y) = \sqrt{w(x)w(y)}\,\frac{\kappa_{N-1}}{\kappa_N}\,\frac{q_N(x)q_{N-1}(y) - q_{N-1}(x)q_N(y)}{x-y}.$$

An *integrable kernel* is a kernel of the form

$$K(x,y) = \frac{f(x)^\mathsf{T} g(y)}{x-y}, \qquad f(x)^\mathsf{T} g(x) = 0.$$

Exercise 36.10 (Projection and resolvent identities)**.** Derive the Christoffel–Darboux formula from the three-term recurrence. Prove

$$\int K_N(x,z)K_N(z,y)d\mu(z) = K_N(x,y), \qquad \int K_N(x,x)d\mu(x) = N.$$

For an integrable trace-class kernel K, prove that the resolvent $R = K(I-K)^{-1}$ is again integrable and express its numerator using $(I-K)^{-1}f$ and $(I-K^\mathsf{T})^{-1}g$.

36.5 Bulk, soft-edge, and hard-edge universality

Definition 36.11 (Universal limiting kernels)**.** The bulk, soft-edge, and hard-edge kernels are respectively

$$K_{\sin}(u,v) = \frac{\sin \pi(u-v)}{\pi(u-v)},$$

$$K_{\mathrm{Ai}}(u,v) = \frac{\mathrm{Ai}(u)\,\mathrm{Ai}'(v) - \mathrm{Ai}'(u)\,\mathrm{Ai}(v)}{u-v},$$

and, for $\alpha > -1$,

$$K_{\mathrm{Bes}}^{(\alpha)}(u,v) = \frac{J_\alpha(\sqrt{u})\sqrt{v}\,J_\alpha'(\sqrt{v}) - \sqrt{u}\,J_\alpha'(\sqrt{u})J_\alpha(\sqrt{v})}{2(u-v)}.$$

The values on the diagonal are defined by continuity.

Exercise 36.11 (Classical kernel universality)**.** Use the Christoffel–Darboux formula and the asymptotics of the Hermite, Laguerre, and Jacobi polynomials to prove convergence of their rescaled kernels, where applicable, to $K_{\sin}$ in the bulk, K_{Ai} at a soft edge, and $K_{\mathrm{Bes}}^{(\alpha)}$ at a hard edge. Deduce convergence of every correlation function and of the associated Fredholm gap probabilities. Recover the Tracy–Widom law for the largest GUE eigenvalue.

37 Integrable Probability

37.1 RSK and last-passage percolation

Definition 37.1 (Partitions, tableaux, and Schur functions)**.** A partition is a sequence $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ with finite sum $|\lambda| = \sum_i \lambda_i$. A semistandard Young tableau of shape λ has weakly increasing rows and strictly increasing columns. For variables $a = (a_1, \ldots, a_m)$, define

$$s_\lambda(a) = \sum_T \prod_{i=1}^m a_i^{m_i(T)},$$

where $m_i(T)$ is the number of entries equal to i.

Definition 37.2 (RSK correspondence and last-passage time)**.** The Robinson–Schensted–Knuth construction associates

$$A = (a_{ij}) \longmapsto (P, Q)$$

to a matrix of nonnegative integers a pair of semistandard Young tableaux of the same shape λ, with the row and column sums of A giving the contents of P and Q. For weights $a_{ij} \geq 0$, define

$$G(m,n) = \max_{\pi:(1,1)\nearrow(m,n)} \sum_{(i,j)\in\pi} a_{ij}.$$

Exercise 37.1 (Schensted and Greene theorems)**.** Prove the RSK bijection and Greene's theorem. For $A \in M_{m,n}(\mathbb{Z}_{\geq 0})$, deduce

$$\lambda_1 = G(m,n)$$

and identify $\lambda_1 + \cdots + \lambda_r$ with the maximal weight of r disjoint up-right paths. For independent geometric weights with parameters $a_i b_j$, use RSK to derive

$$\mathbb{P}(\lambda) = \prod_{i,j}(1 - a_i b_j) s_\lambda(a) s_\lambda(b)$$

and prove the Cauchy identity

$$\sum_\lambda s_\lambda(a) s_\lambda(b) = \prod_{i,j}(1 - a_i b_j)^{-1}.$$

37.2 Schur and Macdonald processes

Definition 37.3 (Schur measures and processes)**.** For $\mu \subseteq \lambda$, the *skew Schur function* $s_{\lambda/\mu}$ is the tableau sum over the skew diagram $\lambda \setminus \mu$. A *specialization* is an algebra homomorphism from the algebra of symmetric functions to $\mathbb{C}$; it is *Schur-positive* when $s_{\lambda/\mu}(\rho) \geq 0$ for all $\mu \subseteq \lambda$. The *Schur measure* associated to two Schur-positive specializations ρ^+, ρ^- is

$$\mathbb{P}(\lambda) = \frac{s_\lambda(\rho^+) s_\lambda(\rho^-)}{\sum_\nu s_\nu(\rho^+) s_\nu(\rho^-)}.$$

An ascending *Schur process* is a probability measure on $\lambda^{(1)} \subseteq \cdots \subseteq \lambda^{(m)}$ proportional to

$$s_{\lambda^{(1)}}(\rho_1^+) \prod_{r=2}^{m} s_{\lambda^{(r)}/\lambda^{(r-1)}}(\rho_r^+) s_{\lambda^{(m)}}(\rho^-).$$

Exercise 37.2 (Schur processes as determinantal processes)**.** Encode a partition by the Maya diagram

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \tfrac{1}{2} : i \geq 1\}.$$

Use the Cauchy and skew-Cauchy identities to prove that the induced space–time point process is determinantal. Derive its double-contour correlation kernel and express the distribution of λ_1 as a Fredholm determinant. Recover geometric last-passage percolation as a one-time marginal.

Definition 37.4 (Macdonald measures and processes)**.** Let $P_\lambda(\,\cdot\,; q, t)$ be the Macdonald polynomials and let $Q_\lambda = b_\lambda(q,t)P_\lambda$ be their dual basis. Define the skew functions by

$$P_\lambda(x, y) = \sum_\mu P_{\lambda/\mu}(x)P_\mu(y), \qquad Q_\lambda(x, y) = \sum_\mu Q_{\lambda/\mu}(x)Q_\mu(y).$$

For finite specializations a, b, define

$$\Pi(a; b) = \prod_{i,j} \frac{(ta_ib_j; q)_\infty}{(a_ib_j; q)_\infty}.$$

The *Macdonald measure* is

$$\mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a; q, t)Q_\lambda(b; q, t)}{\Pi(a; b)}.$$

The ascending *Macdonald process* is

$$\frac{P_{\lambda^{(1)}}(a^{(1)}) \prod_{r=2}^m P_{\lambda^{(r)}/\lambda^{(r-1)}}(a^{(r)})Q_{\lambda^{(m)}}(b)}{\Pi(a^{(1)}, \dots, a^{(m)}; b)}.$$

Exercise 37.3 (Macdonald observables and degenerations)**.** Apply the commuting Macdonald difference operators to the Cauchy identity and derive contour-integral formulas for the joint moments of

$$\sum_i q^{\lambda_i} t^{N-i}.$$

Prove that $q = t$ gives the Schur process, $q = 0$ gives the Hall–Littlewood process, and $t = 0$ gives the q-Whittaker process. Obtain Jack processes from the limit $q = t^\alpha$, $t \uparrow 1$.

37.3 Dyson Brownian motion and KPZ-type dynamics

Definition 37.5 (Brownian motion and space–time white noise)**.** A *standard Brownian motion* is a continuous real process $(B_t)_{t\geq 0}$ with $B_0 = 0$ and independent increments satisfying

$$B_t - B_s \sim N(0, t - s) \qquad (0 \leq s < t).$$

Itô differentials satisfy

$$dB_i\, dB_j = \delta_{ij}\, dt, \qquad dB_i\, dt = dt^2 = 0.$$

Space–time white noise on $\mathbb{R}_+ \times \mathbb{R}$ is the centered Gaussian generalized process ξ satisfying

$$\mathbb{E}[\xi(\varphi)\xi(\psi)] = \int_{\mathbb{R}_+\times\mathbb{R}} \varphi(t, x)\psi(t, x)\, dt\, dx$$

for test functions φ, ψ.

Definition 37.6 (Dyson Brownian motion)**.** For $\beta \geq 1$ and a confining potential V, *Dyson Brownian motion* is the interacting diffusion

$$d\lambda_i = \sqrt{\frac{2}{\beta N}}\, dB_i + \left(\frac{1}{N}\sum_{j\neq i} \frac{1}{\lambda_i - \lambda_j} - \frac{1}{2}V'(\lambda_i)\right) dt.$$

Its reversible density is

$$Z^{-1} \prod_{i<j} |\lambda_i - \lambda_j|^\beta \prod_i e^{-\beta N V(\lambda_i)/2}.$$

Exercise 37.4 (Matrix diffusion and eigenvalue dynamics)**.** Apply Itô calculus to Hermitian and real-symmetric Ornstein–Uhlenbeck matrix processes and derive Dyson Brownian motion for $\beta = 2$ and $\beta = 1$. Prove noncollision, compute the generator in divergence form, and verify the stated reversible measure. Derive the complex Burgers equation for the large-N Stieltjes transform and, in the unitary case, the sine- and Airy-kernel local equilibria.

Definition 37.7 (TASEP and the KPZ equation)**.** In the totally asymmetric simple exclusion process, particles on $\mathbb{Z}$ attempt rate-one jumps from x to $x+1$, and a jump is suppressed when $x+1$ is occupied. Its integrated current defines a height function. The one-dimensional KPZ equation is

$$\partial_t h = \nu\, \partial_x^2 h + \frac{\lambda}{2}(\partial_x h)^2 + \sqrt{D}\,\xi,$$

where ξ is space–time white noise. Its Cole–Hopf solution is $h = (2\nu/\lambda)\log Z$, where Z solves the corresponding stochastic heat equation with multiplicative noise.

Exercise 37.5 (Last passage, particle dynamics, and KPZ scaling)**.** Use the recursion

$$G(m,n) = \max\{G(m-1,n), G(m,n-1)\} + a_{mn}$$

to identify exponential last-passage times with TASEP currents. For i.i.d. exponential weights, prove

$$\frac{G(n,n)-4n}{2^{4/3}n^{1/3}} \Longrightarrow \chi_2, \qquad \mathbb{P}(\chi_2 \le s) = F_2(s).$$

Derive the 1:2:3 KPZ fluctuation scaling and its spatial Airy process limit.

Exercise 37.6 (Macdonald dynamics and KPZ degenerations)**.** At $t = 0$, intertwine the q-Whittaker difference operators with the q-TASEP generator. Derive a Fredholm-determinant formula for the q-Laplace transform of a particle position. Take the $q \uparrow 1$ limit to the O'Connell–Yor semi-discrete polymer and then its intermediate-disorder limit to the stochastic heat equation and the KPZ equation. Identify the corresponding degenerations of the Macdonald-process observables.

38 Free Probability

38.1 Free independence and additive convolution

Definition 38.1 (Free independence and free cumulants)**.** Unital subalgebras $(A_i)_i$ of (A,φ) are *freely independent* if

$$\varphi(a_1 \cdots a_m) = 0$$

whenever $a_j \in A_{i_j}$, $\varphi(a_j) = 0$, and neighboring indices are distinct. Variables are free when their generated unital subalgebras are.

Let $NC(n)$ be the lattice of noncrossing partitions of $\{1,\dots,n\}$. The *free cumulants* κ_n are the multilinear functionals determined recursively by

$$\varphi(a_1 \cdots a_n) = \sum_{\pi \in NC(n)} \kappa_\pi(a_1,\dots,a_n),$$

where κ_π is the product of the cumulants associated to the blocks of π. Freeness is equivalently the vanishing of every mixed free cumulant.

Definition 38.2 (Semicircular variables)**.** A self-adjoint variable S is *semicircular* with variance σ^2 if its distribution is

$$d\mu_{\mathrm{sc},\sigma}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2 - x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x)\,dx.$$

Equivalently, its only nonzero free cumulant is $\kappa_2(S,S) = \sigma^2$.

Exercise 38.1 (Free central limit theorem)**.** Use noncrossing pairings to prove

$$\varphi(S^{2m+1}) = 0, \qquad \varphi(S^{2m}) = C_m\sigma^{2m}, \qquad C_m = \frac{1}{m+1}\binom{2m}{m}.$$

Let $X_1, X_2, \ldots$ be freely independent, identically distributed, centered self-adjoint variables with $\varphi(X_i^2) = \sigma^2$ and moments of every order. Use multilinearity and vanishing mixed cumulants to prove that the moments of $n^{-1/2}(X_1 + \cdots + X_n)$ converge to these semicircular moments. Compare this free central limit theorem with the Gaussian limit for classically independent variables.

Definition 38.3 (Cauchy and R-transforms)**.** For a compactly supported probability measure μ on $\mathbb{R}$, its *Cauchy transform* is

$$G_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(t)}{z-t}, \qquad z \in \mathbb{C} \setminus \operatorname{supp}\mu.$$

Near infinity it has a local compositional inverse $K_\mu(w) = G_\mu^{\langle -1\rangle}(w)$ near $w = 0$. The *R-transform* is defined by

$$K_\mu(w) = \frac{1}{w} + R_\mu(w), \qquad R_\mu(w) = \sum_{n\geq 1} \kappa_n(\mu) w^{n-1}.$$

If freely independent self-adjoint variables X, Y have laws μ, ν, then the law of $X+Y$ is their *free additive convolution* $\mu \boxplus \nu$, characterized by

$$R_{\mu\boxplus\nu} = R_\mu + R_\nu.$$

Exercise 38.2 (Free convolution and the semicircle equation)**.** Derive the additivity of the R-transform from the vanishing of mixed free cumulants. Show that a centered semicircular law of variance σ^2 has $R(w) = \sigma^2 w$ and hence

$$\sigma^2 G(z)^2 - zG(z) + 1 = 0.$$

Select the solution satisfying $zG(z) \to 1$ at infinity and recover the semicircle density from the Stieltjes inversion formula. Deduce that the free sum of semicircular variables of variances σ_1^2 and σ_2^2 is semicircular of variance $\sigma_1^2 + \sigma_2^2$, and rederive the free central limit theorem at the level of transforms.

38.2 Random matrices and asymptotic freeness

Definition 38.4 (Asymptotic freeness)**.** Random matrix families $A_N^{(1)}, \ldots, A_N^{(r)}$ are *asymptotically free* when their normalized expected joint trace moments converge to the moments of freely independent variables:

$$\lim_{N\to\infty} \frac{1}{N}\mathbb{E}\operatorname{Tr} P(A_N^{(1)}, \ldots, A_N^{(r)}) = \varphi(P(a_1, \ldots, a_r))$$

for every noncommutative polynomial P.

Exercise 38.3 (Asymptotic freeness of independent GUE matrices)**.** For independent GUE matrices $H_N^{(1)}, \dots, H_N^{(r)}$, apply Wick's rule to normalized mixed traces of centered polynomials. Prove that crossing pairings are lower order and that mixed free cumulants vanish in the limit. Conclude that the family converges in joint moments to freely independent semicircular variables.

39 Partitions and Generating Functions

39.1 Partition generating functions and finite boxes

Definition 39.1 (Partitions and q-Pochhammer symbols)**.** An *integer partition* of $n \geq 0$ is a finite nonincreasing sequence $\lambda = (\lambda_1, \dots, \lambda_r)$ of positive integers with $|\lambda| = \sum_i \lambda_i = n$. Let $p(n)$ be the number of partitions, with $p(0) = 1$. For a formal variable q, set

$$(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1 - aq^j).$$

Analytically, the infinite product converges for $|q| < 1$. Euler's partition generating function is

$$P(q) = \sum_{n\geq 0} p(n)q^n = \frac{1}{(q;q)_\infty}.$$

Exercise 39.1 (Bosonic and fermionic occupation numbers)**.** Consider oscillator modes with one-particle energies $\hbar\omega, 2\hbar\omega, 3\hbar\omega, \dots$. Show that a bosonic Fock state of excitation energy $n\hbar\omega$ is a partition of n, and derive

$$\prod_{m\geq 1} \frac{1}{1 - q^m} = \sum_{n\geq 0} p(n)q^n.$$

For fermionic occupation numbers in $\{0, 1\}$, show that $\prod_{m\geq 1}(1 + q^m)$ counts partitions into distinct parts. Construct a bijection, by separating powers of two in the multiplicities, between partitions into distinct parts and partitions into odd parts, and conclude

$$\prod_{m\geq 1}(1 + q^m) = \prod_{m\geq 1} \frac{1}{1 - q^{2m-1}}.$$

Interpret the two products as different quasiparticle descriptions with the same exact degeneracies.

Definition 39.2 (Gaussian binomial coefficients)**.** For $0 \leq r \leq n$, the *Gaussian binomial coefficient* is

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_r (q;q)_{n-r}},$$

and it is zero outside this range.

Exercise 39.2 (Gaussian binomial theorem)**.** Derive the recurrence

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n-1 \\ r \end{bmatrix}_q + q^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_q.$$

Prove that $\left[\begin{smallmatrix} n \\ r \end{smallmatrix}\right]_q$ is a polynomial in q with nonnegative integer coefficients and that the coefficient of q^m counts partitions of m whose Young diagrams fit inside an $r \times (n-r)$ rectangle.

39.2 Rogers–Ramanujan identities and exclusion rules

Definition 39.3 (Rogers–Ramanujan sum and product series)**.** For $|q| < 1$, define

$$G_{\mathrm{sum}}(q) = \sum_{r\geq 0} \frac{q^{r^2}}{(q;q)_r}, \qquad H_{\mathrm{sum}}(q) = \sum_{r\geq 0} \frac{q^{r(r+1)}}{(q;q)_r},$$

$$G_{\mathrm{prod}}(q) = \frac{1}{(q;q^5)_\infty (q^4;q^5)_\infty}, \qquad H_{\mathrm{prod}}(q) = \frac{1}{(q^2;q^5)_\infty (q^3;q^5)_\infty}.$$

Exercise 39.3 (Difference conditions and Rogers–Ramanujan counting)**.** Fix the number r of parts. Subtract the staircase $(2r-1, 2r-3, \ldots, 1)$ to prove that $q^{r^2}/(q;q)_r$ generates partitions with exactly r parts whose successive parts differ by at least two. Use the staircase $(2r, 2r-2, \ldots, 2)$ for the same difference condition with smallest part at least two. Interpret the product sides as partitions into parts congruent to 1 or 4 modulo 5, and to 2 or 3 modulo 5, respectively. Compute all four series through q^{30} using finite products and finite sums, and verify coefficientwise that $G_{\mathrm{sum}} = G_{\mathrm{prod}}$ and $H_{\mathrm{sum}} = H_{\mathrm{prod}}$ to that order. Translate the coefficient equalities into finite tests of the two partition identities and identify which further step would be required for a proof in all degrees.

40 CFT Characters and Integrable Models

40.1 Virasoro representations and character formulas

Definition 40.1 (The Virasoro algebra and its characters)**.** The *Virasoro algebra* has generators L_n for $n \in \mathbb{Z}$ and a central element acting by c, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

A highest-weight vector $|h\rangle$ satisfies $L_0|h\rangle = h|h\rangle$ and $L_n|h\rangle = 0$ for $n > 0$. For a positive-energy module H_a with finite-dimensional L_0 eigenspaces, its *character* is

$$\chi_a(\tau) = \mathrm{Tr}_{H_a}\, q^{L_0 - c/24}.$$

The shift by $-c/24$ is the Casimir energy produced by mapping the plane to the cylinder.

Exercise 40.1 (Descendants, partitions, and null states)**.** Use the Virasoro commutation relations to move positive modes to the right and prove directly the triangular Poincaré–Birkhoff–Witt statement that the ordered monomials form a basis of the Verma module:

$$L_{-\lambda_1} \cdots L_{-\lambda_r}|h\rangle, \qquad \lambda_1 \geq \cdots \geq \lambda_r \geq 1.$$

Before quotienting by null vectors, deduce the Verma-module character

$$\chi_{c,h}^{\mathrm{Verma}}(\tau) = \frac{q^{h-c/24}}{(q;q)_\infty}.$$

For the vacuum, impose $L_{-1}|0\rangle = 0$ and find the resulting generic vacuum descendant product. Explain why additional null vectors subtract states and turn unrestricted partition numbers into the more structured q-series of minimal models.

40.2 Modular invariance and exact state counting

Definition 40.2 (Rational characters and modular invariants)**.** A finite family of positive-energy sectors H_a is a *rational modular character system* when its characters span a representation of the modular group,

$$\chi_a(-1/\tau) = \sum_b S_{ab}\chi_b(\tau), \qquad \chi_a(\tau+1) = \sum_b T_{ab}\chi_b(\tau).$$

A full torus partition function has the form

$$Z(\tau,\bar\tau) = \sum_{a,b} M_{ab}\chi_a(\tau)\overline{\chi_b(\tau)}$$

with nonnegative integer multiplicities. It is modular invariant when M commutes with S and T.

Exercise 40.2 (Exact state counting in the Ising CFT)**.** For the Ising theory with $c = 1/2$ and primary weights $0, 1/2, 1/16$, take

$$\begin{aligned}
\chi_0 &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}}\right),\\
\chi_{1/2} &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}}\right),\\
\chi_{1/16} &= \frac{1}{\sqrt{2}}\sqrt{\frac{\vartheta_2}{\eta}}.
\end{aligned}$$

Expand each as $q^{h-c/24}$ times a power series and interpret its first coefficients as exact descendant degeneracies. Derive

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \qquad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Verify the modular invariance of $|\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$. Using the Verlinde formula

$$N_{ab}{}^{c} = \sum_x \frac{S_{ax}S_{bx}\overline{S_{cx}}}{S_{0x}},$$

recover the Ising fusion rules. Relate these towers to the critical Ising spectroscopy exercise in Quantum Topology.

Exercise 40.3 (The Cardy density of states)**.** Let a modular-invariant unitary CFT have discrete spectrum, unique vacuum, and left and right central charges $c, \bar c$. Set $\tau = i\beta/(2\pi)$ and use S-invariance to show that the high-temperature partition function is controlled by the vacuum at inverse temperature $4\pi^2/\beta$. A saddle-point inverse Laplace transform then gives, for large left and right excitation numbers,

$$\log\rho(N,\bar N) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{\bar c\bar N}{6}}.$$

State how c is replaced by $c_{\text{eff}} = c - 24h_{\min}$ in a nonunitary theory. For a nonrotating state with $c = \bar c$ and $N = \bar N$, recover the thermal entropy used in the BTZ exercise and hence reproduce the horizon entropy from boundary-state counting; see Cardy, *Nuclear Physics B* **270** (1986).

40.3 Integrable lattice models

Definition 40.3 (Transfer matrices and integrability)**.** For a two-dimensional lattice model on a cylinder, a *row transfer matrix* $T_N(u)$ propagates one width-N row to the next and depends on a spectral parameter u. The model is *integrable* when it admits a nontrivial commuting family

$$[T_N(u), T_N(v)] = 0 \qquad \text{for every } u, v,$$

usually obtained from a Yang–Baxter or star–triangle relation. Commuting transfer matrices permit simultaneous spectral analysis, but integrability alone does not guarantee that every finite-size degeneracy has a simple closed formula.

Exercise 40.4 (Hard hexagons and exact finite-volume counting)**.** In the hard-hexagon model, occupied vertices of a triangular lattice may not be adjacent. On an $N \times M$ torus define

$$Z_{N,M}(z) = \sum_{I \text{ independent}} z^{|I|}.$$

Index a row transfer matrix by cyclic binary rows and set its matrix element to zero when two consecutive rows violate nearest-neighbor exclusion, and otherwise to $z^{(|a|+|b|)/2}$. Prove $Z_{N,M}(z) = \mathrm{Tr}(T_N(z)^M)$ and compute the polynomial for several small sizes. Show that the coefficients are exact state counts, whereas the largest eigenvalue controls the thermodynamic free energy.

Verify the algebraic identity

$$z_c = \frac{11 + 5\sqrt{5}}{2} = \left(\frac{1 + \sqrt{5}}{2}\right)^5,$$

and, for the finite matrices already constructed, evaluate the largest eigenvalue and its first two z-derivatives near z_c. Plot their change with N and explain why finite matrices remain analytic even when their large-size limit may develop a singularity.

41 Topological Structures in Holography

41.1 Ryu–Takayanagi surfaces and entanglement wedges

Definition 41.1 (Relative homology and the Ryu–Takayanagi surface)**.** Let Σ be a static bulk time slice with conformal boundary B, and let $A \subset B$ be a spatial region. A codimension-one surface $\gamma \subset \Sigma$ satisfies the *RT homology constraint* for A if

$$\partial\gamma = \partial A \quad \text{and} \quad \partial r_A = A \cup \gamma$$

with the appropriate orientations for some bulk region $r_A \subset \Sigma$. Equivalently, after regulating the conformal boundary, the cycle $A - \gamma$ is null-homologous in the bulk.

In a static state with a semiclassical asymptotically anti-de Sitter dual, the *Ryu–Takayanagi prescription* at leading order is

$$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N},$$

where γ_A has least area among surfaces satisfying the anchoring and homology conditions. The cutoff used to regulate its area is the geometric counterpart of the field-theory ultraviolet cutoff.

Exercise 41.1 (An interval in AdS_3)**.** On the time-zero slice of Poincaré AdS_3, use

$$ds^2 = L^2 \frac{dz^2 + dx^2}{z^2}, \qquad z > 0,$$

and regulate the boundary at $z = \epsilon$. Show that the geodesic anchored at the endpoints of an interval of length ℓ is a semicircle orthogonal to $z = 0$, and compute

$$\mathrm{Length}(\gamma_A) = 2L \log \frac{\ell}{\epsilon} + O(\epsilon^2/\ell^2).$$

Using the Brown–Henneaux relation $c = 3L/(2G_3)$, derive

$$S(A) = \frac{c}{3} \log \frac{\ell}{\epsilon} + O(1).$$

Compare this with the replica calculation for a vacuum interval in a two-dimensional CFT. State clearly that the equality is a test of the holographic dictionary in its semiclassical regime, not a derivation of AdS/CFT for arbitrary CFTs; see Ryu–Takayanagi, *Physical Review Letters* **96** (2006) and Brown–Henneaux, *Communications in Mathematical Physics* **104** (1986).

Definition 41.2 (Entanglement wedges, HRT surfaces, and quantum corrections)**.** For a static RT surface, the *entanglement wedge* of A is the domain of dependence of the bulk region r_A bounded by $A \cup \gamma_A$. In a time-dependent spacetime, the *Hubeny–Rangamani–Takayanagi surface* is instead an extremal codimension-two spacetime surface anchored at ∂A and homologous to A, with least area among the appropriate extremal candidates.

Beyond leading classical order, a *quantum extremal surface* extremizes the generalized entropy

$$S_{\mathrm{gen}}(\gamma) = \frac{\mathrm{Area}(\gamma)}{4G_N} + S_{\mathrm{bulk}}(r_A),$$

and the entropy is obtained from the least admissible extremum.

Exercise 41.2 (Topology change for two intervals)**.** On a constant-time slice of Poincaré AdS_3, let $A = [x_1, x_2]$ and $B = [x_3, x_4]$ with $x_1 < x_2 < x_3 < x_4$. The two admissible pairings of boundary endpoints have regulated lengths

$$L_{\mathrm{disc}} = 2L \log \frac{x_2 - x_1}{\epsilon} + 2L \log \frac{x_4 - x_3}{\epsilon},$$

$$L_{\mathrm{conn}} = 2L \log \frac{x_4 - x_1}{\epsilon} + 2L \log \frac{x_3 - x_2}{\epsilon}.$$

Show that the RT surface changes topology when these two expressions cross, and derive

$$I(A : B) = \max\left\{0, \frac{c}{3} \log \frac{(x_2 - x_1)(x_4 - x_3)}{(x_4 - x_1)(x_3 - x_2)}\right\}.$$

Explain why vanishing leading-order mutual information corresponds to a disconnected classical entanglement wedge, while finite-G_N bulk correlations need not vanish exactly.

Exercise 41.3 (Entropy inequalities from cutting and regluing)**.** For disjoint A and B in a static bulk slice, draw minimal representatives for the regions needed in subadditivity and strong subadditivity.

Cut the surfaces where they intersect and reglue the pieces into admissible, not necessarily minimal, surfaces for the opposite sides of

$$S(A) + S(B) \geq S(A \cup B),$$

and

$$S(AB) + S(BC) \geq S(B) + S(ABC).$$

Use minimality to prove the inequalities. Identify where the relative homology constraint is needed to ensure that the reglued surfaces are admissible. Explain why this argument is geometric and classical and why the bulk-entropy term must be included at quantum order.

41.2 Black-hole topology and entropy

Definition 41.3 (The BTZ horizon and thermofield-double topology)**.** The nonrotating BTZ black hole has metric

$$ds^2 = -\frac{r^2 - r_+^2}{L^2}\, dt^2 + \frac{L^2}{r^2 - r_+^2}\, dr^2 + r^2 d\phi^2, \qquad \phi \sim \phi + 2\pi,$$

temperature $T = r_+/(2\pi L^2)$, and horizon entropy

$$S_{\mathrm{BH}} = \frac{2\pi r_+}{4G_3}.$$

Its maximally extended geometry has two asymptotic boundaries and is dual, in the standard semiclassical dictionary, to a thermofield-double state of two CFT copies. The bifurcation circle is homologous to either complete boundary separately, but not to the union of both boundaries.

Exercise 41.4 (Homology, horizons, and black-hole entropy)**.** For the pure thermofield-double state on $H_L \otimes H_R$, prove for its reduced states that $S(\rho_L) = S(\rho_R)$ while the joint entropy vanishes. Apply the RT homology constraint to show that the surface for one complete boundary is the bifurcation circle, while the empty surface is admissible for the union of both boundaries. Recover $S(H_L) = 2\pi r_+/(4G_3)$ and interpret the result as entanglement between the two boundary theories. For a boundary circle of circumference $2\pi L$, use $\beta = 2\pi L^2/r_+$ and $c = 3L/(2G_3)$ to verify that the high-temperature CFT expression

$$S_{\mathrm{thermal}} = \frac{\pi c}{3}\frac{2\pi L}{\beta}$$

equals S_{BH}. In the Arithmetic part this thermal formula will be recovered from modular invariance and asymptotic state counting. Consult Hubeny–Rangamani–Takayanagi, *Journal of High Energy Physics* **07** (2007) for the covariant prescription and Faulkner–Lewkowycz–Maldacena, *Journal of High Energy Physics* **11** (2013) for the bulk quantum correction, and Engelhardt–Wall, *Journal of High Energy Physics* **01** (2015) for quantum extremal surfaces.